Measurement of $\nu_\mu + \bar{\nu}_\mu$ Charged-Current Single Pion Production Differential Cross Sections on Argon with the MicroBooNE Detector in the NuMI Beam

Analysis Note, Full FHC, RHC, and Combined Exposure

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`d5c929d · 51 extractions · blind`

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Abstract

We present a measurement of $\nu_\mu + \bar{\nu}_\mu$ charged-current single charged-pion (CC$\pi^\pm$) differential cross sections on argon-40 using the full MicroBooNE NuMI exposure, in three configurations: forward horn current (FHC; Runs 1, 2, 4, 5; 8.857×10^{20} protons on target, POT), reverse horn current (RHC; Runs 1, 2, 3, 4a, 4b, 4c; 11.082×10^{20} POT), and their combination (19.939×10^{20} POT). The analysis is blind: results are extracted on Poisson-thrown central-value fake data. Events are selected requiring a reconstructed muon candidate track and exactly one contained pion candidate track emerging from a fiducial neutrino vertex, with zero reconstructed electromagnetic showers and a muon–pion opening angle less than 2.6 rad. In the measured phase space the selection achieves $\approx$55% signal purity and $\approx$15% efficiency (Table 10). Differential cross sections are extracted for each configuration via Wiener-SVD unfolding in six kinematic variables: muon momentum p_μ, pion momentum p_π, muon polar angle $\cos\theta_\mu$, pion polar angle $\cos\theta_\pi$, the muon–pion opening angle $\theta_{\mu\pi}$, and the muon polar angle in its θ parameterisation. Each configuration is normalised with its own integrated $\nu_\mu + \bar{\nu}_\mu$ flux (FHC 6.81159, RHC 6.44646, combined 6.60865×10^{-10} cm^{-2} POT^{-1}; new-G4 GEANT4 values above 60 MeV, Sec. 4); the RHC value is lower because its beam is $\bar{\nu}_\mu$-dominant. The resulting Wiener-SVD integrated cross sections are of order 0.58–0.86×10^{-38} cm^2/Ar (Table 18), with RHC *below* FHC in every observable. That deficit is a property of the injected sample rather than a normalisation artefact: it is present in the fake-data truth (RHC/FHC = 0.785) and in the GENIE tune (0.779), two independent model inputs which agree, while the exposures sum exactly and each configuration carries its own flux. The measured ratio (0.900) in fact *understates* the deficit, because the extraction distorts it toward unity. For the current fake-data extraction the injected central-value sample corresponds to $\approx$0.75$\times$ the standalone GENIE v3 prediction in this phase space, consistent with the known over-prediction of CC$\pi^\pm$ production in current neutrino interaction generators. No beam-on data is unblinded in this note: every “data” point shown is Poisson-thrown from the central-value Monte Carlo (§5), so no statement here is a measurement of a real

deficit. The extraction is validated by a same-sample self-closure, and cross-checked against the D'Agostini iterative-Bayesian method: its flat, iteration-stable integral (27–33% above Wiener-SVD, and flat to $\leq 2.1\%$ across observables where the Wiener-SVD spread is 16–37%) confirms that the observable-to-observable spread of the Wiener-SVD integrated cross sections is the additional-smearing matrix A_C, not a physical or acceptance effect (Sec. 4.5 of the technical supplement). All results use the XSecAnalyzer framework on the resolution-optimised binning of Sec. 6 of the technical supplement. The unfolded integral reproduces the fake-data truth to 1.007 (identity regularisation) and 1.001 (second derivative) over the eighteen inclusive extractions. Earlier versions of this note carried a 13% excess whose origin was unknown; it has since been traced to a double application of the central-value weight to the externally-thrown fake data, and correcting it moved every one of the fifty-one extractions by between 3.8% and 12.9% (§15). **Scope.** The primary results are the total cross section, the angular observables, $d\sigma/dp_\mu$, and a deliberately coarse two-bin $d\sigma/dp_\pi$. The proton-tagged subsample is secondary scope because the exposure supports an integrated cross section rather than a resonance shape, so the differential $W_{\pi p}$ is withdrawn; its systematics are otherwise complete. All fifty-one extractions carry a full covariance including detector variations, the proton-tagged ones having been rebuilt and re-unfolded for that purpose on 2026-08-31 — they previously carried none, the detector variation samples having been processed but never referenced by the proton-tagged configuration. Sec. 11.1 gives a per-result status table. **Validation.** Two results bound what the extraction can be trusted to do. An ensemble of thirty-two independent Poisson realisations, each carried through the whole chain, gives per-bin pull widths of 1.04 ($d\sigma/dp_\mu$) and 0.96 ($d\sigma/d\cos\theta_\mu$) against the statistical covariance, with 68% and 95% interval coverage recovered: the statistical uncertainties are correctly sized, and the observable whose additional smearing is most severe behaves no differently from the best-conditioned one. Separately, all fifty-one extractions rebuild bit-identically from the processed ntuples with every cached universe deleted, so the released numbers are regenerable from the raw inputs rather than from a preserved intermediate. The same ensemble shows a small negative bias, 2.5% on the integral, which is stated rather than corrected. **Companion documents.** This note describes the current state of the analysis. The validation and diagnostic studies behind it — the pion-momentum estimator survey, the binning and threshold optimisation, and the closure programme — are in the technical validation supplement. Every correction that changed a result, with the values before and after, is in the change log. The additional-smearing and covariance matrices for all fifty-one extractions are released as text in `report/data_release/`; a prediction cannot be compared with these results without the former, nor used quantitatively without the latter.

Contents

1 Introduction

Charged-current single charged-pion (CC$\pi^{\pm}$) production,

$$\nu_\mu(\bar{\nu}_\mu) + \mathrm{Ar} \longrightarrow \mu^{\mp} + \pi^{\pm} + X, \tag{1}$$

is among the most frequent neutrino interaction channels at energies of 1–5 GeV and constitutes a significant background for oscillation analyses at the same energy scale, most notably for ν_e appearance measurements at experiments such as NOvA [3] and T2K [4], as well as the future DUNE [5]. Despite its importance, the cross section on nuclear targets remains poorly constrained: major neutrino interaction generators (GENIE v3, NuWro, NEUT) disagree by 20–30% in absolute rate and show significant shape differences for individual kinematic variables.

MicroBooNE is a single-phase liquid argon time-projection chamber (LArTPC) at Fermilab that operated from 2015–2021, providing the world's highest-statistics ν_μ-on-argon dataset below 5 GeV during its NuMI running period. This note presents a simultaneous measurement of five differential cross sections in the CC$\pi^{\pm}$ channel using the full NuMI exposure in FHC, RHC, and combined configurations, employing Pandora pattern recognition [6] and Wiener-SVD unfolding [7].

2 Theoretical Background and Prior CCπ^+ Measurements

2.1 Production Mechanisms

At neutrino energies of 0.5–5 GeV, charged-current single charged-pion production proceeds through three mechanisms.

Baryon resonance production dominates at $E_\nu \lesssim 2$ GeV. The neutrino excites the struck nucleon to a baryon resonance, most prominently the $\Delta(1232)$ isobar, which then decays to a nucleon and pion:

$$\nu_\mu n \rightarrow \mu^- \Delta^{++} \rightarrow \mu^- \pi^+ p, \qquad \nu_\mu p \rightarrow \mu^- \Delta^{+} \rightarrow \mu^- \pi^0 p \text{ (or } \pi^+ n). \tag{2}$$

The standard model is due to Rein and Sehgal [14], which uses the Feynman–Kislinger–Ravndal (FKR) quark model to compute form factors for all baryon resonances up to $W \lesssim 2$ GeV. The modern implementation used in GENIE for resonance (RES) production is the Kuzmin–Lyubushkin–Naumov Berger–Sehgal (KLN-BS) model [15, 16, 17, 18], which combines the lepton-mass-corrected vector and axial form factors of Kuzmin *et al.* [15] with the updated $\Delta(1232)$ electromagnetic form factors of Graczyk and Sobczyk [16] and the improved treatment of the Δ line shape, the N–Δ interference with non-resonant amplitudes, and lepton-mass effects of Berger and Sehgal [17]. Higher resonances ($N(1440)P_{11}$, $N(1520)D_{13}$, etc.) are included up to $W \lesssim 2$ GeV and contribute at $E_\nu \gtrsim 1.5$ GeV.

Non-resonant and DIS backgrounds. A non-resonant continuum contributes at all W and interferes (constructively or destructively by isospin) with the resonant amplitude. Deep-inelastic scattering (DIS) becomes significant above $W \approx 1.6$ GeV. Together these processes represent ~20% of the CC$\pi^{\pm}$ rate in the energy range of this analysis and are modelled in GENIE via the Bodek–Yang formalism.

Coherent pion production. The neutrino can scatter coherently off the entire nucleus, $\nu_\mu A \rightarrow \mu^- \pi^+ A^*$, leaving the nucleus in its ground state. The cross section is small ($\lesssim 10\%$ of the total) below 5 GeV and is treated as a background in this analysis.

2.2 Nuclear Effects and Final-State Interactions

On a heavy nucleus such as argon ($A = 40$), two classes of nuclear correction modify the CC$\pi^{\pm}$ rate and kinematics significantly.

Initial-state effects. Fermi motion of the struck nucleon broadens kinematic distributions, and Pauli blocking suppresses transitions to low-momentum nucleon final states near threshold. Nuclear binding also shifts the effective nucleon mass relative to free-nucleon predictions.

Final-state interactions (FSI). The produced pion must traverse the nuclear medium before being detected. Intranuclear rescattering occurs via:

- *Pion absorption* ($\pi A \to NN\ldots$): probability $\sim$25–30% on Ar; converts CC$\pi^{\pm}$ events into CC0π topology.
- *Charge exchange* ($\pi^{+}n \to \pi^{0}p$): produces a CCπ^{0} event.
- *Quasi-elastic scattering*: pion survives but changes direction; modifies p_π and θ_π distributions.
- *Inelastic production* ($\pi N \to \pi\pi N$): rare below $p_\pi \approx 400$ MeV/c.

FSI are modelled in GENIE via the INTRANUKE (hA) cascade. Because absorption and charge exchange mix CC$\pi^{\pm}$, CCπ^{0}, and CC0π topologies at the $\sim$30% level, the signal in any CCπ analysis is necessarily defined by the final-state particle content observed in the detector, not by the underlying production process (Section 6).

2.3 Generator Predictions and Known Tensions

Major neutrino event generators reproduce the inclusive CC cross section to 10–15% but show systematic spread in the CC$\pi^{\pm}$ rate, owing to differences in the resonance model, the treatment of the transition region between resonance and DIS, and the FSI cascade.

GENIE v3 [9]. For resonance production GENIE implements the KLN-BS model [15, 16, 17, 18] for the full tower of 18 baryon resonances up to $W \lesssim 2$ GeV, including the $\Delta(1232)P_{33}$, $N(1440)P_{11}$ (Roper), $N(1520)D_{13}$, $N(1535)S_{11}$, $\Delta(1620)S_{31}$, $\Delta(1700)D_{33}$, $N(1680)F_{15}$, $\Delta(1950)F_{37}$, and their isospin partners. The hadronic invariant mass range $1.0 < W < 1.7$ GeV is bridged by a resonance–DIS transition using the AGKY (Andreopoulos–Gallagher–Kehayias–Yang) model, which applies KNO scaling in the transition region and the Bodek–Yang DIS model above $W \approx 1.7$ GeV. Non-resonant single-pion production (via the Rein–Sehgal non-resonant amplitude) is included and interferes with the resonant contributions. Coherent π^{+} production uses the Rein–Sehgal coherent model. FSI are handled by the INTRANUKE hA empirical cascade, which propagates hadrons step-by-step using free-nucleon πN cross sections scaled by an effective nuclear density; the alternative hN microscopic cascade is also available. GENIE is documented to over-predict the CC$\pi^{\pm}$ rate by 20–35% relative to bubble-chamber and MiniBooNE data when normalised to the BNL dataset; the over-prediction is reduced but not eliminated in the AR23 tune.

NuWro. NuWro models resonance production via the $\Delta(1232)$ only; no higher baryon resonances are included. The Δ is treated with the Paschos–Yu formalism for the Δ excitation amplitude, retaining the full angular structure of the pion decay. A smooth analytic interpolation connects the single-Δ resonance region to the DIS regime at higher hadronic invariant mass ($W \gtrsim 1.4$ GeV), using Bodek–Yang structure functions for the deep inelastic contribution; the transition avoids a hard boundary between resonance and continuum. Non-resonant single-pion production is included as a separate amplitude. FSI are implemented using the microscopic Salcedo–Oset model, which treats pion propagation in nuclear matter through the pion self-energy in the random-phase approximation, naturally accounting for medium modifications such as Δ broadening and pion absorption via two-body processes ($\pi NN \to NN$) that are absent in purely cascade-based approaches. NuWro has historically shown better agreement with MiniBooNE and bubble-chamber data than GENIE, partly attributable to the absence of overestimated higher-resonance contributions.

NEUT [19]. NEUT, the generator used by T2K and Super-Kamiokande, implements resonance production via a modified Rein–Sehgal model including 18 resonances up to $W \lesssim 2$ GeV, with Graczyk–Sobczyk [16] electromagnetic form factors for the $\Delta(1232)$. The axial mass M_A^{RES} is a tunable parameter; T2K applies a data-driven tune that reduces the overall CC$\pi^{\pm}$ normalisation by $\sim$10–20% relative to the default. The transition to DIS is handled similarly to GENIE, using GRV98 LO parton distribution functions with Bodek–Yang corrections above $W \approx 2$ GeV. Coherent pion production follows the Rein–Sehgal coherent model with a Berger–Sehgal-style correction for finite lepton mass. FSI are modelled by a custom NEUT intranuclear cascade that tracks each produced hadron through the nucleus using step-by-step interactions sampled from πN cross sections, with nuclear density and mean-free-path corrections derived from the optical potential.

GiBUU. The Giessen Boltzmann–Uehling–Uhlenbeck (GiBUU) generator takes a fundamentally different approach: instead of a probabilistic step-by-step cascade, it solves a coupled set of kinetic transport equations for the full hadronic system inside the nuclear medium. Resonance production includes the $\Delta(1232)$ and higher resonances whose in-medium spectral functions are broadened by collisional damping; the Δ width in the nuclear medium is substantially larger than its free-space value of $\Gamma_0 \approx 118$ MeV, suppressing threshold production. Pion propagation after production is governed by a mean-field potential (the pion optical potential), which shifts pion momenta as they traverse the nucleus – an effect absent in cascade models and particularly important for low-momentum pions ($p_\pi \lesssim 200$ MeV/c) near the pion production threshold. The transition to DIS uses PYTHIA for string fragmentation above $W \approx 1.6$ GeV. GiBUU predictions for pion angular distributions and for the backward-hemisphere $d\sigma/d\cos\theta_\pi$ can differ substantially from cascade-based generators, making it a valuable cross-check of FSI systematics.

The $\sim$25–27% data/MC deficit observed in the present analysis (Sec. B of the technical supplement) is consistent with the GENIE over-prediction pattern seen across all of the experimental results surveyed below.

2.4 Previous Experimental Measurements

Table 1 summarises published CCπ^+ measurements on nuclear targets ordered chronologically.

Table 1: Selected CCπ^+ cross-section measurements on nuclear targets. $\langle E_\nu \rangle$ is the approximate mean or peak neutrino energy. "Type" indicates whether a total integrated (T) or differential (D) cross section was reported. "Target" gives the nucleus primarily used for cross-section extraction.

Experiment	Target	$\langle E_\nu \rangle$ (GeV)	Type	Key result / reference
ANL 12-ft [20]	D	1.0	T	Bubble chamber; foundational σ vs E_ν
BNL 7-ft [21]	D	1.5	T	Higher statistics; 15% normalisation tension with ANL
K2K [22]	C	1.3	T	First accelerator CCπ^+ on carbon near detector
MiniBooNE [23]	CH_2	0.7	D	Q^2, W, p_μ, θ_μ; 20% excess over NUANCE
MINERvA [24]	CH	3.6	D	p_π, θ_π, Adler angles; NuMI medium-energy
T2K ND280 [27]	CH	0.85	D	p_μ, $\cos\theta_\mu$; off-axis beam similar to this work
ArgoNEUT [28]	Ar	9.6	T	First CCπ^+ on Ar; 115 ν + 158 $\bar{\nu}$ events (1.25×10^{20} POT)
MicroBooNE BNB [30]	Ar	0.7	D	Full BNB dataset (1.11×10^{21} POT); $\sigma = (3.75 \pm 0.80) \times 10^{-38}$ cm^2/Ar
MicroBooNE NuMI ν_e [29]	Ar	0.73	D	First $\nu_e + \bar{\nu}_e$ CCπ^+ on Ar; $\sigma = (0.93 \pm 0.28) \times 10^{-39}$ cm^2/nucleon

Deuterium bubble chambers: ANL and BNL (1979–1986)

The ANL 12-foot bubble chamber [20] and BNL 7-foot bubble chamber [21] provided the first high-statistics measurements of $\nu_\mu d \to \mu^- \pi^+ pp$ in the 1–3 GeV range. These measurements are important because the deuterium target is close to a free nucleon, minimising nuclear corrections relative to heavier targets. The ANL and BNL total cross sections disagree by $\sim$15% in absolute normalisation, a discrepancy long attributed to flux uncertainties; the BNL flux is now believed to be better constrained. This ANL–BNL tension directly propagated into the spread of generator predictions: models tuned to ANL tend to predict lower rates than those tuned to BNL. The MINERvA experiment [26] subsequently resolved much of this tension by simultaneously constraining the flux and the cross section.

K2K near detector (2008)

The K2K experiment [22] made the first accelerator-based total CCπ^+ cross-section measurement on a carbon target at $\langle E_\nu \rangle = 1.3$ GeV using its fine-grained near detector. The result was consistent with NEUT predictions and established the nuclear-target correction from deuterium to carbon at intermediate energies.

MiniBooNE (ν_μ on CH_2, BNB beam, 2011)

The MiniBooNE detector (818-tonne mineral-oil Cherenkov detector) published differential CCπ^+ cross sections on CH_2 in the Booster Neutrino Beam ($E_\nu^{\text{peak}} \approx 0.7$ GeV) [23]. MiniBooNE was the first experiment to provide high-statistics differential distributions as a function of Q^2, the invariant hadronic mass W, and the outgoing muon and pion kinematics. The flux-integrated total cross section was measured to be $\sigma \approx 3.4 \times 10^{-38}$ cm^2 per CH_2 molecule, or equivalently

$\sim 2.0\times10^{-38}$ cm^2 per carbon nucleus at $\langle E_\nu \rangle \approx 1$ GeV [30]. The absolute rate exceeded NUANCE predictions by ∼20%, establishing an empirical over-prediction that has since been seen across all subsequent measurements. The hydrogen content of CH_2 complicates the extraction of the carbon-only cross section, and the spherical Cherenkov geometry makes it difficult to measure low-energy pions or reconstruct proton recoils. These limitations drove subsequent adoption of finer-grained tracking detectors and liquid-argon targets.

MINERvA (ν_μ on CH, NuMI medium energy, 2015–)

The MINERvA experiment [24] uses a finely-segmented scintillator-strip tracking calorimeter in the NuMI medium-energy beam ($\langle E_\nu \rangle \approx 3.6$ GeV). It has published the most kinematically complete CCπ^+ measurements to date, reporting differential cross sections as a function of: pion momentum p_π, pion angle θ_π, muon momentum p_μ, muon angle θ_μ, and the full set of Adler angles describing the pion in the hadronic centre-of-mass frame. MINERvA also measured CCπ^+ on iron and lead targets [25], providing direct constraints on nuclear A-dependence at the same beam energy. GENIE consistently over-predicts MINERvA by 20–35% across all kinematic variables in the $\Delta(1232)$ peak, the same qualitative tension seen at lower energies. The higher beam energy at MINERvA means that the $p_\pi > 0.5$ GeV/c region is well-populated, making it complementary to the MicroBooNE NuMI result which is statistics-limited at high pion momentum.

T2K ND280 (ν_μ on CH, off-axis beam, 2017)

T2K measured CCπ^+ at its near detector ND280 in the off-axis T2K beam (OAB), which peaks at $\langle E_\nu \rangle \approx 0.85$ GeV [27]. The T2K OAB spectrum closely overlaps with the MicroBooNE NuMI off-axis spectrum in the kinematically-sensitive 0.3–1.5 GeV range, making this the most directly comparable prior measurement to the present work. T2K reported differential cross sections in p_μ and $\cos\theta_\mu$, finding a data/MC deficit of ∼30% relative to NEUT. The carbon target ($A = 12$) differs from argon ($A = 40$), but the overlapping beam spectrum enables direct comparison of nuclear-mass dependence at fixed neutrino energy.

ArgoNEUT ($\nu_\mu + \bar{\nu}_\mu$ on Ar, NuMI, 2017)

ArgoNEUT [28] was a 175-litre LArTPC placed upstream of the MINOS near detector in the NuMI low-energy beam ($\langle E_\nu \rangle \approx 9.6$ GeV). Using 1.25×10^{20} POT, ArgoNEUT selected 115 ν_μ and 158 $\bar{\nu}_\mu$ CCπ^+ candidate events after background subtraction [30], publishing the first CCπ^+ total cross section on argon as a function of neutrino energy. The result was consistent with GENIE predictions at these high energies, where sensitivity to the $\Delta(1232)$ resonance is suppressed. The LArTPC technology and argon target make ArgoNEUT the direct predecessor to MicroBooNE, but the factor ∼10 higher beam energy means that extrapolation to the $E_\nu < 2$ GeV regime relevant for oscillation experiments requires generator assumptions and limits the constraining power on FSI models in the resonance region.

MicroBooNE BNB channel: inclusive precedent (2022)

The same MicroBooNE detector used in this analysis published the first measurement of energy-dependent inclusive ν_μ charged-current cross sections on argon in the Booster Neutrino Beam [11] ($E_\nu^{\text{peak}} \approx 0.7$ GeV), establishing the detector's absolute cross-section capability on argon together with the BNB flux modelling and reconstruction performance relevant to the present work. The dedicated exclusive CCπ^+ measurement in the same beam followed on the full BNB dataset (below), and it is that exclusive result to which the present NuMI CCπ^+ measurement is most directly compared.

MicroBooNE BNB channel: full dataset (2026)

Reference [30] reports CCπ^+ differential cross sections using the complete MicroBooNE BNB dataset of 1.11×10^{21} POT [30], selecting 6816 events after background subtraction and measuring differential cross sections in muon momentum, muon angle, pion momentum, and pion angle – the same four observables as the present analysis. The flux-integrated total cross section is

$$\sigma = (3.75 \pm 0.07_{\text{stat}} \pm 0.80_{\text{syst}}) \times 10^{-38}\ \text{cm}^2/\text{Ar}$$

at mean neutrino energy $\langle E_\nu \rangle \approx 0.8$ GeV. Most generators reproduce the data within uncertainties; a notable exception is the muon angle with respect to the beam, where all tested generators overestimate the cross section at very forward angles ($\cos\theta_\mu \to 1$) [30]. The pion momentum differential cross section reported in this analysis represents the first such measurement on an argon target.

The BNB and NuMI analyses probe complementary phase space: BNB is on-axis with a narrow ν_μ spectrum peaked at $\sim$700 MeV and negligible $\bar{\nu}_\mu$ content, while NuMI FHC at 135 mrad off-axis provides a softer, broader spectrum with a $\sim$38% $\bar{\nu}_\mu$ admixture and extends sensitivity to $E_\nu \sim 0.1$–0.5 GeV.

MicroBooNE NuMI $\nu_e + \bar{\nu}_e$ CCπ^+ (2025)

Reference [29] presents the first measurement of $\nu_e + \bar{\nu}_e$ CC single charged-pion differential cross sections on argon, using the full MicroBooNE NuMI dataset (2.0×10^{21} POT, FHC and RHC combined). The observable set – electron energy E_e, electron angle θ_e, pion angle θ_π, and electron–pion opening angle $\theta_{e\pi}$ – mirrors the muonic observables of the present analysis. The flux-integrated total cross section is measured to be

$$\sigma = (0.93 \pm 0.13_{\text{stat}} \pm 0.27_{\text{syst}}) \times 10^{-39}\ \text{cm}^2/\text{nucleon}$$

at a mean ν_e energy of 730 MeV, in good agreement with all tested generator predictions (GENIE, NuWro, NEUT, GiBUU) within the combined uncertainties. Prior to this measurement, CCπ^+ production by electron neutrinos on argon had never been measured; the result is a key input to the DUNE ν_e-appearance analysis, where CCπ^+ events are a background to the ν_e quasi-elastic signal.

3 Detector

3.1 Operating Principle

A liquid argon time projection chamber records three-dimensional ionisation images of charged-particle tracks by exploiting two properties of ultra-pure liquid argon:

1. Ionisation electrons produced by a traversing charged particle are not captured by argon atoms and can drift centimetres or metres under an applied electric field without significant recombination, provided the purity is sufficient (O_2-equivalent $\lesssim 100$ ppt).
2. Argon scintillates at 127 nm (vacuum ultraviolet), producing $\sim$24 000 photons per MeV of deposited energy, enabling nanosecond-scale timing.

An applied electric field drifts the ionisation electrons to a set of sense-wire planes at the anode, which record three two-dimensional projections of each track. The third coordinate (along the drift direction) is reconstructed from the known drift velocity and the timing of the signals relative to the beam-gate start, combining the two-dimensional wire images into a full 3D reconstruction.

MicroBooNE operated from October 2015 to July 2021 at the Fermilab Accelerator Laboratory, collecting data in both the Booster Neutrino Beam (BNB) and, as a parasitic experiment, the NuMI beam [12]. It was the world's largest operating LArTPC until the start of SBND and ICARUS data-taking.

3.2 Active Volume and Electric Field

The MicroBooNE active volume has rectangular geometry: 256 cm (drift, x) $\times$ 233 cm (height, y) $\times$ 1037 cm (beam, z), containing ≈ 87 tonnes of liquid argon at ≈ 87 K. The cathode is a stainless-steel plane at $x = 256$ cm held at -70 kV; the anode wire planes are at $x = 0$. The resulting 273 V/cm drift field gives an electron drift velocity of ≈ 1.11 mm/μs, so the maximum drift time across the full gap is approximately 2.3 ms.

Liquid argon purity is maintained by continuous recirculation through oxygen and water absorbers, keeping the electron lifetime well above 5 ms (corresponding to O_2-equivalent contamination below $\sim$100 ppt). At this purity level, charge attenuation over the full 2.56 m drift is less than 10%.

3.3 Wire Readout and Signal Formation

Three wire planes at the anode sample the drifting charge:

- **U plane** (first induction): 2400 wires at $+60°$ from the vertical, 3 mm pitch.
- **V plane** (second induction): 2400 wires at $-60°$ from the vertical, 3 mm pitch.
- **Y plane** (collection): 3456 vertical wires, 3 mm pitch.

Induction-plane wires are biased to be transparent to the drifting electrons, producing bipolar (derivative-like) induced signals; collection-plane wires absorb the charge and produce unipolar signals. The combination of two stereo induction views at $\pm 60°$ and a vertical collection view provides sufficient 2D information in each plane to resolve ambiguities during 3D reconstruction.

Front-end electronics (application-specific integrated circuits, ASICs) are mounted inside the cryostat on the wire planes and cold-multiply each channel at ≈ 87 K, reducing thermal noise. The analogue signals are digitised at 16 MHz inside the cryostat by 12-bit ADCs and transmitted over $\sim$110 m of cold cables to warm readout electronics. The data stream is sampled into a circular buffer; beam triggers and cosmic triggers select 4.8 ms readout windows (corresponding to $\approx$two full drift windows) for permanent storage.

A dead-wire region at $z \in [675.1, 775.1]$ cm interrupts the collection plane over a 100 cm span due to a shorted wire bundle. Tracks crossing this region have degraded hit-counting and range-based momentum estimation; the fiducial volume (Section 7) excludes this region.

3.4 Photon Detection System

Thirty-two 8-inch Hamamatsu R5912-MOD photomultiplier tubes (PMTs) are installed behind the collection-plane wire frame facing the active volume. The PMTs detect the 127 nm argon scintillation light via a tetraphenyl-butadiene (TPB) wavelength-shifting coating that converts VUV photons to visible ($\sim$430 nm) light. The system serves two purposes in this analysis:

1. **Beam-gate synchronisation**: the PMT flash time identifies which 1.6 μs NuMI spill window produced the neutrino interaction.
2. **Flash-matching for cosmic rejection**: the Pandora reconstruction associates PFParticle hypotheses with the PMT flash closest in time and position, suppressing out-of-time cosmic-ray muons that overlap with beam-induced events in the long 2.3 ms drift window.

3.5 Calibration and Corrections

Three independent calibration inputs are used:

- **UV laser system** [10]: two pulsed laser beams produce straight ionisation tracks at precisely known locations. Deviations of the reconstructed track positions from the known trajectories are used to build a three-dimensional space-charge distortion map, which is applied to all reconstructed hits as a position correction. Distortions reach $\sim$10 cm near the anode and cathode edges.

- **Stopping cosmic-ray muons**: the Bragg peak of cosmic muons stopping inside the active volume calibrates the absolute energy scale and the effective recombination factor as a function of local electric field and drift distance. Through-going muons calibrate relative wire-by-wire gains and the electron lifetime.
- **Michel electrons**: $\mu^- \to e^- \bar{\nu}_e \nu_\mu$ decay products from stopped muons provide a check of the calorimetric reconstruction at $E_e \lesssim 55$ MeV.

3.6 Cosmic Ray Environment and Reconstruction

Operating at the surface with minimal overburden, MicroBooNE is exposed to a cosmic-ray muon rate of ≈ 5 kHz through the active volume. Each 4.8 ms readout window contains on average $\sim$10 cosmic-ray tracks unrelated to the beam interaction. The 1.6 μs NuMI spill itself contains $\lesssim 0.1$ expected cosmics on average, but the drift-window pile-up is the dominant background at early selection stages (Section 8).

Cosmic rejection proceeds in stages:

1. Flash-matching (PMT timing): candidate neutrino slices must have a consistent optical flash within the beam window.
2. Topological score: the Pandora-based topological neutrino score discriminates between neutrino-like vertex structures and through-going/stopping cosmic topologies.
3. Dedicated BDT scores: the `bdt_mip` and `bdt_cosmic` scores trained on track-level and event-level features further suppress residual cosmics that passed the topological cut.

The CRT (Cosmic Ray Tagger) sub-detector [13], consisting of scintillator panels surrounding the cryostat, provides an additional cosmic veto but is not used in the present analysis because the relevant branch variables are zero-filled in the NuMI data ntuples.

Reconstruction uses the Pandora [6] multi-algorithm pattern recognition framework, which groups hits into PFParticles organised in a hierarchy with the neutrino interaction as the top-level particle. Track candidates are fitted with a Kalman-filter-based track fitter to obtain momentum from track curvature (negligible for <300 MeV/c muons at 273 V/cm) and from range for contained tracks. Log-likelihood ratio particle identification (LLR PID) using the dE/dx profile along the track is the primary tool for muon/pion/proton separation (Section 8).

4 Beam and Neutrino Flux

4.1 NuMI beam and flux composition

The NuMI beam is produced by 120 GeV protons from the Fermilab Main Injector striking a graphite target. Charged secondary hadrons are focused by two magnetic horns whose polarity selects the sign of the focused mesons. In Forward Horn Current (FHC) mode the horns focus positive pions and kaons, giving a beam with predominantly ν_μ content; in Reverse Horn Current (RHC) mode the polarity is reversed to focus negatives, giving a predominantly $\bar{\nu}_\mu$ beam. MicroBooNE is located 135 mrad (7.71°) off-axis from the NuMI beam axis[1], reducing the peak neutrino energy to the 0.5–2.5 GeV range and narrowing the energy spectrum. Because MicroBooNE sits off-axis and at comparatively low energy, the horn sign-selection is imperfect and both modes carry a sizeable wrong-sign component, most notably RHC, whose wrong-sign ν_μ fraction is large.

The flux composition at the MicroBooNE site is taken from the new-G4 central-value flux histograms (`flux_newg4.root`), integrated over $E_\nu > 60$ MeV to exclude an unphysical sub-threshold pile-up near $E_\nu = 0$ (removed by the flux group's `RemoveFluxPeak` step; below

[1] From the detector position in beam coordinates, (5502, 7259, 67270) cm: 91.1 m transverse at 672.7 m downstream. Earlier versions of this note quoted 110 mrad. See §4.2.

the reaction threshold and irrelevant to the measurement). The muon-flavour split of these central-value histograms matches the author's authoritative integrated fluxes exactly (Sec. 4.4):

Table 2: NuMI flux composition at the MicroBooNE site by neutrino type, for the FHC and RHC horn modes (new-G4 central-value flux `flux_newg4.root`, integrated over $E_\nu > 0.1$ GeV). FHC is ν_μ-dominant; RHC is $\bar{\nu}_\mu$-dominant with a large wrong-sign ν_μ fraction. The combined row weights each mode by its data exposure (FHC 8.857, RHC 11.082 $\times 10^{20}$ POT; Table 3), giving a near-balanced $\nu_\mu/\bar{\nu}_\mu$ mix. The signal is charge-blind ($|\nu_{\rm pdg}| = 14$), so both ν_μ and $\bar{\nu}_\mu$ contribute in every mode.

Horn mode	ν_μ	$\bar{\nu}_\mu$	ν_e	$\bar{\nu}_e$
FHC	63.4%	34.0%	1.7%	0.9%
RHC	44.2%	53.3%	1.4%	1.2%
Combined	53.0%	44.4%	1.5%	1.1%

Flux corrections from the Package to Predict the FluX (PPFX) [8] are applied per event via the central-value weight `ppfx_cv` stored in the analysis ntuples. (The absolute flux normalisation used for the cross-section in Sec. 4.4 is still taken from the previous flux histograms; moving that normalisation onto the new-G4 flux is a separate, tracked update.)

4.2 Beam direction in detector coordinates

Two different angles describe the MicroBooNE–NuMI geometry, and they must not be confused (Fig. 1). The *off-axis angle* is a property of the beamline: the detector sits 91.1 m off the NuMI axis at 672.7 m downstream of the target, a baseline of 678.8 m and an off-axis angle of 7.71° (134.6 mrad).[2] That angle says nothing about how the TPC is oriented.

The angle that enters the measurement is a different one: the direction in which the neutrinos travel *relative to the detector z axis*. Averaged over interactions in the active volume, the true neutrino direction in detector coordinates is (0.4624, 0.0489, 0.8853), which lies 28.6° away from z. The detector z axis is therefore a poor stand-in for the beam: it is not the direction the neutrinos came from, nor the NuMI beamline axis (itself 23.2° from z).

[2]Computed from the detector position in beam coordinates, (5502, 7259, 67270) cm, taken from the same rotation matrix used for the NuMI beamline geometry weights.

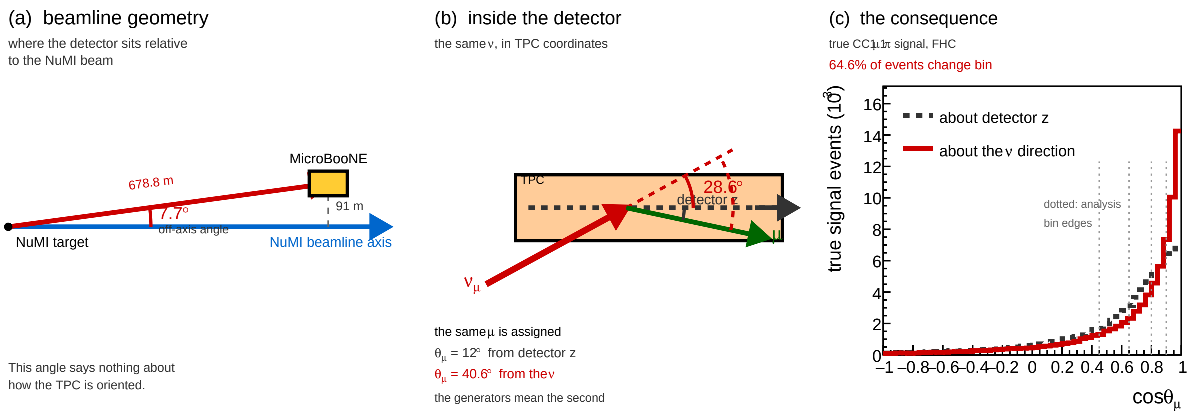


Figure 1: The NuMI geometry seen by MicroBooNE. (a) Beamline geometry, to scale: the detector sits 91.1 m off the NuMI axis at a baseline of 678.8 m, an off-axis angle of 7.7°. (b) The same neutrinos in TPC coordinates, to scale (the x–z projection; the quoted 28.6° is the full three-dimensional angle): the beam arrives 28.6° away from the detector z axis, so one and the same muon is assigned $\theta_\mu = 12°$ about z but $\theta_\mu = 40.6°$ about the neutrino. The generators mean the second. (c) The resulting shift in the true $\cos\theta_\mu$ distribution; dotted lines mark the analysis bin edges. Under the two conventions 64.6% of signal events fall in a different bin (44.6% for $\cos\theta_\pi$).

This matters because every polar angle and every transverse-kinematic-imbalance variable is defined with respect to the neutrino direction. In the generator predictions the neutrino defines $+z$ by construction — the GENIE event records used here have $p^\nu_x = p^\nu_y = 0$ and $p^\nu_z = E_\nu$ exactly — so a measurement that refers its angles to the detector z axis is not being compared with the prediction it appears to be compared with. The effect is not small: 64.6% of signal events change $\cos\theta_\mu$ bin between the two conventions, and 44.6% change $\cos\theta_\pi$ bin. For the TKI variables the axis error exceeds the observable itself, the mean δp_T being 0.742 GeV/c about z against 0.264 GeV/c about the beam.

Accordingly, all angular and TKI observables in this analysis are referred to the neutrino direction, following the β convention of the NuMI ν_e CC1π internal note [1] and of the earlier NuMI ν_e cross-section measurement it derives from [2]. At truth level the exact per-event neutrino direction is used, which reproduces the generator convention exactly. At reconstruction level the neutrino direction is approximated by the vector from the NuMI target — at $(-31388, -3316, -60100)$ cm in detector coordinates, a 684 m baseline — to the reconstructed interaction vertex. Adopting the same β as the ν_e analysis keeps the two NuMI measurements directly comparable.

The choice of reconstruction-level axis is in any case not a sensitive one. A single fixed axis and the per-event β differ by a median of 2.4 mrad (maximum 7.3 mrad), which moves only 0.42% of muons across a $\cos\theta_\mu$ bin edge — to be compared with the 64.6% moved by the detector-z convention. Against the true per-event direction the two perform almost identically, with a mean opening angle of 1.95° for β and 2.00° for a fixed axis. What neither can recover is the physical divergence of the beam: neutrinos reaching the detector come from decays distributed along the decay pipe, not from the target, and over a 684 m baseline a 2.5 m detector subtends only $\sim$4 mrad, so the vertex position carries almost no information about the direction. That residual is a resolution effect, absorbed by the response matrix like any other.

Quantities that are invariant under this choice of axis are unaffected: p_μ, p_π, the muon–pion opening angle $\theta_{\mu\pi}$, the invariant mass $W_{\pi p}$, and all integrated cross sections. The event selection is likewise unaffected, as no selection cut and no BDT input variable uses a polar angle.

4.3 Data exposure

The analysis is *blind*: the real beam-on data are not used. Each configuration is extracted on Poisson-thrown central-value fake data, generated and scaled *per run* to that run's data POT (Table 3). The FHC measurement uses the full FHC exposure (Runs 1, 2, 4 and 5); RHC uses Runs 1, 2, 3, 4a, 4b and 4c; the combined measurement uses all fourteen overlay files. Every run is normalised independently to its own data exposure, so the prediction carries the data-POT mixture of run periods and the framework is ready for real per-run beam-on data (each fake-data sample is a drop-in replacement for its beam-on counterpart).

Table 3: Run-period exposure accounting (final, full FHC+RHC). "MC POT" is each period's native simulated exposure; "Data POT" is the per-run exposure to which the (blind, per-run) fake data is scaled. *Each run is normalised independently* by its own factor, Scale = (run Data POT)/(run MC POT), so the prediction is the data-POT–weighted mixture of run periods; the normalisation a real per-run beam-on dataset produces. This replaces an earlier single per-mode pooling factor: because the per-run MC/data POT ratios vary by up to a factor of three (Scale ranges 0.051–0.141 within FHC alone), a single factor would weight the runs by their MC statistics rather than by the data exposure. The RHC Run 4 row sums the 4a/4b/4c sub-files and Run 3 sums aa–ae; sub-files sharing a run's data POT are scaled by the group MC total so they pool within that run. The combined measurement applies each run's own factor. Run 5 uses the reweighted-PPFX ν_μ overlay. The scheme is validated by the FHC closure (unfolded vs. smeared truth $\chi^2 = 1.6/7$, $p = 0.98$).

Run period	MC POT ($\times 10^{20}$)	Data POT ($\times 10^{20}$)	Scale
Run 1 (FHC)	23.282	3.283	0.1410
Run 2 (FHC)	24.934	1.268	0.0509
Run 4 (FHC)	28.335	2.075	0.0732
Run 5 (FHC)	19.300	2.231	0.1156
FHC total	**95.85**	**8.857**	*per run*
Run 1 (RHC)	8.997	0.605	0.0673
Run 2 (RHC)	57.865	2.591	0.0448
Run 3 (RHC, aa–ae)	55.185	5.003	0.0907
Run 4 (RHC, 4a–4c)	32.586	2.883	0.0885
RHC total	**154.63**	**11.082**	*per run*
Combined (FHC+RHC)	**250.49**	**19.939**	*per run*

The full exposure substantially reduces the data-statistical term (Table 17 of the technical supplement–Table 19 of the technical supplement), leaving the measurement flux- and detector-systematic limited. When the analysis is unblinded, the real beam-on POT (and gate counts) replace the data-POT values above.

4.4 Integrated flux

The absolute flux normalisation uses the new-G4 GEANT4 NuMI flux (`flux_newg4.root`), integrated over $E_\nu > 60$ MeV (the flux group's `RemoveFluxPeak` cut, which removes an unphysical ~25–30 MeV artifact spike). The signal is charge-blind ($|\nu_{\rm pdg}| = 14$, covering both $\nu_\mu \to \mu^-\pi^+$ and $\bar\nu_\mu \to \mu^+\pi^-$), so the normalisation uses the combined $\nu_\mu + \bar\nu_\mu$ flux $\Phi = \Phi_{\nu_\mu} + \Phi_{\bar\nu_\mu}$. Because RHC is $\bar\nu_\mu$-dominant with a different absolute flux than FHC, each configuration uses its *own* integrated flux:

Table 4: Authoritative integrated ν_μ and $\bar{\nu}_\mu$ fluxes per POT (new-G4 flux, $E_\nu > 60$ MeV), and the combined $\nu_\mu + \bar{\nu}_\mu$ value used as the per-configuration normalisation. The combined row is the data-POT-weighted average of the two modes (weights 8.857 : 11.082; Table 3). The FHC value 6.81159×10^{-10} is the previously confirmed normalisation; RHC and combined were previously normalised with the FHC flux and are corrected here (raising the RHC cross section by 5.7% and the combined by 3.1%).

Config	Φ_{ν_μ}	$\Phi_{\bar{\nu}_\mu}$	$\Phi_{\nu_\mu+\bar{\nu}_\mu}$	$\bar{\nu}_\mu$ frac.
	$(10^{-10}\,\mathrm{cm}^{-2}\,\mathrm{POT}^{-1})$			
FHC	4.43515	2.37644	6.81159	34.9%
RHC	2.92348	3.52298	6.44646	54.6%
Combined	3.59497	3.01368	6.60865	45.6%

The mean $\bar{\nu}_\mu$ energy is 0.500 GeV (FHC) and 0.417 GeV (RHC), the lower RHC value reflecting the softer wrong-sign-enhanced spectrum. These fluxes are set per configuration in `integrated_numu_flux_in_FV()` (`include/XSecAnalyzer/FiducialVolume.hh`). The correction relative to the superseded flux (`uboone_numi_flux_histograms.root`, whose combined value 2.3725×10^{-9} was $\sim$3.48× too large and made every extracted cross section $\sim$3.5× too small) was validated four independent ways, standalone NuWro and GENIE on the old vs. corrected flux, the flux author's numbers, and a GENIE closure, documented in `generator_predictions/`.

The number of argon target nuclei in the fiducial volume is computed from first principles as

$$N_{\mathrm{Ar}} = \frac{\rho_{\mathrm{LAr}} \cdot V_{\mathrm{FV}} \cdot N_A}{A_{\mathrm{Ar}}} = 8.710 \times 10^{29}, \tag{3}$$

with $\rho_{\mathrm{LAr}} = 1.3836$ g/cm^3, $A_{\mathrm{Ar}} = 39.948$ g/mol and $N_A = 6.02214076 \times 10^{23}$ mol^{-1}. The fiducial box is $10 < x < 246$, $-101 < y < 101$, $10 < z < 986$ cm (4.6528×10^7 cm^3), from which the $675.1 < z < 775.1$ cm dead region is subtracted, leaving $V_{\mathrm{FV}} = 4.1761 \times 10^7$ cm^3. Without that subtraction the same expression would give 9.705×10^{29}. This is the value the code uses, computed by `num_Ar_targets_in_FV(FV, 675.1, 775.1)` in `CrossSectionExtractor.hh`, and it is the value entering the conversion factor of §10 and the cut-and-count normalisation of §11.3. and the per-configuration normalisation factor is $N_{\mathrm{norm}} = \Phi_{\mathrm{mode}} \cdot N_{\mathrm{POT,mode}} \cdot N_{\mathrm{Ar}}$, evaluated with each mode's flux and data POT (Table 3).

5 Monte Carlo Samples

Monte Carlo events are generated with GENIE v3 [9] on argon-40 and overlaid with real cosmic-ray data (the "overlay" technique) so that simulated events experience in-situ detector noise and cosmogenic activity. Table 5 summarises the input samples.

Table 5: Input samples (final, full FHC+RHC). "n" is the number of overlay files; "MC POT" is the summed native simulated exposure ($\times 10^{20}$). The blind Poisson-thrown fake data stands in for the (unused) beam-on data; the detector variations use the native Run-4 FHC and RHC samples. The per-file lists and POT/trigger values are in the `configs/file_properties_numi_*` tables.

Sample	Composition	n	MC POT
ν_μ overlay (FHC)	Runs 1, 2, 4 (pandora) + Run 5 (reweighted-PPFX)	4	95.85
ν_μ overlay (RHC)	Runs 1, 2, 3, 4a, 4b, 4c (pandora)	10	154.63
Dirt MC	GENIE NuMI dirt overlay (Run 1)	1	16.74
EXT (beam-off)	cosmic data, per run, horn-mode-independent	7	n/a
Detector variations	Run-4 FHC + Run-4 RHC, CV + 8 knobs each	18	n/a
Fake data (blind)	Poisson-thrown central value, per configuration	n/a	n/a
Beam-on data	blinded, not used until unblinding	n/a	n/a

Event weights. Each MC event is weighted by

$$w = \text{Scale[run]} \times w_{\text{PPFX}} \times w_{\text{tune}}, \tag{4}$$

where Scale[run] = (run Data POT)/(run MC POT) is the *per-run* POT factor of Table 3: each run period is scaled independently to its own data exposure, so the prediction reproduces the data-POT mixture of runs rather than the MC-statistics mixture (see the caption of Table 3). Operationally this is exactly what the framework does natively each overlay file carries its run's data POT via a matching per-run beam-on (here, per-run fake-data) sample, and is scaled by run_bnb_pot/summed_pot. w_{PPFX} is the PPFX central-value flux correction (`ppfx_cv`) and w_{tune} is the GENIE/GEANT4 reweighting correction (`weightTune`). Events with non-finite, zero, or unphysically large weights ($w > 30$) are assigned unit weight. At unblinding the per-run fake-data samples are replaced by the corresponding per-run beam-on files (same runs, same POT), so the normalisation is already correct.

EXT scaling. The same combined beam-off (cosmics) sample is attached to every run and scaled by that run's beam-on/beam-off trigger ratio, bnb_trigs[run]/ext_gates. Because the beam-on/beam-off ratio is run-independent, the per-run contributions sum to the mode-total scale; for the full FHC exposure,

$$\text{Scale}^{\text{FHC}}_{\text{EXT}} = \frac{\sum_{\text{run}} \text{bnb\_trigs[run]}}{\text{ext\_gates}} = \frac{22\,667\,109}{3\,821\,593} = 5.93. \tag{5}$$

(The framework requires each run carrying a beam-on sample to carry a beam-off sample as well; the single beam-off file is therefore attached per run through symlinks so that run bookkeeping is complete without duplicating the data.) The uncertainty on this ratio is a leading systematic on the EXT background.

A directional cut against residual cosmics: considered and rejected. Part of the surviving beam-off sample is thought to be broken cosmic muon tracks, where the break is reconstructed as a vertex and the track direction bears no relation to it. That leaves a directional signature, and it is present in the data: among selected events the beam-off sample is enriched at backward reconstructed muon angles by nearly an order of magnitude relative to signal.

reco $\cos\theta_\mu$	EXT (beam-off)	MC signal	enrichment
< 0	31.8%	4.1%	7.8×
< -0.5	9.1%	1.4%	6.4×

The enrichment is real but the arithmetic does not support a cut. At the final cut the FHC sample is 974 signal, 128 EXT and 1 746 predicted in total (Tab. 9). A $\cos\theta_\mu < 0$ requirement would remove ≈ 41 EXT events and ≈ 40 signal events, close to one for one. Because the beam-off sample is subtracted from the data directly, and that subtraction is unbiased by construction, removing part of it buys only a reduction in the subtraction's *statistical* uncertainty, from $\sqrt{128} \approx 11$ to $\sqrt{87} \approx 9$ events. Spending 40 signal events to gain ≈ 2 events of uncertainty is a net loss.

The same bound applies to any cosmic cut, not just this one: EXT is 7.3% of the selected sample and is already subtracted without bias, so removing *all* of it for *zero* signal loss would gain about 11 events of statistical uncertainty on a sample of 1 746. Any cut costing more than about 1% of the signal is a net loss regardless of how efficiently it tags cosmics. The residual beam-off contribution is therefore left in place and handled by subtraction.

A more surgical variant remains untested. The CC0π analysis tags mis-directed tracks with the calorimetric flag `trk_bragg_mu_fwd_preferred_v` in coincidence with a poor forward-muon fit, $\chi^2_\mu > 6$, which fires on tracks whose dE/dx profile disagrees with the assigned direction rather than on backward-going tracks as such. That would not cut genuine backward muons, which outnumber mis-reconstructed ones here by roughly 8:1. Both branches exist in the input ntuples but are not currently propagated by `ProcessNTuples`; the study is scoped in `report/BACKWARD_MUON_PLAN.md`. It is bounded by the same 11-event ceiling, so it is worth measuring but cannot become a significant improvement.

Software trigger. Data and EXT files are pre-filtered to require the software trigger (`swtrig == 1`). MC events failing `swtrig = 0` are removed at the NeutrinoSlice cut stage. In practice this has no measurable effect at the current working point: all MC events passing the topological cut are verified to have `swtrig == 1` (46 810/46 810 events confirmed numerically).

6 Signal Definition

The signal is defined at generator level and is used only to construct the selection efficiency and response matrix. It is *never* applied as a reconstruction cut. The definition is *charge-blind*: both $\nu_\mu \to \mu^-\pi^+$ and $\bar{\nu}_\mu \to \mu^+\pi^-$ are treated as signal on the same footing.

Neutrino flavour
: $|\nu \text{ PDG}| = 14$ (ν_μ or $\bar{\nu}_\mu$).

Interaction type
: Charged-current (`ccnc == 0`).

True vertex
: Inside the fiducial volume (Section 7).

Primary multiplicity
: Exactly one muon ($|\text{PDG}| = 13$), exactly one charged pion ($|\text{PDG}| = 211$), zero neutral pions ($|\text{PDG}| = 111$), zero kaons. Any number of protons is allowed.

Kinematic thresholds
: $p_\mu > 0.15\,\text{GeV}/c$, $p_\pi > 0.175\,\text{GeV}/c$, and $\theta_{\mu\pi} < 2.6$ rad. These define the *measured phase space*, chosen from the efficiency turn-on curves (Section 9). The momentum efficiency turns on sharply at $\sim$0.15–0.20 GeV/c(set by the 10 cm muon and 20 cm pion track-length cuts); below that it is only $\sim$1–3% with a strongly model-dependent correction. The opening-angle upper bound (2.6 rad) excludes the large-angle region, which is dominated by cosmic and dirt background ($\sim$7% purity in the $\theta_{\mu\pi} > 2.513$ rad bin). No *lower* opening-angle cut is applied: the small-angle region is high-purity ($\sim$58%), so cutting it would only remove

signal (it is low-efficiency but not low-purity). The same cut is applied at reco level so the signal and selection acceptances match. The first p_μ/p_π bins and the $\theta_{\mu\pi}$ upper edge start/end at these thresholds.

(Earlier iterations used $p_\mu, p_\pi > 0.1\,\mathrm{GeV}/c$ with the full 0–π angle; that left near-dead efficiency bins at low momentum and at the large-angle extreme. The phase space above raises the efficiency from 12.1% to 15.6% at ∼55% purity, see Section 9.)

Pion threshold and the sibling ν_e measurement. The companion MicroBooNE NuMI ν_e CC 1π analysis defines its pion signal by kinetic energy, $\mathrm{KE}_\pi > 40$ MeV, equivalent to $p_\pi > 0.113\,\mathrm{GeV}/c$. Our $p_\pi > 0.175\,\mathrm{GeV}/c$ ($\mathrm{KE}_\pi > 84$ MeV) is therefore a higher threshold. Lowering it toward the ν_e value is possible, the signal definition is a truth-level choice and the unfolding maps reco to truth but the pion momentum reconstruction (track range under the muon hypothesis, mass-corrected) degrades rapidly below $0.175\,\mathrm{GeV}/c$. The table below is the study that led to *rejecting* a lower threshold; it is not part of the measurement. The measured phase space is $p_\pi > 0.175\,\mathrm{GeV}/c$ throughout this note, and no cross section is reported below it:

Table 6: Pion momentum reconstruction versus true p_π (correctly reconstructed signal pions). Below the $0.175\,\mathrm{GeV}/c$ threshold the bias and resolution blow up and the reconstructed fraction halves.

True p_π [GeV/c]	Bias $\langle (p^{\mathrm{reco}} - p^{\mathrm{true}})/p^{\mathrm{true}} \rangle$	RMS resolution	Reco fraction
0.080–0.113	+131%	73%	3.4%
0.113–0.150	+66%	56%	3.4%
0.150–0.175	+30%	51%	4.3%
0.175–0.250 (threshold)	**−8**%	**31**%	**10.0**%
0.250–0.400	−30%	21%	10.4%

At the threshold the resolution is 31%; in the $[0.113, 0.175]\,\mathrm{GeV}/c$ region it is 51–56% with a +30–66% bias and only ∼ 3–4% reconstructability. Extending the measurement to the ν_e threshold would thus add bins carrying large, strongly model-dependent unfolding systematics. For consistency with the ν_e measurement the threshold could be lowered to $0.113\,\mathrm{GeV}/c$ with the $[0.113, 0.175]\,\mathrm{GeV}/c$ region kept in a single coarse bin to contain those systematics, or both phase spaces reported; this is a systematics-versus-cross-analysis-consistency trade rather than a reconstruction limitation.

Why the threshold was not lowered. Extending to the sibling ν_e threshold was investigated and rejected: the $[0.113, 0.175]\,\mathrm{GeV}/c$ region has 2.2% selection efficiency and a strongly model-dependent reconstruction correction. The study is recorded in the technical supplement. It is not part of this measurement, and no cross section is reported below $p_\pi > 0.175\,\mathrm{GeV}/c$.

7 Fiducial Volume

The fiducial volume (FV) is defined in `CC1mu1piXp.cxx` (the independent cross-check selection carries the same numbers in `FV_new.h`):

$$x \in [10.0,\ 246.0]\,\mathrm{cm}, \quad y \in [-101.0,\ +101.0]\,\mathrm{cm}, \quad z \in [10.0,\ 986.0]\,\mathrm{cm}, \tag{6}$$

with the dead-wire region $z \in [675.1,\ 775.1]$ cm excluded. The total FV volume is

$$\begin{aligned} V_{\mathrm{FV}} &= (246 - 10) \times 202 \times (976 - 100)\,\mathrm{cm}^3 \\ &= 236 \times 202 \times 876\,\mathrm{cm}^3 \approx 4.17 \times 10^7\,\mathrm{cm}^3. \end{aligned} \tag{7}$$

A wider *containment* region (2 cm inset from the physical TPC walls) is used to require that the pion candidate track endpoint is contained:

$$x \in [2.0,\ 254.35]\,\text{cm}, \quad y \in [-113.53,\ +107.47]\,\text{cm}, \quad z \in [2.1,\ 1034.9]\,\text{cm}. \tag{8}$$

7.1 Which volume applies where

Three distinct volumes appear above and it is worth stating plainly which is used for what, since "contained" and "in the fiducial volume" are not the same requirement.

volume	extent	used for
fiducial (FV)	$[10, 246] \times [-101, 101] \times [10, 986]$ cm minus $z \in [675.1, 775.1]$ cm	neutrino vertex, true *and* reco
containment	$[2, 254.35] \times [-113.53, 107.47] \times [2.1, 1034.9]$ cm	track endpoints
active TPC	$[0, 256.35] \times [-115.5, 117.5] \times [0, 1036.9]$ cm	(reference only)

True and reco use identical dimensions. The framework keeps the two as separate quantities, `true_FV()` and `reco_FV()`, and other selections in the code base use that freedom. Here they are defined as the same box (`CC1mu1piXp.cxx`, `define_true_FV` / `define_reco_FV`). The signal definition tests the *true* neutrino vertex and the selection tests the *reconstructed* vertex, both through the same helper, so both inherit the dead-region veto as well. There is no asymmetry between the two, and the FV therefore does not itself generate any migration between signal and background.

The dead-region veto applies to the vertex only. An event whose vertex lies outside the unresponsive-wire slab is kept even if one of its tracks crosses that slab. This is the conventional choice, but it is an assumption rather than a necessity: tracks crossing dead wires have degraded calorimetry and range reconstruction, and the associated inefficiency is absorbed into the response matrix rather than removed by a cut.

The dead slab is also removed from the target count. The argon target number is computed from the FV with the slab subtracted, not from the full box. Retaining the slab would inflate the target count by $100/976 \approx 10\%$ and bias every cross section low by the same factor. Note that this subtraction is applied on the NuMI code path specifically; the generic path computes the target number without it, so the two must not be mixed.

Relation to the active volume. The FV is inset from the active TPC by approximately 10 cm in x, 15 cm in y, 10 cm at the upstream z face and 51 cm at the downstream face. The containment box, by contrast, is inset by only 2 cm: track endpoints are required to be inside the detector, not inside the fiducial region, which is the correct requirement for range-based momentum and for the pion-containment cut.

8 Event Selection

The event selection is implemented in the framework selection class `src/selections/CC1mu1piXp.cxx` (which produces the response matrices and spectra for all results in this note); the standalone `selection/ccpi_selection.C` pipeline used for the cross-pipeline validation of Sec. C of the technical supplement implements the byte-identical cut sequence. The selection comprises 11 sequential cuts.

8.1 BDT-based particle identification

Three gradient-boosted decision trees (TMVA), trained on the MicroBooNE overlay MC, supply the calorimetric particle identification used in the muon- and pion-candidate cuts. All three share a common set of per-track inputs: the three Bragg-peak dE/dx likelihoods under the proton, muon and MIP hypotheses (`trk_bragg_{p,mu,mip}_v`), the log-likelihood-ratio PID score (`trk_llr_pid_score_v`), the Pandora track-vs-shower score (`trk_score_v`), and the space-charge-corrected track end-point coordinates (`trk_sce_end_{x,y,z}_v`):

- **MIP BDT** (`bdt_mip`; weights `dataset_MIP_BDT_no_len`): separates minimum-ionising tracks (muons, pions) from stopping protons. Both the muon and the pion candidate are required to have `bdt_mip` > -0.1 (Cuts 4–5).
- **Pion BDT** (`bdt_pi`; weights `dataset_pion_BDT_no_len`): a dedicated pion-versus-proton discriminant applied to the pion candidate, `bdt_pi` > -0.1 (Cut 5).
- **Muon BDT** (weights `dataset_muon_BDT`): the same inputs plus the track length (`trk_len_v`). Rather than a threshold, it *ranks* the muon-candidate tracks; the highest-scoring track is chosen as the muon.

The two candidate-PID BDTs deliberately omit track length so the score does not bias the momentum spectra. The weight files are resolved from `$CC1MU1PIXP_BDT_WEIGHTS_DIR`, falling back to the in-tree `booster_decision_tree/` copy. A fourth score, `bdt_cosmic`, is available for cosmic rejection but is disabled by default (no benefit in an A/B test; the Pandora topological score already provides the cosmic rejection at Cut 3).

8.2 Cut definitions

Cut 0. EventsInTrueFV. All events. Reference denominator for truth-level studies.

Cut 1. NeutrinoSlice. ≥ 1 reconstructed track (`ntracks > 0`), `bdt_cosmic > 0.0` (currently disabled; no benefit found in A/B test), and `swtrig == 1`.

Cut 2. InFiducialVol. The SCE-corrected reconstructed neutrino vertex lies inside the FV.

Cut 3. Topological. `topological_score > 0.67`. This Pandora-produced multivariate score distinguishes cosmic-ray topologies from neutrino-like topologies and provides the primary cosmic rejection: EXT events are reduced by a factor of $\sim$33 between Cut 2 and Cut 3.

Cut 4. MuonCandidate. At least one primary PFP track satisfies the muon PID criteria of Table 7. The longest qualifying track is the muon candidate.

Cut 5. ContainedPion. Exactly one primary track (other than the muon candidate) satisfies the pion PID criteria of Table 8 and has its endpoint inside the containment volume. The track with the highest LLR PID score is the pion candidate.

Cut 6. MuonIn3Planes. The muon candidate has ≥ 1 hit on each of the U, V, Y planes.

Cut 7. PionIn3Planes. The pion candidate has ≥ 1 hit on each of the U, V, Y planes.

Cut 8. ShowerCut. Zero primary-level EM showers, unless exactly one shower is present with fewer than 50 Y-plane hits and a vertex distance > 10 cm (consistent with a delta-ray or low-energy nuclear photon).

Cut 9. OpeningAngle. The reconstructed μ–π opening angle satisfies $\theta_{\mu\pi} < 2.6$ rad ($\approx 149°$), rejecting the large-angle cosmic/dirt background. No lower bound is applied (the small-angle region is high-purity). The same cut is imposed on the signal definition (Section 6).

Cut 10.

Nonprotons (final cut). Fewer than 4 primary tracks have LLR PID > 0.1 (non-proton-like), and the total number of primary reconstructed tracks is less than 5.

Table 7: Muon candidate PID thresholds (applied at Cut 4).

Variable	Threshold	Description
`trk_score_v`	> 0.80	Track-like PFP
Vertex distance	≤ 4.0 cm	Attached to ν vertex
Track length	> 10.0 cm	Minimum track length
LLR PID (`trk_llr_pid_score_v`)	> 0.20	Muon-like
MIP BDT (`bdt_mip`)	≥ -0.10	Minimum-ionising-particle

Table 8: Pion candidate PID thresholds (applied at Cut 5). Endpoint containment requires the track endpoint inside the containment volume. `trk_bragg_pion_v`: pion-hypothesis Bragg-peak likelihood score (rejects stopping protons).

Variable	Threshold	Description
LLR PID	> 0.10	Pion / non-proton-like
Vertex distance	< 4.0 cm	Attached to ν vertex
Track length	> 20.0 cm	Minimum track length
MIP BDT (`bdt_mip`)	> -0.10	MIP-like
Pion BDT (`bdt_pi`)	> -0.10	Pion-like (vs. proton)
Bragg-pion score	≥ 0.08	Proton rejection
Endpoint contained	yes	Inside containment volume

8.3 Kinematic reconstruction

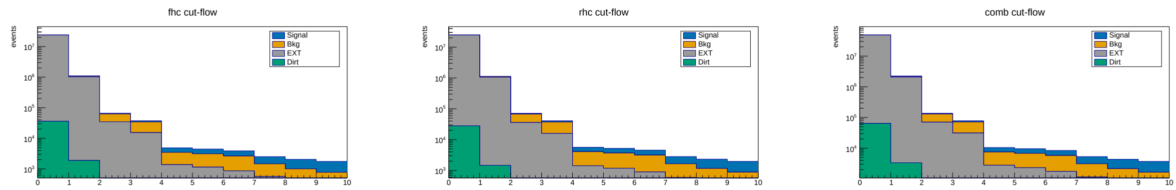
Muon momentum. A *hybrid* estimator, switched on containment of the muon-candidate track: the range-based momentum `trk_range_muon_mom_v` is used when the track ends inside the detector, and the multiple-Coulomb-scattering momentum `trk_mcs_muon_mom_v` when it exits (range is meaningless for an exiting track). The reported p_μ is therefore `candidate_muon_mom_reco`, not either branch alone. In the selected FHC sample the split is 37% contained (range) and 63% exiting (MCS), i.e. the measurement is *MCS-dominated*: the NuMI muons are energetic and the MicroBooNE TPC is small, so most of them leave it. This matters for the p_μ resolution and for the detector systematic, since the range and MCS estimators carry different uncertainties.

Pion momentum. Taken from the *muon-hypothesis* range momentum (`trk_range_muon_mom_v`). The muon mass (105.7 MeV) is close to the pion mass (139.6 MeV), so the range-to-momentum conversion is a good proxy, and, crucially it is a *momentum*, directly comparable to the true p_π it is binned against. An earlier version of this analysis used the proton-hypothesis range *kinetic energy* (`trk_energy_proton_v`) instead. That is a different physical quantity from the momentum used for the true axis with the same bin edges, so the migration matrix was systematically shifted rather than near-diagonal: a true 0.2 GeV/c pion gives $KE \approx 0.104$ GeV, two bins low, and a true 0.175 GeV/c pion falls below the first reco bin entirely. Range momentum is meaningful only for contained tracks, which the containment requirement in the selection guarantees.

Angles. Polar angles are defined with respect to the *neutrino direction* $\hat{u}_\nu$, not the detector z-axis: $\cos\theta_\mu = \hat{u}_\mu \cdot \hat{u}_\nu$, $\cos\theta_\pi = \hat{u}_\pi \cdot \hat{u}_\nu$. The two are far from equivalent — the NuMI beam reaches MicroBooNE 28.6° away from z, and referring the angles to z would place 64.6% of signal events in a different $\cos\theta_\mu$ bin than the generator predictions they are compared with (Sec. 4.2). For reconstructed quantities $\hat{u}_\nu$ is the β direction of Refs. [1, 2], from the NuMI target to the reconstructed vertex; at truth level it is the exact per-event neutrino direction. The opening angle $\theta_{\mu\pi} = \arccos(\hat{u}_\mu \cdot \hat{u}_\pi)$ is computed from reconstructed track directions and, being an angle between two measured tracks, is independent of this choice.

9 Cut-Flow and Selection Performance

Figure 2 shows the full-exposure cut-flow yields for all three configurations as stacked plots, and Table 9 gives the detailed per-category, per-cut breakdown for each.



(a) FHC. (b) RHC. (c) Combined.

Figure 2: Cut-flow yields at *full exposure* for FHC, RHC and combined (log scale; MC-prediction stack: signal blue, background orange, EXT grey, dirt green; colourblind-safe Okabe-Ito), signal and background POT-scaled per mode. Data is blind, so this shows the MC-prediction stack; the detailed per-category breakdown is in Table 9.

Table 9: Cut-flow yields at *full exposure* for FHC, RHC and combined (MC prediction; data is blind, so Pred = Data by construction for the Poisson-thrown fake data). Signal and "Bkg" (all non-signal MC: OOFV, NC, CC1π^0, CC0π, CC$N\pi$, CCK, CCother) are from the ν_μ overlays, POT-scaled *per run* (each run by its own $D_{\text{run}}/\text{MC}_{\text{run}}$, Table 3); EXT (beam-off cosmic) is scaled by the (per-run summed) data/EXT trigger ratio times the 2% NuMI beam-occupancy factor (FHC 0.98×5.93, RHC 0.98×6.16, combined 0.98×12.09); dirt from the GENIE NuMI dirt overlay. Pred. = Signal + Bkg + EXT + Dirt. Eff. is the cumulative signal efficiency relative to the generated signal (the None stage, where all true signal is counted); Pur. = Signal/Pred.

Cut	Signal	Bkg MC	EXT	Dirt	Pred.	Eff. [%]	Pur. [%]
FHC (Runs 1,2,4,5; 8.857×10^{20} *POT)*							
None	5 580	279 180	23 921 822	36 142	24 242 724	100.0	0.02
InFiducialVol	4 632	62 146	1 020 804	1 890	1 089 473	83.0	0.43
Topological	3 501	28 040	34 481	204	66 225	62.7	5.3
MuonCandidate	3 143	18 980	15 206	137	37 466	56.3	8.4
ContainedPion	1 395	2 094	1 348	25	4 862	25.0	28.7
MuonIn3Planes	1 336	1 975	1 122	17	4 450	23.9	30.0
PionIn3Planes	1 239	1 770	854	7.0	3 871	22.2	32.0
ShowerCut	1 048	898	558	6.2	2 510	18.8	41.7
OpeningAngle	1 040	813	186	2.0	2 041	18.6	51.0
Nonprotons	**974**	**643**	**128**	**1.3**	**1 746**	**17.5**	**55.8**
RHC (Runs 1,2,3,4a,4b,4c; 1.108×10^{21} *POT)*							
None	6 130	304 347	24 837 750	28 032	25 176 259	100.0	0.02
InFiducialVol	5 124	69 102	1 059 889	1 466	1 135 581	83.6	0.45
Topological	3 960	31 629	35 801	158	71 548	64.6	5.5
MuonCandidate	3 559	21 443	15 788	106	40 896	58.1	8.7
ContainedPion	1 577	2 649	1 400	19	5 646	25.7	27.9
MuonIn3Planes	1 512	2 500	1 165	13	5 189	24.7	29.1
PionIn3Planes	1 409	2 258	887	5.5	4 559	23.0	30.9
ShowerCut	1 170	1 061	579	4.8	2 815	19.1	41.6
OpeningAngle	1 162	965	193	1.5	2 321	19.0	50.1
Nonprotons	**1 081**	**746**	**133**	**1.0**	**1 961**	**17.6**	**55.1**
Combined FHC+RHC (1.994×10^{21} *POT)*							
None	11 710	583 527	48 759 975	64 174	49 419 386	100.0	0.02
InFiducialVol	9 756	131 248	2 080 711	3 357	2 225 072	83.3	0.44
Topological	7 461	59 669	70 282	362	137 773	63.7	5.4
MuonCandidate	6 702	40 423	30 994	243	78 362	57.2	8.6
ContainedPion	2 972	4 743	2 749	44	10 508	25.4	28.3
MuonIn3Planes	2 847	4 475	2 287	30	9 639	24.3	29.5
PionIn3Planes	2 648	4 028	1 742	12	8 430	22.6	31.4
ShowerCut	2 218	1 959	1 137	11	5 325	18.9	41.7
OpeningAngle	2 202	1 778	379	3.5	4 363	18.8	50.5
Nonprotons	**2 055**	**1 389**	**261**	**2.3**	**3 707**	**17.5**	**55.4**

Summary at the final cut (reference phase space, $p_\mu, p_\pi > 0.1\,\mathrm{GeV}/c$).

- Signal MC (POT-weighted): 324.7 events
- Total background: 242.6 = 198.5 (ν-bkg) + 40.1 (EXT) + 3.8 (dirt)
- True signal generated (efficiency denominator): 2577–2684 events (observable-dependent)
- **Signal purity: 57.2%**
- **Signal efficiency: 12.1–12.6%**
- **Measured phase space (reported result): efficiency** 15.6%**, purity** 54.9% (reference phase-space study: selected signal 311.9, generated 1999, background 253.7; Table 10)

Per-run scaling. The yields above and in Table 9 use the per-run POT factors of Table 3 (each run scaled by its own $D_{\rm run}/{\rm MC}_{\rm run}$), and the EXT by the per-run trigger ratios summing to 5.93 (FHC). Relative to the earlier single per-mode factor this shifts the FHC final-cut signal by -1.9% ($993 \to 974$ events) and rescales the beam-off by the updated ratio; because the shift is a small, nearly uniform re-weighting of the run mixture, the efficiency and purity move by $< 0.5\%$ and the quoted percentages are unchanged at this precision. The per-observable figures are finalised with the full per-run re-extraction; the FHC p_μ closure (unfolded vs. smeared truth $\chi^2 = 1.6/7$, $p = 0.98$) confirms the scheme is unbiased.

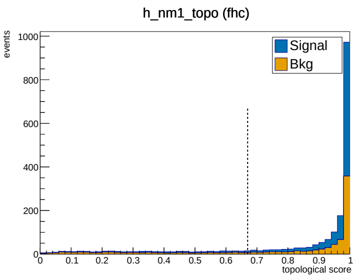
Measured phase space and efficiency turn-on. The reference selection above has near-dead efficiency at low momentum and at the large opening-angle extreme: from the fine truth distributions, the per-bin efficiency is only ∼2–5% for $p_\mu, p_\pi < 0.15\,\mathrm{GeV}/c$ (the 10/20 cm track-length cuts) and the region above the reco $\theta_{\mu\pi} = 2.6$ rad cut is dominated by cosmics (∼7% purity). The measured phase space $p_\mu > 0.15$, $p_\pi > 0.175\,\mathrm{GeV}/c$, $\theta_{\mu\pi} < 2.6$ rad (Section 6) raises and flattens the efficiency to ∼13–20% (Fig. 19). Table 10 compares the selection across the iterations. Two lessons: (i) the 2.6 rad *upper* cut is essential, dropping it admits 3.4× more cosmic background and cuts the purity to 45%; (ii) a *lower* opening-angle cut is counterproductive, the small-angle region is high-purity (∼58%), so cutting it lowers the selected signal without improving purity, which is why no lower cut is used.

Table 10: Selection performance across the phase-space iterations (NuWro fake data). "no upper cut" drops the $\theta_{\mu\pi} < 2.6$ cut; "[0.314, 2.6]" adds a lower OA cut. The final choice keeps the 2.6 rad upper cut with no lower cut.

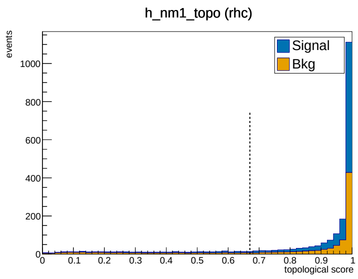
	Reference (p>0.1)	no upper cut (p>.15/.175)	[0.314, 2.6] (lower cut)	**Final** (θ<2.6)
Generated signal	2684	2055	1916	1999
Selected signal	324.7	320.3	301.8	311.9
Efficiency	12.1%	15.6%	15.8%	**15.6%**
EXT (cosmic)	40.1	136.2	38.0	40.1
Total bkg	243.6	385.4	252.3	256.4
Purity	**57.1%**	45.4%	54.5%	**54.9%**
sig × purity	185.6	145.4	164.4	**171.2**

9.1 Key N−1 distributions

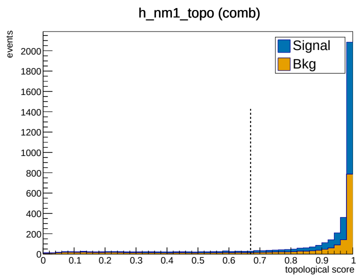
Figure 3 shows the N−1 distributions (all cuts applied except the one under study) for the two cuts instrumented per event; the topological score (Cut 3) and the μ–π opening angle (Cut 9), for all three configurations. The shapes are consistent across horn modes, as expected since the selection is identical; RHC and combined simply carry the larger exposure. Signal (blue) and background (orange) are stacked and the dashed line marks the applied threshold. The remaining cut variables are shown at the final cut in Sec. 9.3.



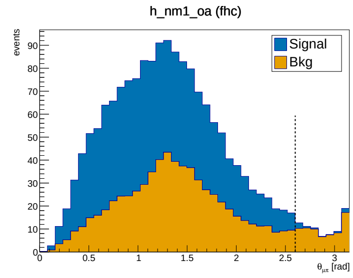
(a) Topo. score, FHC.



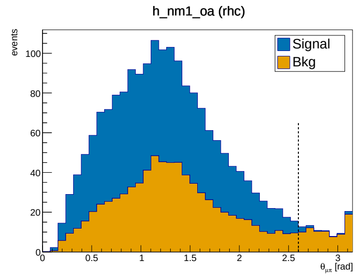
(b) Topo. score, RHC.



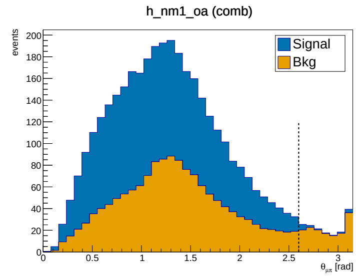
(c) Topo. score, comb.



(d) Opening angle, FHC.



(e) Opening angle, RHC.

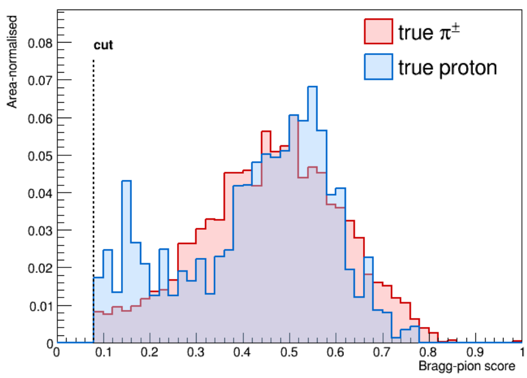


(f) Opening angle, comb.

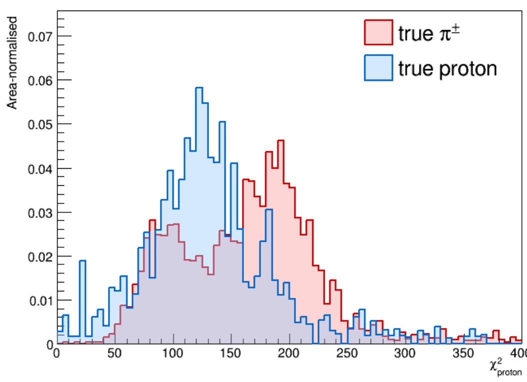
Figure 3: N−1 distributions for the topological score (Cut 3, top) and μ–π opening angle (Cut 9, bottom), for FHC, RHC and combined. Signal (blue) and background (orange) stacked; dashed line = applied threshold. Colourblind-safe Okabe-Ito palette.

9.2 Pion proton-rejection variables

Figure 4 shows the two main proton-rejection variables on the pion candidate, evaluated on MC truth-labelled pions ($\pi^{\pm}$) and protons at the final cut stage, used to tune the working-point thresholds.



(a) Bragg-pion score (`trk_bragg_pion_v`). Working-point threshold: ≥ 0.08.

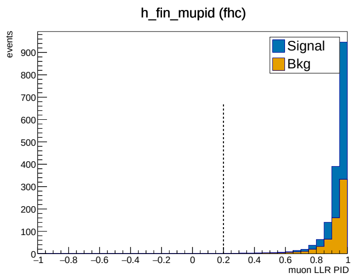


(b) χ^2 proton PID (`trk_pid_chipr_v`).

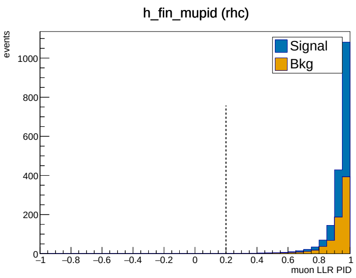
Figure 4: Pion-candidate PID distributions for true $\pi^{\pm}$ (red) and true protons (blue) at the final cut stage. The Bragg-pion score provides the primary proton rejection; χ^2_p is studied as cross-check.

9.3 Final-cut data/MC distributions

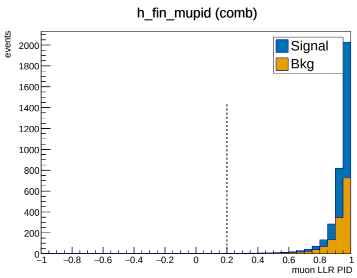
Figures 5 and 6 show the candidate-level distributions at the final selection cut for the four instrumented PID and track-length variables, for all three configurations. Signal (blue) and background (orange) are stacked (colourblind-safe Okabe-Ito palette).



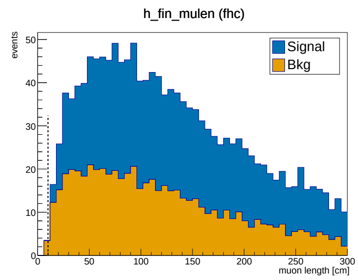
(a) Muon LLR PID, FHC.



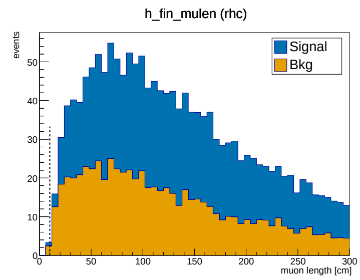
(b) Muon LLR PID, RHC.



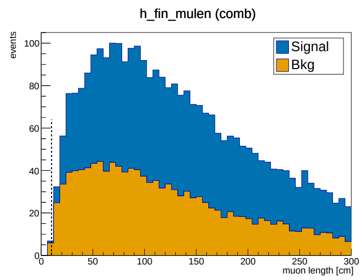
(c) Muon LLR PID, comb.



(d) Muon length, FHC.

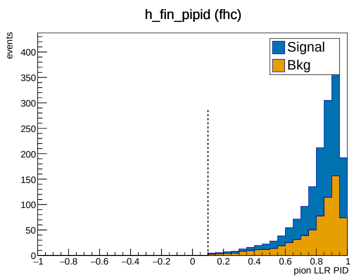


(e) Muon length, RHC.

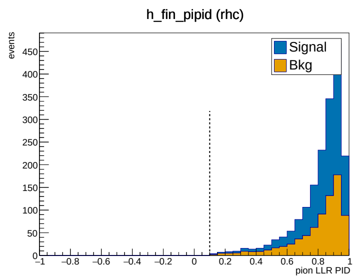


(f) Muon length, comb.

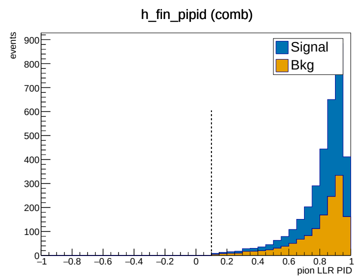
Figure 5: Muon-candidate LLR PID (top) and track length (bottom) at the final cut, for FHC, RHC and combined. Signal (blue) / background (orange) stacked.



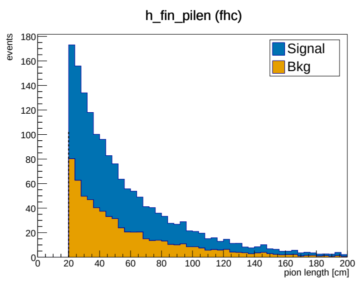
(a) Pion LLR PID, FHC.



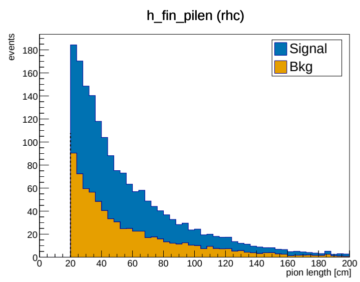
(b) Pion LLR PID, RHC.



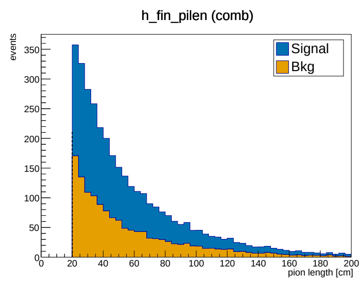
(c) Pion LLR PID, comb.



(d) Pion length, FHC.

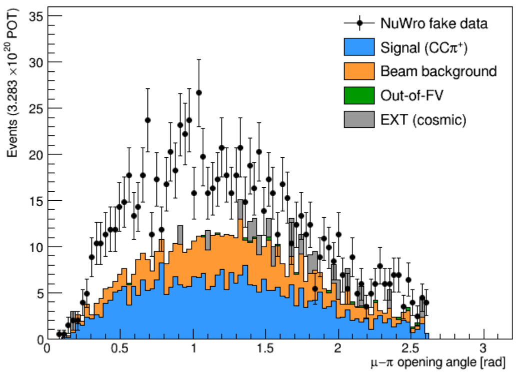


(e) Pion length, RHC.

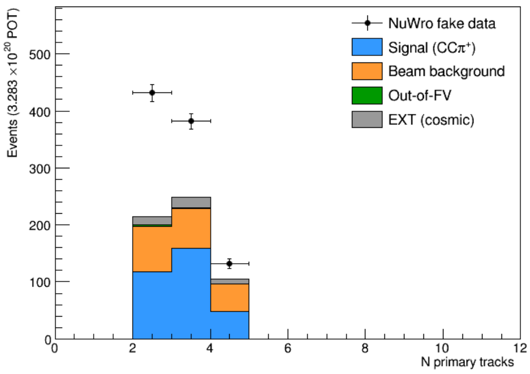


(f) Pion length, comb.

Figure 6: Pion-candidate LLR PID (top) and track length (bottom) at the final cut, for FHC, RHC and combined. Signal (blue) / background (orange) stacked.



(a) μ–π opening angle.



(b) Number of primary tracks.

Figure 7: Data/MC comparison for topological variables at the final cut.

9.4 Background decomposition

Table 11 shows the truth-level background breakdown at the final cut.

Table 11: Background composition at the final cut (truth-level MC categories, POT-weighted).

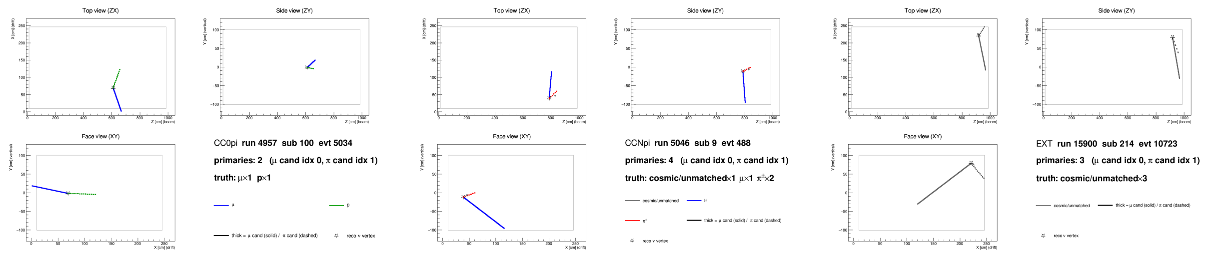
Category	Events	Fraction
CC0π (proton mis-ID as pion)	~ 82	$\sim 41\%$
CC$N\pi$ (≥ 2 pion events)	~ 67	$\sim 34\%$
OOFV (vertex outside FV)	~ 18	$\sim 9\%$
NC (neutral current)	~ 14	$\sim 7\%$
CC1π^0	~ 8	$\sim 4\%$
CC other / CC kaon	~ 6	$\sim 3\%$
ν_e / other flavour	~ 4	$\sim 2\%$
EXT (cosmic)	40.1	n/a
Dirt	3.8	n/a

The dominant background ($\sim 41\%$) is CC0π interactions in which a proton is reconstructed as the pion candidate. This mis-identification was verified directly by 2D event displays (truth-labelled by backtracked PDG): the pion candidate track shows a green (proton) truth colour with a stopping Bragg peak, while the genuine muon track is correctly identified.

At the current working point (`trk_bragg_pion_v` ≥ 0.08), the pion candidate is truth-matched to a proton in $\approx 18\%$ of cases. Tighter Bragg-score thresholds (0.20–0.30) were studied but shown not to improve the figure of merit: the pion-to-proton ratio in CC$\pi^\pm$ signal is $\approx 4:1$, so rejecting protons removes genuine pions at nearly the same rate.

9.5 Event displays

Figure 8 shows representative 2D truth-labelled event displays for the two dominant backgrounds and the EXT cosmic sample. Tracks are coloured by backtracked truth PDG: μ (blue), $\pi^\pm$ (red), p (green), cosmic/unmatched (grey). The muon candidate is drawn solid-thick; the pion candidate dashed-thick; the reconstructed ν vertex is shown as a star.



(a) CC0π background: a proton (green, dashed) is reconstructed as the pion candidate.

(b) CC$N\pi$ background: two genuine charged pions present; one is selected as the pion candidate.

(c) EXT cosmic event: all tracks are grey (no truth match), faking a neutrino vertex topology.

Figure 8: Truth-labelled 2D event displays (ZX projection) for representative background events at the final cut. Three views (ZX top, ZY side, XY face) are generated per event by `selection/event_display.C`.

10 Unfolding

10.1 The extraction chain, stage by stage

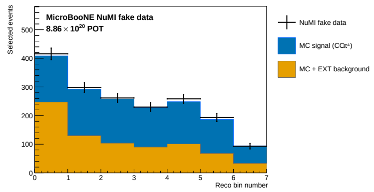
Before the formalism, it is useful to see the measurement assembled in the order it is actually performed. Each stage is shown for all three beam configurations in the flat reco-bin space in which the unfolding operates (every analysis slice concatenated).

Stage 1, selected events, POT-scaled (Fig. 9). The starting point is the selected reconstructed spectrum, scaled to the full exposure of each configuration with the per-run POT factors of Sec. B of the technical supplement. The Monte Carlo is split into signal and background. At the final cut the purity is $\simeq 55\%$ (Table 9), so roughly half of the selected events have to be removed before anything can be interpreted as a signal rate. Because the analysis is blind, the points are the Poisson-thrown fake data rather than beam-on data.

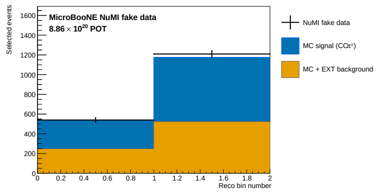
Stage 2, background subtraction (Fig. 10). The estimated background is subtracted bin by bin, leaving the background-subtracted data alongside the central-value MC signal. The neutrino background is taken from the simulation, the cosmic component from beam-off (EXT) data, and the dirt component from the dedicated overlay; the four control regions of §11.4 are what validate this step. This subtracted spectrum, not the raw selection, is the input to the unfolding.

Stage 3, unfolding. The subtracted spectrum still folds in the detector response and the selection efficiency, encoded in the response matrix of §10.3. With 50–99% bin migration a direct inversion is unusable (§10.4), so the measurement is regularised with Wiener–SVD. The price is the additional-smearing matrix A_C: what is recovered is $A_C\times$(true signal), and *every* model must therefore be multiplied by A_C before being compared with the data.

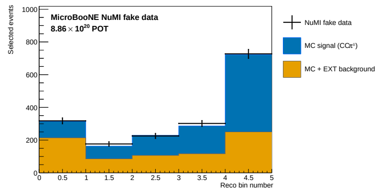
Stage 4, comparison with models. The unfolded result is then compared with the MicroBooNE tune and the four external generators, all A_C-smeared into the same space (§11). It is worth separating two effects that are easily conflated. A_C suppresses the integral of each prediction, but it does so by a *common* factor: for FHC p_μ it is 0.64, 0.64, 0.59 and 0.67 for GENIE, GiBUU, NEUT and NuWro respectively. The truth-level totals of those same generators span 0.79 to 1.26×10^{-38} cm^2/Ar, a factor 1.6. A common $\simeq$0.6 factor cannot produce a $1.6\times$ spread, so the differences between the generators are genuine model differences and not an artefact of the regularisation. The cut-and-count measurement of §11.3 provides the independent check, since it involves no unfolding and hence no A_C at all.



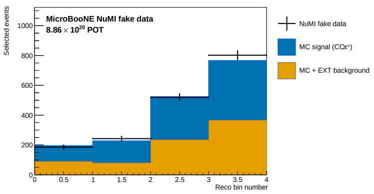
(a) p_μ



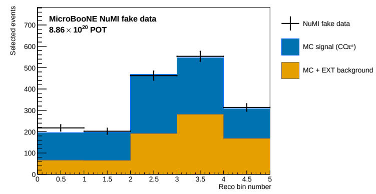
(b) p_π



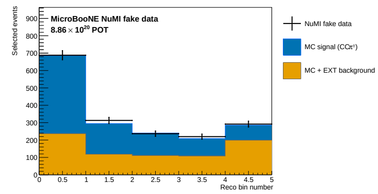
(c) $\cos\theta_\mu$



(d) $\cos\theta_\pi$

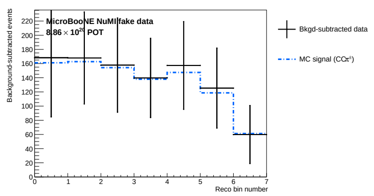


(e) $\theta_{\mu\pi}$

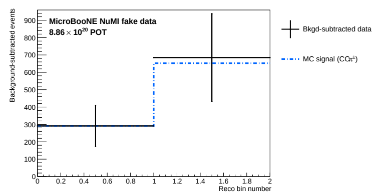


(f) θ_μ

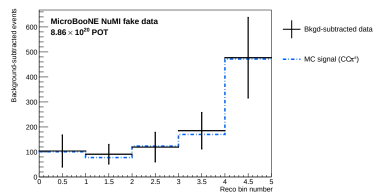
Figure 9: Stage 1, FHC: selected reconstructed spectra scaled to the full exposure, with the MC split into signal and background. Points are the Poisson-thrown fake data. The RHC and combined equivalents are Figs. 11 and 13.



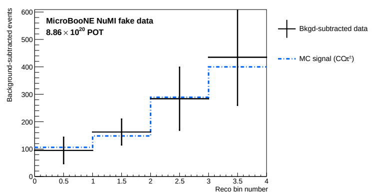
(a) p_μ



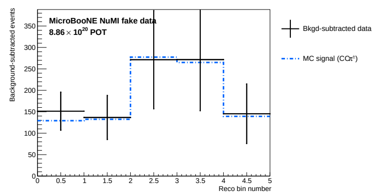
(b) p_π



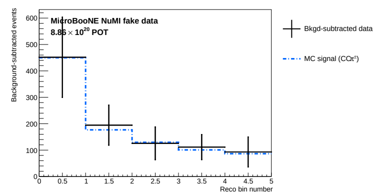
(c) $\cos\theta_\mu$



(d) $\cos\theta_\pi$

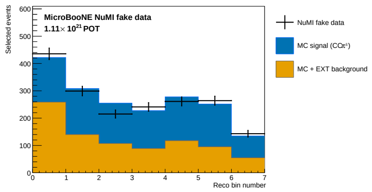


(e) $\theta_{\mu\pi}$

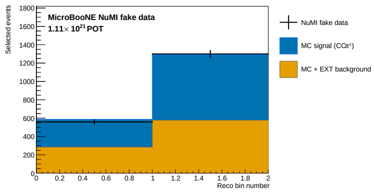


(f) θ_μ

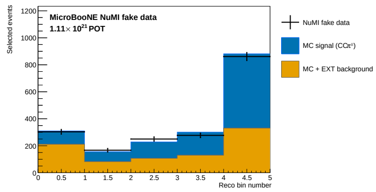
Figure 10: Stage 2, FHC: background-subtracted data compared with the central-value MC signal; the input to the unfolding. RHC and combined: Figs. 12 and 14.



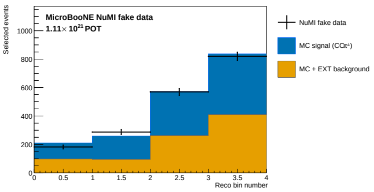
(a) p_μ



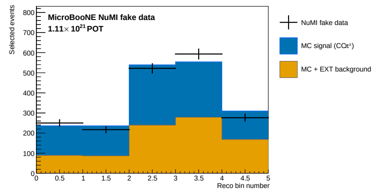
(b) p_π



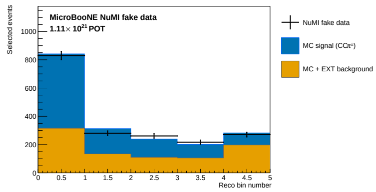
(c) $\cos\theta_\mu$



(d) $\cos\theta_\pi$

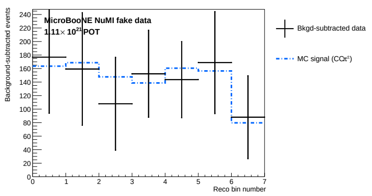


(e) $\theta_{\mu\pi}$

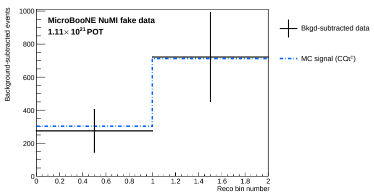


(f) θ_μ

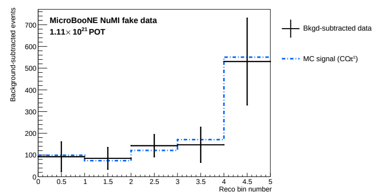
Figure 11: Stage 1, RHC: selected reconstructed spectra scaled to the full RHC exposure, MC split into signal and background.



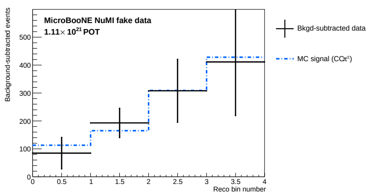
(a) p_μ



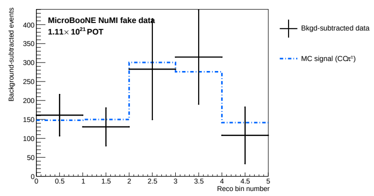
(b) p_π



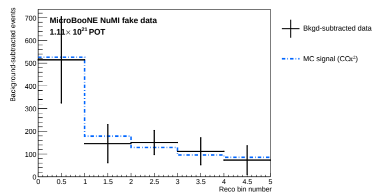
(c) $\cos\theta_\mu$



(d) $\cos\theta_\pi$

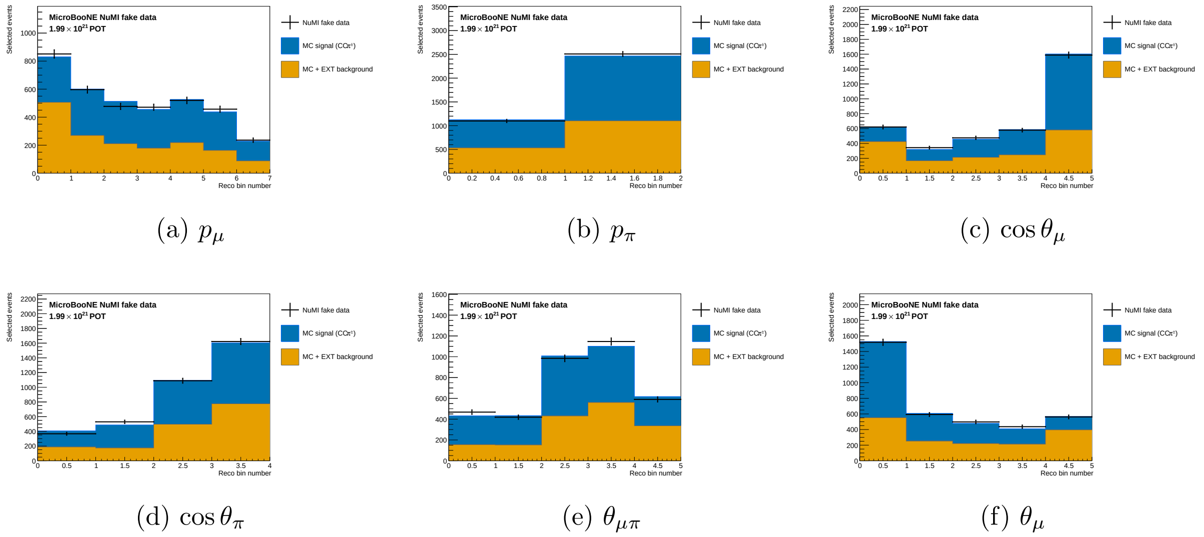


(e) $\theta_{\mu\pi}$



(f) θ_μ

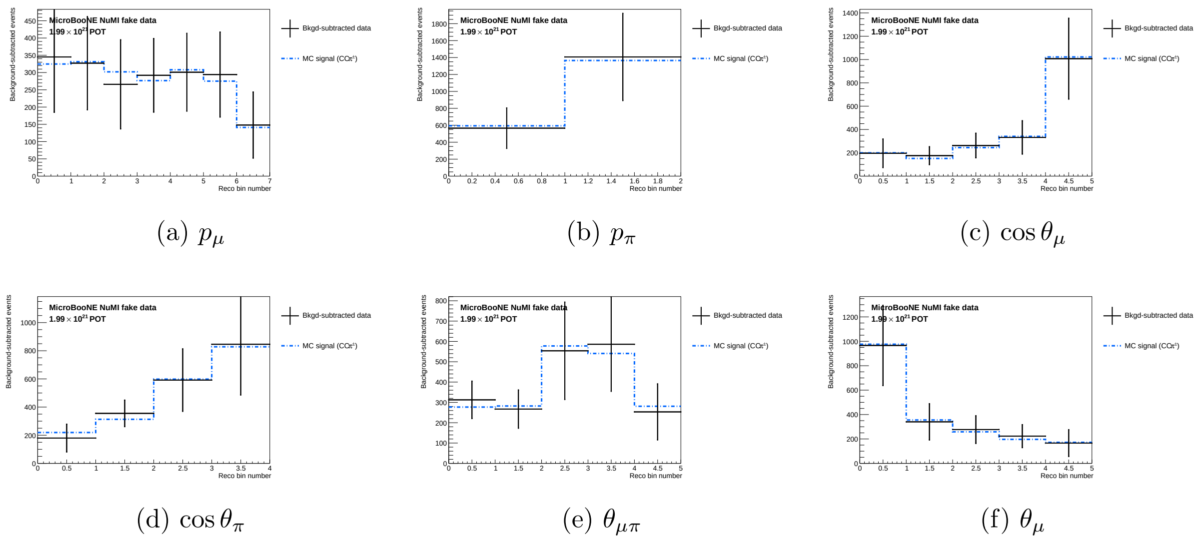
Figure 12: Stage 2, RHC: background-subtracted data compared with the central-value MC signal.



(a) p_μ (b) p_π (c) $\cos\theta_\mu$

(d) $\cos\theta_\pi$ (e) $\theta_{\mu\pi}$ (f) θ_μ

Figure 13: Stage 1, combined: selected reconstructed spectra at the combined FHC+RHC exposure.



(a) p_μ (b) p_π (c) $\cos\theta_\mu$

(d) $\cos\theta_\pi$ (e) $\theta_{\mu\pi}$ (f) θ_μ

Figure 14: Stage 2, combined: background-subtracted data compared with the central-value MC signal.

10.2 Signal extraction

After the event selection, the background-subtracted reco-space signal spectrum for observable x is:

$$d_i^{\text{sig}} = d_i^{\text{data}} - b_i^{\nu\text{-bkg}} - b_i^{\text{EXT}} - b_i^{\text{dirt}}, \tag{9}$$

where i runs over reconstructed bins.

10.3 Response matrix

Two distinct matrices are in play and it is worth separating them, since they differ by exactly the efficiency. The *migration* matrix

$$M_{ij} = P(\text{reco in bin } i \,|\, \text{true in bin } j,\ \text{selected}), \qquad \textstyle\sum_i M_{ij} = 1, \tag{10}$$

describes only where selected events move between bins. The *smearceptance* matrix additionally carries the probability of being selected at all,

$$S_{ij} = \varepsilon_j \, M_{ij}, \qquad \textstyle\sum_i S_{ij} = \varepsilon_j, \tag{11}$$

where ε_j is the ratio of selected (`h_smear_sel`) to generated (`h_true_gen`) signal events in true bin j.

It is S, not M, that enters the unfolding: `UnfolderNuMI.C` passes `get_cv_smearceptance_matrix()` to the unfolder, and the efficiency of Fig. 19 is recovered as its column sums. S is built from selected signal MC using the same binning on both axes. Figure 18 plots the column-normalised M rather than S, because normalising each column to unity makes the migration structure legible independently of how the efficiency varies across the true bins.

10.4 Regularised unfolding: the bias–variance trade-off

Unfolding inverts the folding relation $\vec{m} = A\,\vec{s} + \vec{b}$ to estimate the true signal spectrum $\vec{s}$ from the measured $\vec{m}$. The naive inverse $A^{-1}(\vec{m} - \vec{b})$ is unbiased but useless: bin migration makes A nearly singular, so its inverse amplifies statistical fluctuations into large, anticorrelated bin-to-bin oscillations. Every practical method therefore *regularises*, trading a small, controlled bias for a large reduction in variance. This analysis uses two complementary methods (a fuller pedagogical treatment is given in the companion note `report/unfolding_note.tex`).

Wiener-SVD (primary). Diagonalising the response with an SVD, each mode k is weighted by the Wiener filter $W_k = \sigma_k^2/(\sigma_k^2 + N_k)$; the minimum-mean-squared-error weighting, $\simeq 1$ for well-measured modes and $\to 0$ for noise-dominated ones, so no regularisation strength is tuned by hand. The price is a *known* linear bias: the unfolded data estimates not $\vec{s}$ but the smeared spectrum $A_C\vec{s}$, where the additional-smearing matrix A_C is published so that every theory prediction is smeared by the same A_C before comparison.

D'Agostini iterative Bayesian (cross-check). Bayes' theorem inverts the migration, $P(C_j|E_i) = R_{ij}\pi_j / \sum_l R_{il}\pi_l$, and the unfolded estimate is $\hat{s}_j = \varepsilon_j^{-1} \sum_i P(C_j|E_i)(m_i - b_i)$. The MC prior π_j is updated with the result and iterated; the *number of iterations* is the regularisation knob (few $\Rightarrow$ prior-biased, many $\Rightarrow$ noisy). It carries no A_C, so its integral is essentially unbiased and theory is compared raw.

Table 12: Trade-offs of the two unfolding methods. Neither is "more correct"; they make complementary choices about how to trade bias for variance and how to expose the residual bias.

	Wiener-SVD	**D'Agostini (iter. Bayes)**
Regularisation	Wiener filter (automatic, from S/N)	Iteration count n (chosen)
Residual bias	Explicit A_C, published	Implicit; set by early stopping
Result integral	Suppressed, observable-dependent	Essentially unbiased (physical)
Theory comparison	Smear theory by A_C	Raw theory
Positivity	Not guaranteed	Guaranteed
Uncertainties	Closed-form, linear	Correlated across iters.; needs care

Wiener-SVD is the primary result, for its objective regularisation and its explicit, reportable bias A_C; D'Agostini serves as the quantitative cross-check (Sec. 4.5 of the technical supplement), where its flat, iteration-stable integral, $\sim$ 20–40% above Wiener-SVD (config-dependent), confirms that the observable-to-observable spread of the Wiener-SVD integrated cross sections (Table 18) is the A_C smearing rather than a physical or acceptance effect.

10.5 Wiener-SVD unfolding

The primary method is Wiener-SVD [7], implemented in `xsec/WienerSVD.h` (`RunWienerSVD_FW`) as a faithful replica of the official XSecAnalyzer `WienerSVDUnfolder`: Cholesky whitening of the data covariance, regularisation inside the SVD of $R\,C^{-1}$, and the BNL filter (Eq. 3.24 of Ref. [7]). The method applies a Wiener filter in the SVD basis, providing adaptive regularisation whose strength is set by the data covariance. Because the custom pipeline does not throw systematic universes, the covariance is imported from the framework (the prediction covariance scaled to the custom prediction, plus the custom data Poisson; Sec. B of the technical supplement); a stat-only covariance under-regularises and over-corrects the integral by $\sim 26\%$.

The differential cross section is:

$$\left.\frac{d\sigma}{dx}\right|_i = \frac{\hat{N}_i^{\text{unfold}}}{\varepsilon_i \cdot \Delta x_i \cdot \Phi_{\text{total}} \cdot N_{\text{Ar}}}, \tag{12}$$

where $\hat{N}_i$ is the Wiener-SVD unfolded signal count in true bin i, $\Phi_{\text{total}} = \Phi_{\text{per POT}} \times T_{\text{POT}}$ is the total integrated flux, and N_{Ar} is the number of argon target nuclei in the FV. Equivalently, the events-to-cross-section conversion is $\text{conv} = N_{\text{Ar}}\,\Phi/10^{-38}$ (so $\sigma_i = \hat{N}_i/(\varepsilon_i\,\text{conv})$), with the authoritative flux constant $\phi = 6.81159 \times 10^{-10}$ cm^{-2}/POT ($E > 60$ MeV; Sec. 4). At the fake-data POT the inputs are:

Quantity	Value
N_{Ar} (FV, dead-region excluded)	8.710×10^{29}
Flux constant ϕ ($E > 60$ MeV)	6.81159×10^{-10} cm^{-2}/POT
POT	7.630913×10^{20}
Integrated flux $\Phi = \phi\times$POT	5.198×10^{11} cm^{-2}
$\text{conv} = N_{\text{Ar}}\,\Phi/10^{-38}$	4.527×10^{3}

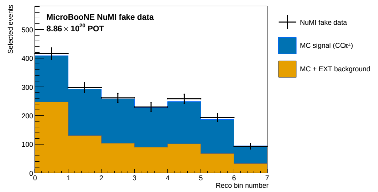
An integrated cross section is computed as the width-weighted sum,

$$\sigma_{\text{int}} = \sum_i \left.\frac{d\sigma}{dx}\right|_i \Delta x_i, \tag{13}$$

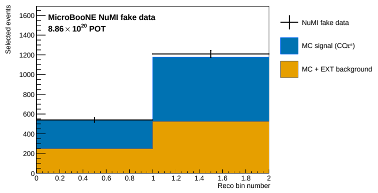
and serves as a closure check: all five observables must agree.

10.6 Reconstructed spectra and response matrices

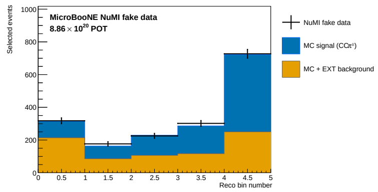
Figure 15 shows the background-subtracted reco-space signal spectra for all six observables, overlaid with the MC signal prediction and the individual background components.



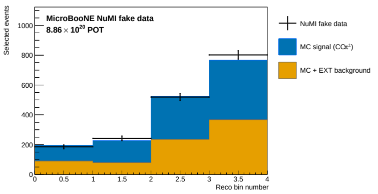
(a) p_μ



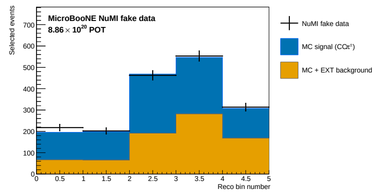
(b) p_π



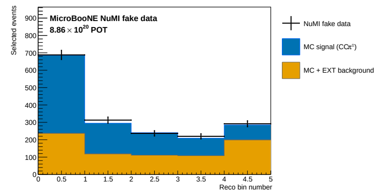
(c) $\cos\theta_\mu$



(d) $\cos\theta_\pi$

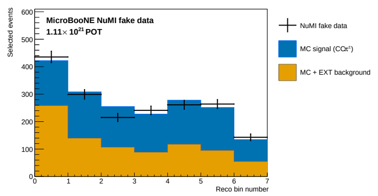


(e) $\theta_{\mu\pi}$

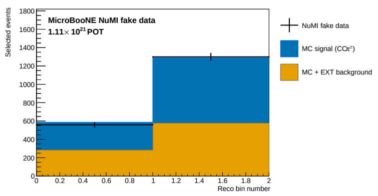


(f) θ_μ

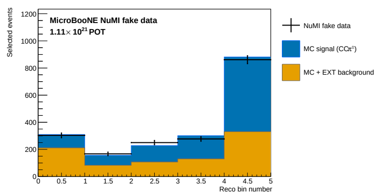
Figure 15: Reconstructed-space spectra at the final cut for all six observables, FHC full exposure (Runs 1+2+4+5): Poisson-thrown central-value fake data (black points), stacked signal (blue), beam background (orange), dirt (green), and EXT (grey). The fake data is drawn from the central-value Monte Carlo and so agrees with the stacked prediction by construction (a reco-level self-closure); the differential results and their closure against the A_C-smeared truth are shown in Fig. 21.



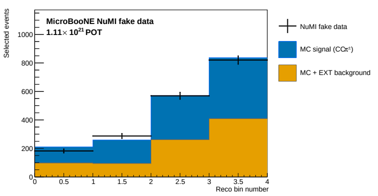
(a) p_μ



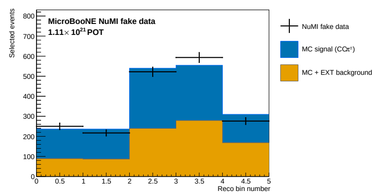
(b) p_π



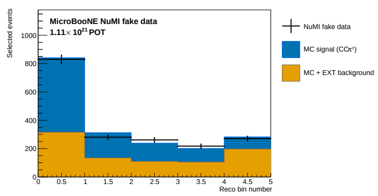
(c) $\cos\theta_\mu$



(d) $\cos\theta_\pi$

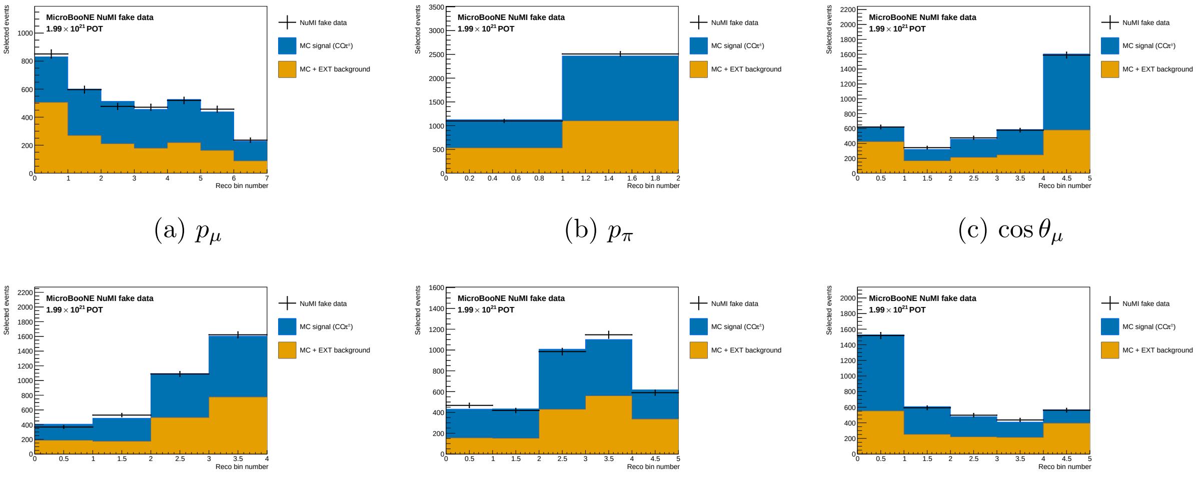


(e) $\theta_{\mu\pi}$



(f) θ_μ

Figure 16: Reconstructed-space spectra, RHC full exposure (Runs 1+2+3+4a+4b+4c); same components as Fig. 15.



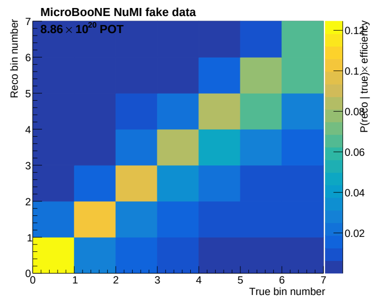
(a) p_μ (b) p_π (c) $\cos\theta_\mu$

(d) $\cos\theta_\pi$ (e) $\theta_{\mu\pi}$ (f) θ_μ

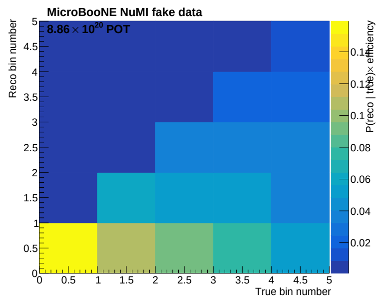
Figure 17: Reconstructed-space spectra, combined FHC+RHC (per-mode normalised); same components as Fig. 15.

Figure 18 shows the response matrices (normalised per true bin) for the six observables. The angular observables and p_μ are near-diagonal; for p_μ the width is set mainly by the MCS estimator, which reconstructs 63% of the selected sample and is intrinsically less precise than range.

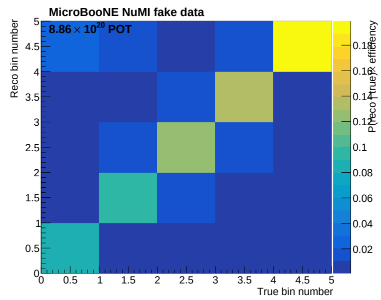
p_π is the outlier. Its response is markedly non-diagonal: reconstructed events pile into the first bin from every true bin, so the diagonal band is largely absent. The likely cause is physical rather than algorithmic, charged pions interact hadronically before coming to rest, so the track range under-estimates the momentum, and no range-based estimator can fully recover it. The consequence is that the unfolded p_π result leans more heavily on the regularisation than the other observables, and it is the observable for which a dedicated resolution and binning study would be most valuable.



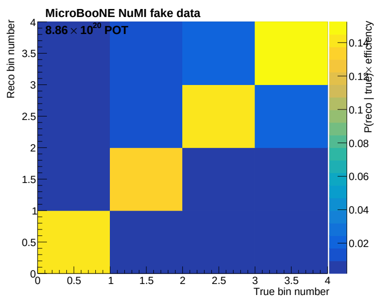
(a) p_μ



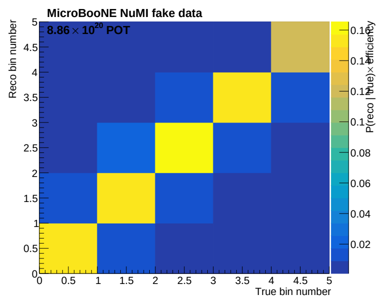
(b) p_π



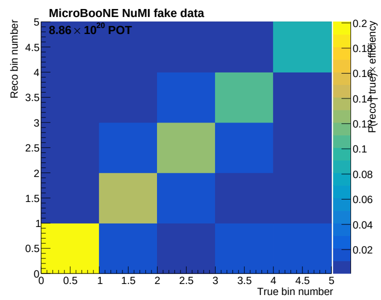
(c) $\cos\theta_\mu$



(d) $\cos\theta_\pi$



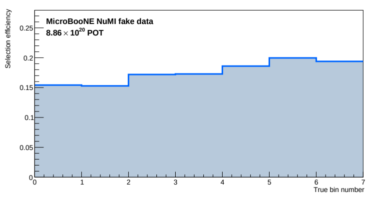
(e) $\theta_{\mu\pi}$



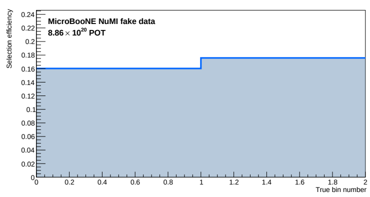
(f) θ_μ

Figure 18: Response matrices A_{ij} for the six observables. Each column is normalised to unity (migration probability). Colour scale: white = 0, dark blue = 1.

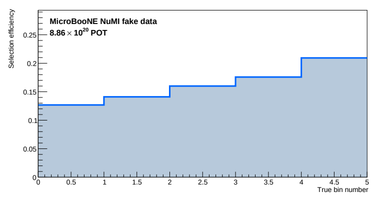
Figure 19 shows the selection efficiency per true bin for all six observables, computed as the ratio of selected to generated signal MC.



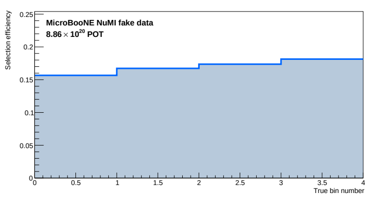
(a) p_μ



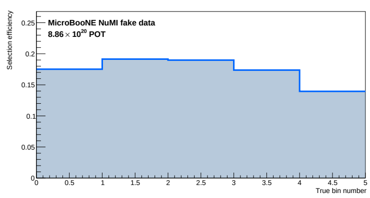
(b) p_π



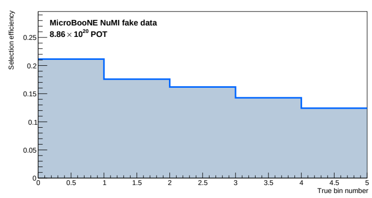
(c) $\cos\theta_\mu$



(d) $\cos\theta_\pi$



(e) $\theta_{\mu\pi}$

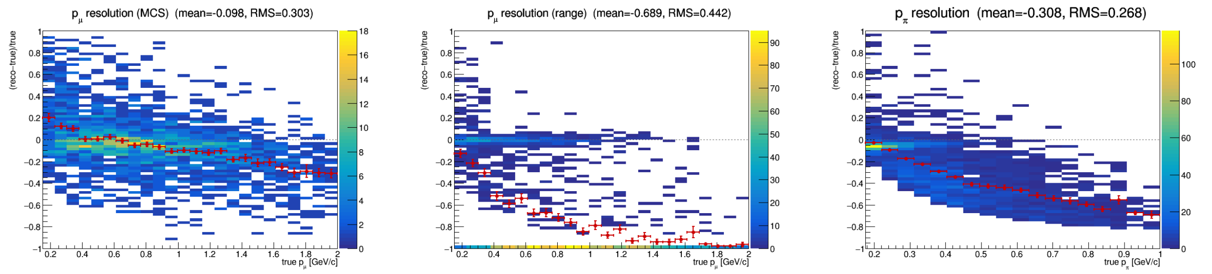


(f) θ_μ

Figure 19: Selection efficiency as a function of true observable value, in the measured phase space ($p_\mu > 0.15$, $p_\pi > 0.175\,\mathrm{GeV}/c$, $\theta_{\mu\pi} < 2.6$ rad). Efficiency is the ratio of selected to generated signal MC ($\varepsilon_i = \sum_j A_{ij}$, integrated over reco bins). With the momentum thresholds and the 2.6 rad opening-angle cut the efficiency is flat at ~13–20% across all observables; the former ~5% low-momentum turn-on bins and the cosmic-dominated large-angle region are removed. The lowest point is the smallest opening-angle bin (~9% efficiency); it is retained because it is high-purity (~58%), not low-purity.

10.7 Reconstruction resolution

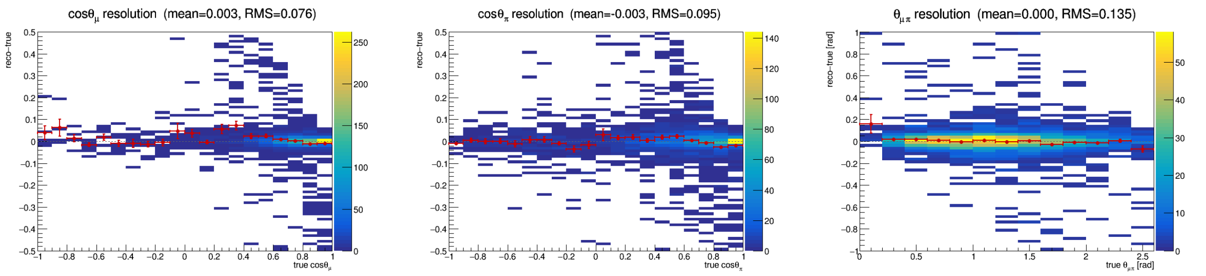
Figure 20 shows the reconstruction resolution of each measured observable, evaluated on selected signal events (GENIE ν_μ overlay, measured phase space). For the momenta the relative resolution (reco − true)/true is shown; for the angular observables the absolute residual (reco − true) is used, since the relative metric diverges as $\cos\theta \to 0$ / $\theta_{\mu\pi} \to 0$. The red points are the mean residual per true bin; the quoted numbers are the overall mean (bias) and RMS (resolution).



(a) p_μ (MCS): bias −9.8%, RMS 30%.

(b) p_μ (range): valid only for contained muons.

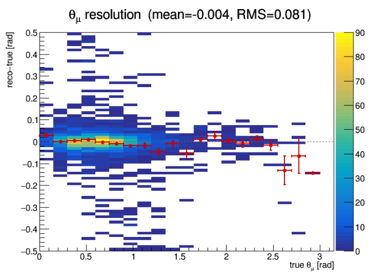
(c) p_π: bias −31%, RMS 27% (analysis estimator).



(d) $\cos\theta_\mu$: RMS 0.076.

(e) $\cos\theta_\pi$: RMS 0.095.

(f) $\theta_{\mu\pi}$: RMS 0.135 rad.



(g) θ_μ: RMS 0.081 rad.

Figure 20: Reconstruction resolution vs. true value, selected signal. Momenta: relative (reco − true)/true; angles: absolute (reco − true). Red points: mean residual per true bin. θ_μ is derived as $\arccos(\cos\theta_\mu)$, exactly as the bin config does it, so the panel shows the quantity the analysis bins on; it is unbiased (−0.004 rad) with RMS 0.081 rad, the drift below zero above ~ 2.5 rad being a handful of backward-going muons. The muon MCS momentum slightly over-estimates at low p_μ and increasingly under-estimates above ∼1 GeV/c; the range estimator is badly biased for *exiting* muons (range = distance to wall), which is why the analysis uses range only for contained muons (range-or-MCS). The pion panel shows the estimator the analysis actually bins on, not the raw `candidate_pion_mom_reco` branch (Sec. 1.1 of the technical supplement): mean bias −31%, RMS 27%. That average is dominated by high momenta; the bias is only −2% in the first true bin and reaches −65% above 0.8 GeV/c; because the reconstructed value saturates near 0.34 GeV/c once pions interact hadronically before stopping. The angular variables are essentially unbiased with narrow resolution.

10.8 Model comparison: χ^2 against truth, tune and generators

Table 13 collects the bin-by-bin χ^2 of the unfolded fake data against each prediction, for every observable and every beam configuration. All predictions are A_C-smeared, so the comparison is made in the measurement space, and the full covariance is used. The *truth* column is the closure test: it is small throughout, confirming the chain, and is visibly larger for RHC than for FHC, consistent with the poorer RHC angular response noted in Sec. 5.1 of the technical supplement.

An anomaly that turned out to be a bug. Earlier versions of this note reported χ^2 values for NEUT and NuWro in $\cos\theta_\mu$ and θ_μ an order of magnitude above every other observable — 40.6/5 and 38.6/5 in FHC θ_μ — and devoted a paragraph to explaining them as an artefact of A_C suppression acting on sharply peaked generators.

The explanation was wrong and so were the numbers. The predictions were being multiplied by A_C *twice*: once in the extractor, which applies the additional smearing to every entry of the prediction map in place, and again on the plotting path. Verified as $A_C^2 \cdot \text{raw} = \text{plotted} \times \text{width}$ to 1.0000 in every bin. The second application has been removed.

With the correction the anomaly disappears: FHC θ_μ gives 8.96/5 for NEUT and 9.43/5 for NuWro, and $\cos\theta_\mu$ gives 4.90/5 and 4.93/5. No observable now sits an order of magnitude apart from the others, and the earlier argument for preferring θ over $\cos\theta$ on χ^2 grounds falls away with the effect that motivated it.

The bug affected only the model comparisons. The unfolded cross sections are unchanged (FHC p_μ integrates to 0.8275 before and after), and the closure test was never affected, because the fake-data truth is the one curve that was not passed through the second multiplication.

Table 13: χ^2/ndf of the unfolded fake data against each A_C-smeared prediction, for the inclusive selection. "truth" is the fake-data truth (the closure test).

Beam	Observable	truth	uB tune	GENIE	GiBUU	NEUT	NuWro
FHC	p_μ	0.10/7	0.09/7	0.57/7	0.13/7	0.74/7	0.53/7
	p_π	0.10/2	0.09/2	0.47/2	1.11/2	0.58/2	0.44/2
	$\cos\theta_\mu$	0.99/5	0.74/5	0.98/5	0.84/5	4.90/5	4.93/5
	$\cos\theta_\pi$	0.75/4	0.90/4	1.67/4	3.07/4	2.02/4	2.01/4
	$\theta_{\mu\pi}$	1.02/5	0.83/5	1.94/5	2.44/5	1.45/5	0.70/5
	θ_μ	0.72/5	0.45/5	2.13/5	2.87/5	8.96/5	9.43/5
RHC	p_μ	1.72/7	2.78/7	2.71/7	2.55/7	2.85/7	3.21/7
	p_π	0.24/2	0.21/2	0.82/2	1.15/2	0.70/2	0.54/2
	$\cos\theta_\mu$	0.42/5	1.69/5	1.57/5	0.90/5	5.32/5	5.59/5
	$\cos\theta_\pi$	1.54/4	1.23/4	2.66/4	2.38/4	1.87/4	2.76/4
	$\theta_{\mu\pi}$	1.78/5	2.31/5	4.40/5	6.65/5	2.81/5	2.96/5
	θ_μ	0.84/5	1.48/5	2.39/5	2.21/5	9.17/5	9.65/5
comb	p_μ	0.67/7	1.39/7	1.78/7	1.56/7	1.99/7	2.40/7
	p_π	0.26/2	0.24/2	0.77/2	1.77/2	0.82/2	0.68/2
	$\cos\theta_\mu$	0.20/5	0.53/5	0.82/5	0.61/5	4.55/5	4.48/5
	$\cos\theta_\pi$	1.78/4	1.71/4	3.38/4	4.17/4	2.72/4	3.25/4
	$\theta_{\mu\pi}$	2.31/5	2.31/5	3.88/5	6.63/5	3.60/5	2.48/5
	θ_μ	0.76/5	0.51/5	2.51/5	3.87/5	10.19/5	10.33/5

Table 14: The same comparison for the muon and pion angle in both parameterisations. The θ bin edges are exactly arccos of the $\cos\theta$ edges, so the phase space is identical; only the variable differs. Values are post-fix: a single application of A_C, on the width-aware regularisation. The two parameterisations give comparable χ^2, and neither shows the order-of-magnitude separation reported in earlier drafts, which was the double-A_C bug rather than a property of either variable.

Beam	Variable	truth	uB tune	GENIE	GiBUU	NEUT	NuWro
FHC	$\cos\theta_\mu$	0.99/5	0.74/5	0.98/5	0.84/5	4.90/5	4.93/5
FHC	θ_μ	0.72/5	0.45/5	2.13/5	2.87/5	8.96/5	9.43/5
RHC	$\cos\theta_\mu$	0.42/5	1.69/5	1.57/5	0.90/5	5.32/5	5.59/5
RHC	θ_μ	0.84/5	1.48/5	2.39/5	2.21/5	9.17/5	9.65/5
Comb	$\cos\theta_\mu$	0.20/5	0.53/5	0.82/5	0.61/5	4.55/5	4.48/5
Comb	θ_μ	0.76/5	0.51/5	2.51/5	3.87/5	10.19/5	10.33/5

11 Full-exposure results: FHC, RHC, and combined

11.1 Scope and status of each result

The analysis has been through a number of corrections, several of which changed results and two of which invalidated conclusions (all recorded, with before-and-after values, in the change log). This subsection states plainly what each result currently is, so that no number in this note has to be interpreted by working out which fixes preceded it.

All fifty-one extractions now carry a full systematic covariance including detector variations. The proton-tagged results previously carried none — the detector variation samples had been processed but never referenced by the proton-tagged configuration — and were rebuilt and re-unfolded on 2026-08-31 to correct that. Their detector terms (10.5%–64.4%, median 23.9%) are of the same order as the inclusive ones (16.9%–42.8%).

The proton-tagged results remain secondary scope for the narrower reason that their model comparisons have not been recomputed since, and because the exposure does not support the differential $W_{\pi p}$ shape at any binning.

Result	Status	Basis
Primary scope		
Total flux-averaged cross section (cut-and-count), all three configurations	Final	Not suppressed by A_C; the most directly interpretable total.
$d\sigma/dp_\mu$	Final	Full covariance including detector variations.
$d\sigma/d\cos\theta_\mu$ and $d\sigma/d\theta_\mu$	Final	Full covariance; discriminates between models after the width-aware regularisation fix.
$d\sigma/d\cos\theta_\pi$ and $d\sigma/d\theta_\pi$	Final	Full covariance.
$d\sigma/d\theta_{\mu\pi}$	Final	Full covariance; the best-conditioned shape observable.
$d\sigma/dp_\pi$, two-bin	Final, coarse	Full covariance, but a two-region partition rather than a spectrum: one near-threshold interval and one broad inclusive interval above 0.205 GeV/c. See below.
Secondary scope — partial covariance		
Proton-tagged integrated cross section	Complete	Detector systematics propagated, central value postdates the weighting fix, closure χ^2 recomputed, A_C and covariance released.
Proton-tagged differential results, including transverse kinematic imbalance	Complete	Detector systematics propagated; beam-frame reconstruction corrected; central values postdate the weighting fix; A_C and covariance released.
Withdrawn — do not quote		
Differential $W_{\pi p}$ shape	Withdrawn	No binning satisfies both the migration and the statistics requirement at this exposure; supports an integrated cross section only.
Five-bin $d\sigma/dp_\pi$	Withdrawn	Four of five bins fall below the 68% diagonal criterion.
Transverse kinematic imbalance about detector z	Withdrawn	Measured and predicted quantities were different observables.
Model comparisons predating the width-aware regularisation	Withdrawn	A_C was near rank one; apparent agreement was an artefact.

On the two-bin p_π result. Charged pions interact hadronically before ranging out, so the reconstructed momentum saturates near 0.34 GeV/c: a pion of true momentum 0.95 GeV/c reconstructs at almost the same value as one of 0.55 GeV/c. The two bins have migration diagonals of 91% and 76% and are sound, but the upper bin integrates over a very wide physical interval and is not a differential point in the usual sense. It is plotted at equal visual width for legibility, which is flagged on the axis, and should be read as a region rather than a spectrum. The estimator survey establishing that this is a physical information limit rather than an unrefined estimator is in the technical supplement.

On the closure χ^2 values. The closure χ^2 quoted for each extraction compares the unfolded result with the fake-data truth using the *full* systematic covariance, and the resulting p-values are almost all above 0.9 (median 0.96 across all fifty-one extractions). That should not be read as excellent agreement with the model. For a self-closure with correctly-sized uncertainties the p-values would be roughly uniform on $[0, 1]$, and they are not.

The reason is that the measurement and the truth it is being compared against share their systematics: both come from the same generator, the same flux and the same exposure, so the flux, cross-section and exposure terms are common-mode and largely cancel in the difference while still inflating the covariance used to judge it. Recomputing the same comparison with the statistical covariance alone raises χ^2/ndf from 0.02–0.2 to 0.5–0.8, much nearer the expected value:

Extraction	full	prediction-only	statistics-only
inclusive FHC p_μ	0.14/7	0.22/7	1.01/7
inclusive comb p_μ	1.43/7	2.87/7	4.71/7
proton-tagged FHC p_μ	1.01/7	2.20/7	3.02/7
proton-tagged comb $W_{\pi p}$	0.48/6	0.77/6	4.88/6

The closure χ^2 is therefore a check that the extraction does not *disagree* with its input beyond the quoted band, which it passes, and not a measure of how well the band is sized. A test of the latter needs either a covariance restricted to the terms that differ between measurement and reference, or an ensemble of throws. Neither is done here, and the p-values should be read with that limitation in mind.

Ensemble coverage: are the uncertainties the right size? The closure test above uses a single throw. It can show that the extraction reproduces its input, but a single residual has no width, so it cannot say whether the quoted uncertainty is correctly sized. To answer that, thirty-two statistically independent Poisson realisations of the FHC data were each put through the *whole* chain — thrown, universes rebuilt from that throw, unfolded — and the per-bin pull

$$p_i = \frac{\hat{x}_i - (A_C\, x_{\mathrm{CV}})_i}{\sigma_i}$$

formed against the fixed central-value truth smeared by that member's own A_C. (The per-throw *fake-data* truth is not a valid reference: it follows the Poisson draw, so numerator and reference fluctuate together and the width comes out too small.)

Two observables were run, each with 32 members, chosen to bracket the conditioning: the well-behaved p_μ, and $\cos\theta_\mu$, whose A_C row sums fall as low as 0.046 so that the regularisation removes up to 95% of a bin's normalisation. If the uncertainties were mis-sized anywhere, the second is where it would show.

Reference covariance	pull mean	pull width	68% / 95% coverage
statistical only			
$d\sigma/dp_\mu$	−0.20	1.04	63.8% / 95.5%
$d\sigma/d\cos\theta_\mu$	−0.18	0.96	73.1% / 94.4%
full			
$d\sigma/dp_\mu$	−0.06	0.32	99.6% / 100%
$d\sigma/d\cos\theta_\mu$	−0.05	0.31	98.8% / 100%
expected	0	1	68% / 95%

Against the statistical covariance both widths are consistent with unity — 1.04 and 0.96, each determined to ±13% with this many members — and the interval coverage is correct. **The statistical uncertainties are correctly sized, and the strongly suppressed observable behaves no differently from the well-conditioned one.** That the severe A_C suppression in $\cos\theta_\mu$ does not distort its uncertainties is the more informative of the two results, since p_μ passing was the expected outcome.

Two further things the ensemble shows. Against the *full* covariance the pull width is 0.31–0.32 for both observables and coverage is essentially 100%: the quoted uncertainty is about three times the throw-to-throw spread, because the ensemble varies only the data statistics while the denominator carries flux, detector and cross-section terms that are identical in every member. That is the same effect as the closure p-values clustered near unity, now measured rather than argued.

Second, the ensemble mean sits 2.5% below the reference (0.7887 against 0.8090), a 2.1σ offset on the mean, and the per-bin pull means are all negative and consistent at -0.10 to -0.25. This is a real if small negative bias in the extraction, around a fifth of the statistical uncertainty and far inside the 41% systematic band, but it is a bias rather than a fluctuation and should be quoted as such rather than absorbed silently.

The study covers two observables in one configuration (FHC). Extending it to RHC and to the proton-tagged family would test whether the same holds where the statistics are sparser; the machinery is in place and each observable costs about four hours.

On the blinding status. Every "data" point in this note is Poisson-thrown central-value fake data. No beam-on signal region has been unblinded. Statements comparing configurations, or comparing the result to a generator, are therefore properties of the injected sample and the extraction machinery, not measurements.

This is the primary result of the analysis: the CC1μ1$\pi^{\pm}$ cross section extracted over the full available exposure in forward horn current (FHC), reverse horn current (RHC), and their combination, all on Poisson-thrown central-value fake data. The FHC measurement uses Runs 1, 2, 4 and 5 (data POT 8.857×10^{20}); RHC uses Runs 1, 2, 3, 4a, 4b, 4c (1.108×10^{21}); the combined measurement uses all fourteen overlay files at the joint POT 1.994×10^{21}. All three close cleanly against the A_C-smeared truth for every observable (Tables 15–17); the differential results are shown in Figs. 21–23.

Table 15: FHC (Runs 1, 2, 4, 5; data POT 8.857×10^{20}): integrated cross section, the two largest systematic terms, the total, and the fake-data closure against A_C-smeared truth. All values are read from the released covariance and results table. The reco-level flux and unfolding-amplification columns of earlier drafts are dropped here and treated in §13.2, since a reco-level flux is not re-derivable from the current extraction logs and mixing passes in one table is how stale numbers survive.

Observable	σ_{int}	Flux	Detector	Total	Closure χ^2/ndf
p_μ	0.828	28.1%	25.0%	41.1%	0.10/7 ($p = 1.00$)
p_π	0.855	31.0%	28.8%	48.1%	0.10/2 ($p = 0.95$)
$\cos\theta_\mu$	0.764	26.8%	26.8%	40.2%	0.99/5 ($p = 0.96$)
$\cos\theta_\pi$	0.751	26.2%	22.2%	38.3%	0.75/4 ($p = 0.94$)
$\theta_{\mu\pi}$	0.857	26.8%	18.6%	34.7%	1.02/5 ($p = 0.96$)
θ_μ	0.846	26.7%	16.9%	33.8%	0.72/5 ($p = 0.98$)

We extend the measurement to the reverse-horn-current beam as both a standalone measurement and a combined $\nu_\mu + \bar{\nu}_\mu$ result. The RHC *flux* is $\bar{\nu}_\mu$-dominant (54.6% $\bar{\nu}_\mu$, versus FHC's 35%), with authoritative > 60 MeV integrals $\Phi^{\text{RHC}}_{\nu_\mu} = 2.923 \times 10^{-10}$, $\Phi^{\text{RHC}}_{\bar{\nu}_\mu} = 3.523 \times 10^{-10}$, total 6.446×10^{-10} cm^{-2}POT^{-1} (the FHC-convention computation reproduces the authoritative FHC values exactly). The *measured event rate*, however, remains ν_μ-dominant (76% ν_μ, 20% $\bar{\nu}_\mu$) because the ν_μ cross section is $\sim$2–3$\times$ larger than the $\bar{\nu}_\mu$ one. The RHC overlays (Runs 1, 2, 3, 4a, 4b, 4c) are all verified as genuine ν_μ overlays and processed, 4.05×10^{6} events, combined MC POT 1.546×10^{22}, mutually consistent (2.62×10^{-16} events/POT) with the PPFX weights, covering the full RHC exposure (Run 3, the largest period at 5.52×10^{21} MC POT / 1.44×10^{6} events, was added last). Native Run-4 RHC detector-variation samples (the standard eight-knob set: light-yield down/Rayleigh, recombination, space charge, and the four wire-modification variations) are now used for the RHC detector systematic, verified as genuine ν_μ overlays with a uniform 3.80×10^{-16} events/POT across all knobs; they replace the Run-1 FHC stand-in used previously. Native RHC dirt is still taken from FHC. The standalone result below uses Poisson-thrown central-value fake data at the RHC data POT 1.108×10^{21}. The generator predictions are re-averaged to the RHC flux composition; the combined FHC+RHC predictions are the fluence-weighted combination of the two horn modes (0.458 FHC / 0.542 RHC, each mode weighted by $\Phi_{\rm tot}\times$POT).

The combined FHC+RHC measurement uses the summed 14-file response over the full exposure (5.6×10^6 events, combined data POT $D_{\rm comb} = 1.994 \times 10^{21} = D_{\rm FHC} + D_{\rm RHC}$), with each horn mode normalised to its *own* data exposure. This is a genuine subtlety: the detector-variation covariance is absolute (POT-scaled reco counts), so a single POT scale applied to all files would weight the modes by their Monte-Carlo POT (0.62:1 FHC:RHC) rather than by data exposure (0.80:1). Each mode's numuMC and detVar samples therefore carry a per-mode `summed_pot` (FHC 2.158×10^{22}, RHC 2.782×10^{22}, $= S_{\rm mode} \times D_{\rm comb}/D_{\rm mode}$) so the framework's $D_{\rm comb}/$`summed_pot` scaling reduces to $D_{\rm FHC}/S_{\rm FHC}$ and $D_{\rm RHC}/S_{\rm RHC}$; the fake data is Poisson-thrown with the matching per-mode rate.

Contrary to an earlier concern, the joint *flux systematic* is correct as built. All fourteen numuMC files carry the *same* 600 PPFX multisim universes, which the universe maker accumulates index-matched across both horn modes in a single chain, so universe u is the same hadron-production draw in FHC and RHC. Per-universe summing therefore propagates one parameter draw through both beam configurations and lets each mode's flux response set the result, capturing the physical (not unit) cross-horn correlation automatically; this is the rigorously correct multisim treatment, superior to any hand-built block covariance. The joint result was in fact blocked by a different, purely technical issue: a framework guard that forbade a second detector-variation file of the same type (needed to carry Run-4 FHC *and* Run-4 RHC as `detVarCV`). That guard now accumulates the per-mode detVar samples exactly as it already did for the alternate-CV samples, matching its own in-code TODO. With both in place the combined closure passes for every observable (Table 17), with an unfolded flux systematic of 22–27% that sits between the standalone FHC and RHC values, as expected for the fluence-weighted mix.

The standalone RHC extraction (native Run-4 RHC detector variations) closes cleanly for every observable (Table 16): the A_C-smeared truth is recovered with $\chi^2/\text{ndf} < 0.3$ and $p > 0.94$ throughout. Two features stand out. First, the reco-level flux systematic is $\sim$15%, genuinely below the FHC $\sim$17% floor; the $\bar{\nu}_\mu$-dominant flux carries a smaller PPFX hadron-production uncertainty. Second, the unfolding amplification (1.29–1.99×) mirrors FHC, so the unfolded flux term (19.5–28%) is comparable, again data-statistics limited.

Table 16: RHC (Runs 1, 2, 3, 4; data POT 1.1082×10^{21}): integrated cross section, the two largest systematic terms, the total, and the fake-data closure against A_C-smeared truth. All values are read from the released covariance and results table. The reco-level flux and unfolding-amplification columns of earlier drafts are dropped here and treated in §13.2, since a reco-level flux is not re-derivable from the current extraction logs and mixing passes in one table is how stale numbers survive.

Observable	$\sigma_{\rm int}$	Flux	Detector	Total	Closure χ^2/ndf
p_μ	0.575	31.9%	25.7%	44.5%	1.72/7 ($p = 0.97$)
p_π	0.763	33.4%	42.8%	59.3%	0.24/2 ($p = 0.89$)
$\cos\theta_\mu$	0.626	29.7%	40.8%	54.1%	0.42/5 ($p = 0.99$)
$\cos\theta_\pi$	0.678	26.3%	31.9%	45.4%	1.54/4 ($p = 0.82$)
$\theta_{\mu\pi}$	0.765	28.0%	18.2%	36.3%	1.78/5 ($p = 0.88$)
θ_μ	0.839	24.0%	21.2%	34.1%	0.84/5 ($p = 0.97$)

Table 17: Combined (data POT 1.9939×10^{21}): integrated cross section, the two largest systematic terms, the total, and the fake-data closure against A_C-smeared truth. All values are read from the released covariance and results table. The reco-level flux and unfolding-amplification columns of earlier drafts are dropped here and treated in §13.2, since a reco-level flux is not re-derivable from the current extraction logs and mixing passes in one table is how stale numbers survive.

Observable	σ_{int}	Flux	Detector	Total	Closure χ^2/ndf
p_μ	0.723	27.5%	25.3%	40.6%	0.67/7 ($p = 1.00$)
p_π	0.799	32.6%	34.2%	52.9%	0.26/2 ($p = 0.88$)
$\cos\theta_\mu$	0.750	27.4%	25.8%	40.1%	0.20/5 ($p = 1.00$)
$\cos\theta_\pi$	0.776	25.0%	22.9%	37.7%	1.78/4 ($p = 0.78$)
$\theta_{\mu\pi}$	0.837	27.1%	18.5%	35.1%	2.31/5 ($p = 0.81$)
θ_μ	0.842	25.0%	20.4%	34.1%	0.76/5 ($p = 0.98$)

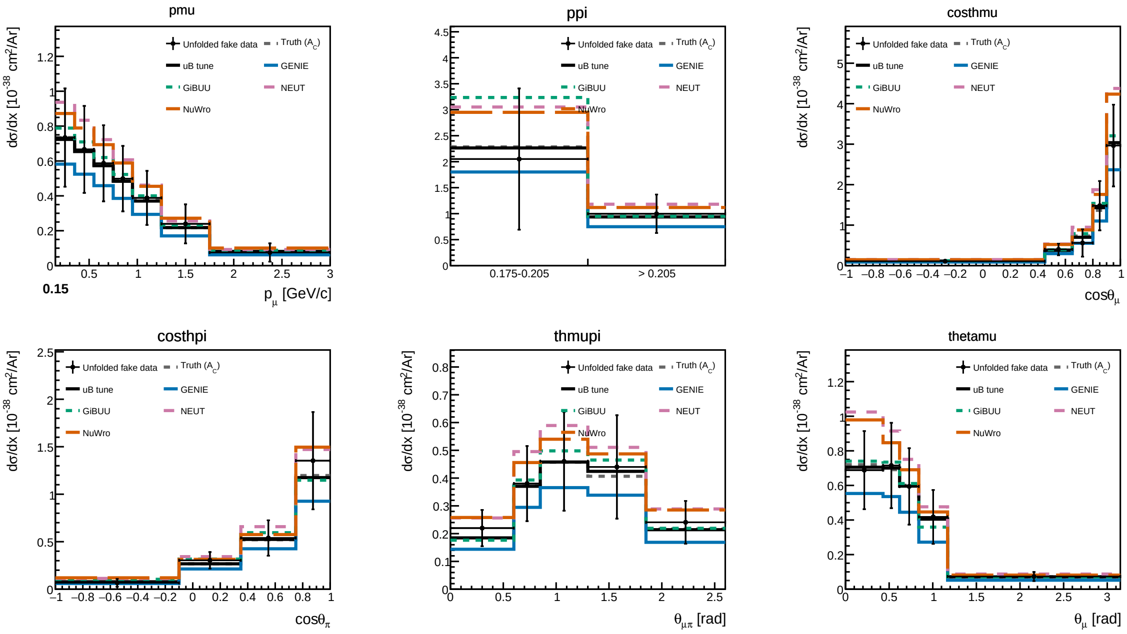


Figure 21: Differential cross sections $d\sigma/dx$ for all five observables on the current Poisson-thrown central-value fake data, FHC full exposure: unfolded fake data (black points), A_C-smeared fake-data truth (grey dashed), the MicroBooNE tune (black solid), and the four generator predictions (GENIE, GiBUU, NEUT, NuWro). The unfolded points recover the injected truth for every observable, the same closure summarised in Tables 16–17.

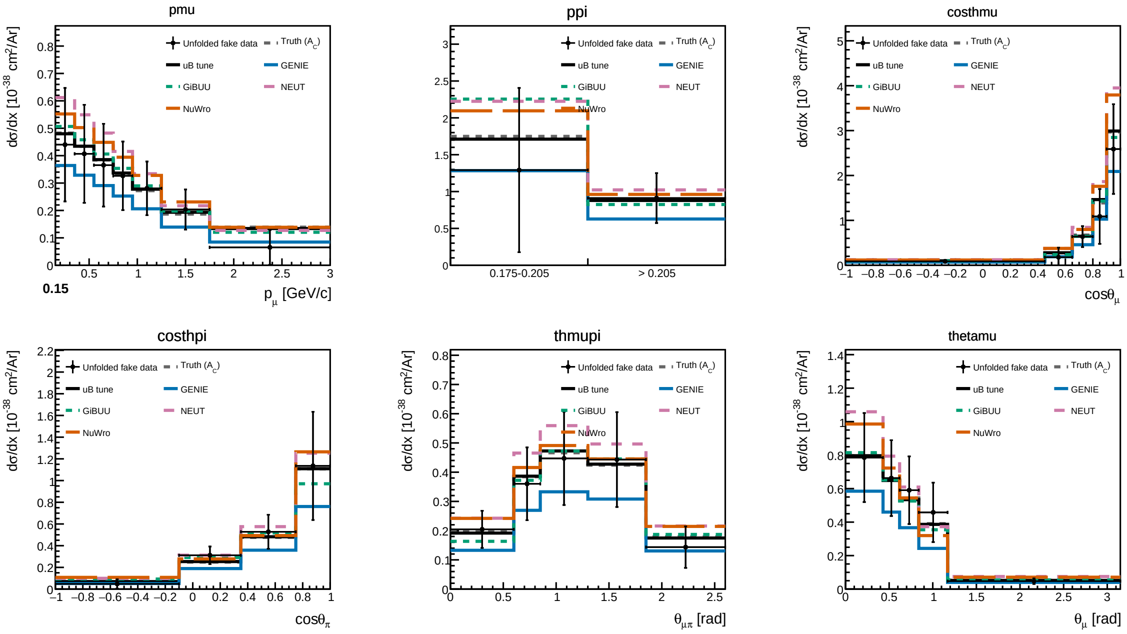


Figure 22: Standalone RHC ($\bar{\nu}_\mu$-dominant) differential cross sections on the current Poisson-thrown fake data, with the RHC-flux-averaged generator predictions. Same layout as Fig. 21.

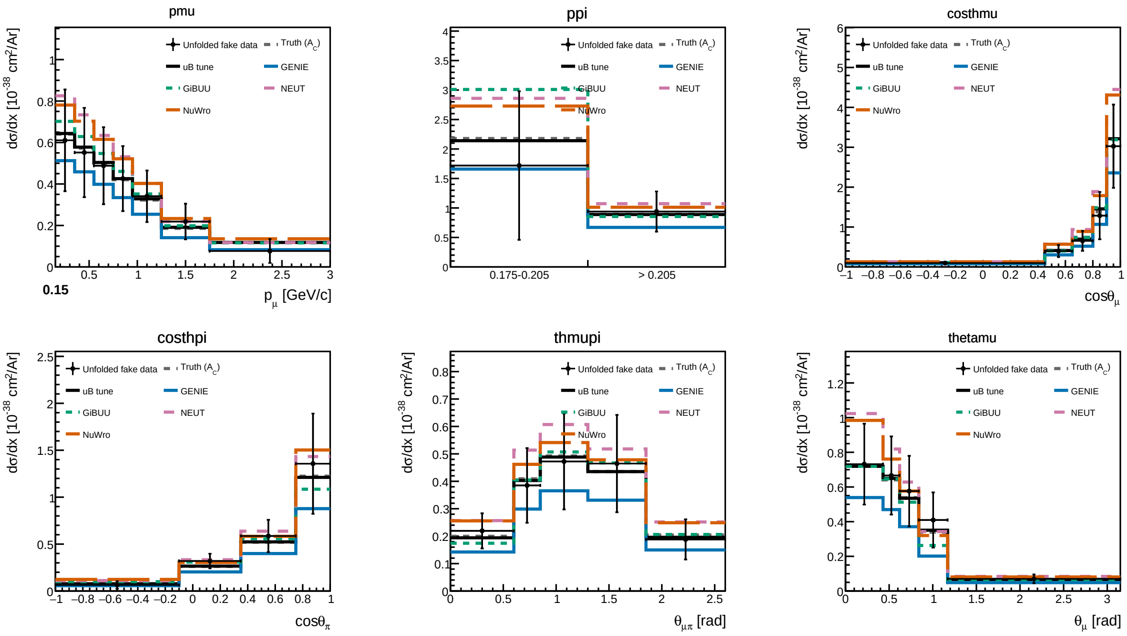


Figure 23: Combined FHC+RHC ($\nu_\mu + \bar{\nu}_\mu$) differential cross sections on the current per-mode-normalised fake data, with the fluence-weighted combined generator predictions.

Table 18: Integrated (unfolded) cross section σ_{int} over the measured phase space, for each observable and configuration (current fake data, coarsened $\cos\theta_\pi/\theta_{\mu\pi}$ binning, per-configuration flux; 10^{-38} cm^2/Ar). The value differs between observables because the Wiener-SVD additional-smearing A_C is not norm-preserving; the D'Agostini cross-check (Sec. 4.5 of the technical supplement) recovers a flatter, ∼20% higher total, confirming the spread is a smearing effect. All entries were recomputed in a single pass with the current code and a forced rebuild of the cached universes. The MC POT normalization scales each ntuple file by its own stored POT ($D_{\text{run}}/\text{POT}_{\text{file}}$): the processing writes the *total* run MC POT into every file of a run, so for the multi-file RHC runs (Run 3 split across 5 ntuple files, Run 4 across 3) the per-file scaling summed over the splits gives the correct total. An intermediate version of this analysis instead divided by the *sum* of the per-file POT values; because those values are already run totals, that over-counted the denominator by the split multiplicity and scaled the RHC/Combined simulation a factor of 5 (Run 3) and 3 (Run 4) too low, which under-subtracted the background and inflated the RHC/Combined cross sections. That change has been reverted and the RHC/Combined columns here are the corrected values, roughly 10–25% below the intermediate ones; FHC (one MC file per run) is unaffected by either convention. The last two rows give the alternative angular parameterisation θ_μ, θ_π (Sec. 5.1 of the technical supplement), now complete for all three configurations. (Values are fake-data closures; the signal is $\nu_\mu + \bar{\nu}_\mu$, so the RHC flux-average is not expected to lie far below FHC.)

Observable	FHC	RHC	Combined
p_μ	0.828	0.575	0.723
p_π	0.855	0.763	0.799
$\cos\theta_\mu$	0.764	0.626	0.750
$\cos\theta_\pi$	0.751	0.678	0.776
$\theta_{\mu\pi}$	0.857	0.765	0.837
θ_μ	0.846	0.839	0.842

Uncertainties shown. Every differential cross section and every integrated value quoted in this note carries *both* statistical and systematic uncertainties. The error bars on the unfolded points are the square root of the diagonal of the *total* covariance matrix, propagated through the unfolding: flux (PPFX multisims), cross-section (GENIE multisims and unisims), detector variations, hadron re-interaction, POT counting and target number, together with the data, MC and EXT statistical terms. For p_μ in FHC the bin-averaged total is 32.9% (Table 17 of the technical supplement), dominated by flux (24.6%), with detector 12.3%, data statistics 11.6% and cross-section 9.2%. The cut-and-count total cross sections of §11.3 quote the same combination, written there as systematic $\oplus$ data-statistical (FHC: 25.5% $\oplus$ 4.3%).

11.2 How far A_C is from preserving normalisation

Because the Wiener-SVD result estimates $A_C x_{\text{true}}$ rather than x_{true}, and A_C is not norm preserving, summing the differential result over its bins does not give the physical total cross section. This subsection quantifies the effect, which has previously only been described qualitatively.

The row sums of A_C are the relevant measure: a row summing to unity passes the normalisation of that bin through unchanged, and a row summing to 0.2 removes four fifths of it.

Observable	FHC	RHC	combined
Inclusive			
p_μ	0.66–0.91	0.58–0.99	0.67–0.99
p_π (2-bin)	0.37–0.53	0.40–0.53	0.30–0.54
$\cos\theta_\mu$	0.20–0.41	0.05–0.41	0.07–0.43
$\cos\theta_\pi$	0.70–0.81	0.78–0.85	0.79–0.92
$\theta_{\mu\pi}$	0.52–1.04	0.59–1.16	0.63–1.22
θ_μ	0.58–0.78	0.17–1.14	0.52–0.66
Proton-tagged			
p_μ	0.45–1.00	0.55–0.89	0.65–0.95
p_π (2-bin)	0.33–0.86	0.43–0.45	0.47–0.49
$\cos\theta_\mu$	0.59–0.69	−0.15–0.58	0.25–0.47
$\cos\theta_\pi$	0.65–0.76	0.63–0.76	0.64–0.75
$\theta_{\mu\pi}$	0.38–0.89	0.32–0.78	0.36–0.80
$W_{\pi p}$	0.44–1.21	−0.13–1.27	0.31–1.24
$W_{\rm had}$	0.55–0.81	0.30–0.88	0.20–0.87
δp_T	0.55–0.86	0.45–0.82	0.46–0.87
$\delta\phi_T$	0.66–0.78	0.54–0.73	0.30–0.68
$\delta\alpha_T$	0.59–0.98	0.57–0.93	0.59–0.96
p_n	0.71–0.86	0.62–0.82	0.61–0.87

Three features matter for interpretation.

The suppression is large: inclusive $\cos\theta_\mu$ retains as little as 5% of a bin's normalisation in RHC. It also differs strongly between observables, which is precisely why the integrated cross section obtained by summing a differential result varies from observable to observable while the underlying physical total cannot. The D'Agostini cross-check, whose integral is flat across observables, is what establishes that this spread is the smearing rather than a physical or acceptance effect.

Two extractions have *negative* row sums, and the pattern identifies their cause. In $W_{\pi p}$ the top bin carries large negative off-diagonal terms in *all three* configurations (−1.08 FHC, −0.94 combined, −0.84 RHC); only RHC tips negative overall, because its diagonal element is the smallest of the three. The same happens in the first $\cos\theta_\mu$ bin of the proton-tagged RHC sample, whose diagonal is 0.075 against off-diagonal terms summing to −0.89.

So this is not an RHC-specific defect but the regularisation pushing back against the least-determined bin of each observable — for $W_{\pi p}$, the same top bin that is only 2% diagonal in the six-bin response and that already led to withdrawing the differential shape. A negative row is not forbidden, since A_C is a regularisation operator rather than a probability matrix, but a prediction smeared through one is being told that raising the truth in that bin *lowers* the measured content, which is a statement about the regularisation and not about the detector. Treat those two rows with corresponding caution.

Numerical data release. All fifty-one extractions are released in `report/data_release/`, indexed by `index_extractions.tsv`: the additional-smearing matrix, the full per-component covariance in cross-section units, and `curves_*.tsv` holding the unfolded data together with every model curve shown here.

A prediction supplied as a per-bin cross section is compared with these results by

$$\left(\frac{d\sigma}{dx}\right)_i^{\rm smeared} = \frac{1}{w_i}\sum_j (A_C)_{ij}\, p_j,$$

with p_j the predicted cross section in true bin j and w_i the width of bin i. This reproduces the published generator curves exactly — the worst per-bin deviation over the released matrices is

1.2×10^{-10} — so an external user can verify their implementation against any `curves_*.tsv` column before applying it to a new model.

That check is what exposed the double-smearing bug described above: before the fix the same expression missed the published curves by up to 55%, because the plotted predictions carried A_C twice.

The practical consequence is stated in the results section and repeated here because it is easy to get wrong: quote the physical total from the cut-and-count extraction, which is not subject to A_C, and treat the sum of a Wiener-SVD differential result as a quantity in smeared space rather than as a cross section.

11.3 Total flux-averaged cross section (cut-and-count)

As a check independent of the unfolding, the total flux-averaged cross section is also extracted directly from the selected event counts,

$$\sigma = \frac{N_{\text{sel}} - N_{\text{bkg}}}{\epsilon\, \Phi\, N_{\text{Ar}}}, \tag{14}$$

where N_{sel} is the number of selected events, N_{bkg} the predicted background (ν-background + EXT + dirt), $\epsilon = N_{\text{sel}}^{\text{sig}}/N_{\text{gen}}^{\text{sig}}$ the selection efficiency in the measured phase space ($\simeq 17.5\%$ for all configurations), Φ the integrated flux ($\Phi = \text{POT} \times \Phi_{\text{POT}}$, Table 4) and $N_{\text{Ar}} = 8.71 \times 10^{29}$ the number of argon nuclei in the fiducial volume (the dead z region removed), giving the same normalisation $N_{\text{Ar}}\,\Phi/10^{38}$ used for the differential result. This uses no matrix inversion; by avoiding the Wiener–SVD additional smearing it recovers the true (un-suppressed) total, consistent with the D'Agostini cross-check (Sec. 4.5 of the technical supplement) and lying above the smearing-suppressed σ_{int} of Table 18, most visibly for RHC. The uncertainty is the quadrature of the systematic breakdown (Table 17 of the technical supplement–Table 19 of the technical supplement, flux-dominated) and the cut-and-count data-statistical term $\sqrt{N_{\text{sel}}}/(N_{\text{sel}} - N_{\text{bkg}})$, which is smaller (3–4%) than the unfolded data-statistical term because no regularisation amplifies it.

11.4 Background-control sidebands

Because the analysis is blind, the background model cannot yet be confronted with beam-on data in the signal region. To exercise it before unblinding, four signal-depleted control regions ("sidebands") are defined by inverting exactly one signal requirement, leaving the rest of the selection intact:

- **CC0π**: the muon selection with no reconstructed charged pion (charged-current quasi-elastic / $2p2h$-enriched);
- **multi-π**: two or more charged pions (deep-inelastic / resonant multi-pion);
- π^0: a reconstructed shower present but the π^0 shower veto failed (CC π^0 / NC π^0);
- **cosmic**: the full signal selection with the muon–pion opening angle inverted ($\theta_{\mu\pi} > 2.6$), enriched in out-of-time cosmic activity.

Each region is evaluated with the same per-run POT scaling and CV event weighting (§5) used for the measurement. Because the current "data" is a Poisson throw of the neutrino Monte Carlo central value only (it contains no EXT or dirt component), the meaningful closure quantity here is the ratio of the pseudo-data to the CV-weighted *neutrino* MC, $N_{\text{data}}/N_{\nu\text{MC}}$; the EXT and dirt yields are shown for scale but enter the background-model test only once real beam-on data are available. Table 19 summarises the four regions for the three beam configurations.

Table 19: Background-control sidebands. "Signal fraction" is the signal contamination of the (CV-weighted neutrino) MC in the region; all four are strongly signal-depleted. $N_{\rm data}/N_{\nu\rm MC}$ is the pseudo-data to neutrino-MC ratio: the high-statistics regions (CC0π, π^0) close to unity across FHC/RHC/combined, validating the per-run weighting and POT-scaling machinery; the low-statistics regions (multi-π, cosmic) scatter within Poisson expectation ($\sqrt{N}\approx$6–10%). The background-model test against the full stack (with EXT and dirt) is performed at control-region unblinding with beam-on data.

	Signal frac.	$N_{\rm data}/N_{\nu\rm MC}$		
Sideband	(of νMC)	FHC	RHC	Combined
CC0π	$\sim 10\%$	1.00	1.00	1.00
multi-π	$\sim 23\%$	1.03	1.11	1.08
π^0	$\sim 10\%$	0.99	1.02	1.01
cosmic	$\sim 8\%$	0.94	1.24	1.10

Yields and composition. Table 20 gives the full breakdown for the three configurations. The regions differ enormously in size and in what they are sensitive to, which is what makes them complementary rather than redundant:

- **CC**0π is by far the largest ($\sim$12k pseudo-data events per beam, $\sim$24k combined). Its neutrino component is dominated by the quasi-elastic and $2p2h$ modes that feed the signal region only through pion *absorption* in the final state. It is therefore the natural handle on the FSI model that converts CC $1\pi^{\pm}$ into CC0π and vice versa; the largest single migration path into and out of the signal definition (§6).
- π^0 is the second largest (6.5–14.7k). It is enriched in CC/NC π^0 events, whose showers are precisely what the shower veto (Cut 8) is designed to remove; it therefore tests the veto efficiency and the π^0 production normalisation together. These are the events that leak into the signal region when a photon is missed.
- **multi-**π is small (127–303 events) but has the highest signal contamination of the four ($\sim$23%), because a true multi-pion event with one pion lost is exactly the wrong-topology background the selection fights. Its size is what limits its constraining power at present.
- **cosmic** is the only region where the beam-off (EXT) contribution *dominates* the prediction: 308 EXT against 68 neutrino-MC events for FHC, a ratio of 4.5, against 0.8 in CC0π. It is consequently the region that constrains the cosmic normalisation, and the one whose interpretation changes most at unblinding, since the current pseudo-data contain no EXT component at all.

What the present numbers do and do not show. With Poisson-thrown neutrino-MC pseudo-data, the ratios of Table 19 test the *machinery*, per-run POT scaling, CV weighting, the selection logic in each inverted region, and not the background model. That test passes: the two high-statistics regions sit at $N_{\rm data}/N_{\nu\rm MC} = 1.00$–$1.02$ in all three configurations, which is a non-trivial check of the per-run normalisation that caused the difficulties described in Sec. B of the technical supplement. The apparent deviations in the small regions are statistical: with 69 cosmic-sideband events for FHC, a $\sqrt{N}$ fluctuation is 12%, so the observed 0.94 and the RHC 1.24 (on 100 events, 10%) are within about one and two standard deviations respectively.

Use at unblinding. Once the control regions are opened with beam-on data the same table becomes a genuine background test, and the statistical precision available is set by the yields above: the CC0π and π^0 regions would constrain their dominant background normalisations to roughly 1% and 1.5% statistically for the combined exposure, well below the $\sim$ 20–30% cross-section systematic they are intended to test. The cosmic region would constrain the EXT normalisation to $\sim$8%, comparable to the 3–5% trigger-ratio uncertainty already assigned. The multi-π region, at $\sim$300 combined events, gives $\sim$6%, useful as a cross-check but not competitive with the others. The intended procedure is to unblind the control regions first, use them to constrain or reweight the dominant background components, and only then open the signal region.

Table 20: Sideband yields at full exposure. "Signal" is the signal contamination of the neutrino MC, "ν bkg" the remaining neutrino MC, EXT the beam-off (cosmic) sample scaled by the trigger ratio, and dirt the out-of-cryostat overlay. The pseudo-data column is a Poisson throw of the neutrino MC only, so it should be compared with Signal + ν bkg; EXT and dirt are listed for scale and enter the comparison only with real beam-on data.

Beam	Sideband	Pseudo-data	Signal	ν bkg	EXT	Dirt	Sig. frac.
FHC	CC0π	11 887	1 122	10 736	8 939	54	9.5%
FHC	multi-π	127	30	94	42	0.1	24.0%
FHC	π^0	6 555	661	5 931	5 148	12	10.0%
FHC	cosmic	69	6	68	308	2.2	8.1%
RHC	CC0π	12 261	1 204	11 034	9 281	42	9.8%
RHC	multi-π	176	35	124	43	0.1	22.0%
RHC	π^0	8 139	778	7 234	5 346	9	9.7%
RHC	cosmic	100	6	74	320	1.7	8.0%
Comb	CC0π	24 148	2 327	21 770	18 219	54	9.7%
Comb	multi-π	303	65	217	85	0.1	22.9%
Comb	π^0	14 694	1 439	13 165	10 494	12	9.9%
Comb	cosmic	169	12	142	629	2.2	8.0%

11.5 Control regions for the proton-tagged subsample

The proton-tagged selection inherits its cut logic from `CC1mu1piXp`, and with it the four sideband definitions of §11.4: the reprocessed proton-tagged files already carry `CC1mu1pi1p_sb_cc0pi`, `_sb_multipi`, `_sb_pi0` and `_sb_cosmic`. No new definitions or reprocessing are required to exercise the background model of the W/TKI measurement; the same four regions can be evaluated either as inherited (the inclusive phase space) or with a reconstructed tagged proton additionally required, which places them in the same phase space as the measurement itself. Table 21 gives both.

Three things follow from the numbers. First, all four regions are *more* signal-depleted for the proton-tagged signal than they were for the inclusive one: 6.0–7.3% signal contamination in CC0π and π^0, against 9.5–10.0% for the same regions in the inclusive phase space and a 48% purity in the signal region, because requiring a proton does not make a pion-less or π^0 event look like signal. Second, the proton requirement costs about a third of the statistics (CC0π: 11 887 $\rightarrow$ 8 268 for FHC) but suppresses the beam-off contribution by roughly a factor six (8 760 $\rightarrow$ 1 418): cosmic-induced events rarely contain a reconstructed proton above threshold, so the proton tag is itself a strong cosmic rejector. This is worth noting for the systematic budget; the EXT normalisation matters far less in the proton-tagged measurement than in the inclusive one. Third, the pseudo-data to neutrino-MC ratios remain consistent with unity in every region and configuration (0.956–1.158 with the tag), so the per-run normalisation machinery is validated in the proton-tagged phase space as well.

What these regions cannot do. All four *apply* the proton tag rather than invert it, so none of them constrains the proton *mis-tag* background; the $\sim$13% of tagged protons that are really a pion or another track, which smears $W_{\pi p}$ and the TKI variables directly. Testing that requires a region defined by inverting the proton PID (selecting candidates that pass the momentum and containment requirements but fail the LLR cut), which does need a new branch in `CC1mu1pi1p` and a reprocess. Slicing a measured W/TKI observable is *not* an alternative: because the signal definition is inclusive in those variables, cutting on them splits the signal region rather than depleting it. Explicitly, on the Run-1 FHC overlay the signal region has 53.7% purity and the slices $W_{\pi p} > 1.5$ GeV, $W_{\pi p} < 1.15$ GeV, $p_n > 0.8$ GeV/c and $\delta p_T > 0.7$ GeV/c give 50.0%, 48.9%, 47.3% and 52.0% respectively; no depletion at all. Even the kinematically forbidden region $W_{\text{had}} < m_\pi + m_p$ retains 53.4% signal, because the calorimetric mass is systematically under-reconstructed and true signal populates it through resolution.

Table 21: Sideband yields for the proton-tagged subsample at full exposure, evaluated with the inherited definitions both as-is and with a tagged proton additionally required. "Sig." is the proton-tagged signal contamination of the neutrino MC. $N_{\text{data}}/N_{\nu\text{MC}}$ compares the Poisson-thrown pseudo-data with the CV-weighted neutrino MC; EXT and dirt are listed for scale.

Beam	Region	N_{data}	Signal	ν bkg	EXT	Sig. frac.	$N_{\text{data}}/N_{\nu\text{MC}}$
inherited definitions (inclusive phase space)							
FHC	CC0π	11 887	1 122	10 736	8 760	9.5%	1.002
FHC	multi-π	127	30	94	41	24.0%	1.030
FHC	π^0	6 555	660	5 931	5 045	10.0%	0.994
FHC	cosmic	69	6.0	68	302	8.1%	0.937
RHC	CC0π	12 261	1 204	11 034	9 095	9.8%	1.002
RHC	multi-π	176	35	124	42	22.0%	1.109
RHC	π^0	8 139	778	7 234	5 239	9.7%	1.016
RHC	cosmic	100	6.4	74	314	8.0%	1.239
Comb	CC0π	24 148	2 326	21 770	17 855	9.7%	1.002
Comb	multi-π	303	64	217	83	22.9%	1.075
Comb	π^0	14 694	1 439	13 165	10 284	9.9%	1.006
Comb	cosmic	169	12	142	616	8.0%	1.095
with a tagged proton required (measurement phase space)							
FHC	CC0π	8 268	484	7 636	1 418	6.0%	1.018
FHC	multi-π	54	9.8	46	17	17.7%	0.978
FHC	π^0	3 988	292	3 730	1 162	7.3%	0.991
FHC	cosmic	24	1.7	23	5.8	6.7%	0.956
RHC	CC0π	8 287	499	7 812	1 473	6.0%	0.997
RHC	multi-π	83	14	58	18	19.8%	1.143
RHC	π^0	5 031	332	4 487	1 207	6.9%	1.044
RHC	cosmic	36	2.3	29	6.0	7.5%	1.158
Comb	CC0π	16 555	983	15 448	2 891	6.0%	1.008
Comb	multi-π	137	24	104	36	18.9%	1.071
Comb	π^0	9 019	624	8 218	2 370	7.1%	1.020
Comb	cosmic	60	4.0	52	12	7.1%	1.067

12 Secondary scope: proton-tagged subsample (hadronic invariant mass and transverse kinematic imbalance)

Scope note. This section is secondary scope, but the reason has changed and the change is substantive.

Detector variations have now been propagated through the proton-tagged selection. All thirty-three proton-tagged extractions were rebuilt on 2026-08-31 with the nine detector variation samples included and re-unfolded from those universes, so every uncertainty quoted here now carries a detector term. Those terms run from 10.5% to 64.4% with a median of 23.9%, against 16.9%–42.8% for the inclusive results: the same order, as expected for the same detector. The central values were regenerated in the same pass and therefore also postdate the central-value weighting fix.

The closure χ^2 values were recomputed in the same pass, so no quantity in this section is carried over from before the fix, and the additional-smearing and covariance matrices are released with the rest (Sec. 11.2).

What remains is a binning limitation that no systematic treatment addresses: the differential $W_{\pi p}$ shape stays withdrawn, because at this exposure no binning satisfies both the migration and the statistics requirement. The proton-tagged sample supports an integrated cross section, not a resonance-shape measurement. See Sec. 11.1.

On the six-bin $W_{\pi p}$ material below. Six-bin $W_{\pi p}$ distributions and their closure appear in this section. They are *diagnostic*: they show why the shape is not measurable, and they are not a result. The differential $W_{\pi p}$ shape is withdrawn (Sec. 11.1), and closure against A_C-smeared truth cannot restore shape information the response does not contain — the top bin is 2% diagonal in that scheme. The proton-tagged deliverable is the integrated cross section.

Requiring an identified proton in the final state, in addition to the muon and charged pion, reconstructs the full visible hadronic system and opens two classes of observable inaccessible to the inclusive CC $1\mu 1\pi Xp$ measurement: the *hadronic invariant mass*, which probes the $\Delta(1232)$ and higher resonances directly, and the *transverse kinematic imbalance* (TKI) between the muon and the hadronic system, which is sensitive to Fermi motion and final-state interactions. This subsample is implemented as a dedicated selection, `CC1mu1pi1p`, that inherits the full CC $1\mu 1\pi Xp$ selection (§8) and additionally requires a leading proton candidate; the most proton-like track (log-likelihood-ratio PID score below the proton cut), contained in the fiducial volume and above the 0.3 GeV/c tracking threshold. The signal is correspondingly narrowed to the inclusive signal plus at least one true proton above 0.3 GeV/c. On the Run-1 FHC sample this yields a selection efficiency of 11.5% and a purity of 52% (versus 17.5% and $\sim$ 56% for the inclusive selection); the proton tag trades signal statistics for a fully reconstructed hadronic final state.

12.1 Extracted cross sections

The proton tag costs efficiency and buys exclusivity: it lowers the selection efficiency from $\approx$17.5% to $\approx$12% and roughly halves the selected signal, while leaving purity at $\approx$47–48% under the more restrictive exclusive truth definition. About 87% of tagged events contain a true proton above threshold, so the mis-tag rate is $\approx$13%. Its clearest benefit is cosmic rejection: requiring a contained proton above 0.3 GeV/c removes beam-off background by up to a factor of six. The full cut-flow, the per-observable binning study and the response tables are in the technical supplement; only the extracted cross sections are given here. The full-exposure cut-flow of the proton-tagged subsample is the inherited CC $1\mu 1\pi Xp$ sequence followed by the leading-proton identification ("IdentProton"), which defines the subsample. Table 6 of the technical supplement gives the per-category, per-cut breakdown for the three beam configurations and Fig. 5 of the technical supplement the corresponding stacks; both use the same per-run POT scaling as the inclusive cut-flow (Table 9), and "signal" here is the proton-tagged signal (inclusive signal plus a true proton above 0.3 GeV/c). The proton-identification cut is the key discriminator: at FHC it rejects $\sim$64% of the neutrino background while retaining $\sim$71% of the signal, lifting the purity from 33% at the inclusive final cut (Nonprotons) to 48% in the proton-tagged sample.

Proton-tag performance. Beyond the yields, the leading-proton identification (log-likelihood-ratio PID score below the proton cut, $p_p \geq 0.3$ GeV/c, contained) is a reliable tag. Truth-matched on the FHC ν_μ overlay, 87% of the tagged events contain an energetic true proton ($p_p > 0.3$ GeV/c); a $\sim$13% mis-tag rate, dominated by a charged pion or other track being taken for the proton, and 52% are the full $\mu\pi p$ signal topology. The reconstructed proton momentum, from the proton-hypothesis range kinetic energy, carries a -9% bias (underestimate) and a 21% resolution. In cross-section terms the tag rejects $\sim$64% of the neutrino background for a $\sim$30% signal loss, improving the signal-to-background figure of merit $S/\sqrt{B}$ by $\sim$17%; the residual mis-tag and momentum smearing are folded into the response matrix and corrected by the unfolding.

Eleven differential observables are measured in the proton-tagged subsample: the six hadronic-mass and transverse-kinematic-imbalance (W/TKI) variables specific to this selection, plus the five muon/pion kinematic observables of the inclusive analysis (p_μ, p_π, $\cos\theta_\mu$, $\cos\theta_\pi$, $\theta_{\mu\pi}$) re-measured on the proton-tagged phase space so that the two selections share a common set of kinematic distributions. The two hadronic-mass variables are the *directly measured* πp invariant mass, $W_{\pi p} = \left[(E_\pi+E_p)^2-(\vec{p}_\pi+\vec{p}_p)^2\right]^{1/2}$, which makes no assumption on the neutrino direction, and the *calorimetric* hadronic mass $W_{\rm had} = \left[m_p^2 + 2m_p(E_{\rm cal} - E_\mu) - Q^2\right]^{1/2}$ built from the reconstructed calorimetric energy. The four TKI observables; the transverse-momentum imbalance δp_T, its two angles $\delta\alpha_T$ and $\delta\phi_T$, and the inferred struck-nucleon momentum p_n, are computed from the muon versus the summed $(\pi + p)$ hadronic system using the framework STV tool. Proton momenta are obtained from the proton-hypothesis range-based kinetic energy. The two W variables are measured in six equal-population bins; the four TKI variables use the coarsened two-bin scheme adopted after the finer binning was found to sit inside the resolution (Sec. 4.2 of the technical supplement). The per-run POT normalisation (Sec. B of the technical supplement) removes the background-subtraction bias that had previously forced a coarser three-bin binning for the lower-statistics reverse-horn-current sample, while the five inherited kinematic observables use the inclusive-analysis binning (four to seven bins). All bin edges are set from the validated true-signal distributions (Table 7 of the technical supplement).

Reconstruction from input quantities

All six observables are built from the three candidate tracks (muon, charged pion, proton) using only reconstructed quantities stored in the input ntuples: the track direction cosines $\hat{n} = (\texttt{track\_dirx}, \texttt{track\_diry}, \texttt{track\_dirz})$, the muon-hypothesis range and multiple-scattering momenta (`track_range_mom_mu`, `track_mcs_mom_mu`) and the proton-hypothesis calorimetric kinetic energy (`track_kinetic_energy_p`).

Candidate three-momenta. The muon momentum magnitude is range-based when the muon track ends inside the fiducial volume, $p_\mu = \texttt{track\_range\_mom\_mu}[\mu]$, and multiple-Coulomb-scattering based otherwise, $p_\mu = \texttt{track\_mcs\_mom\_mu}[\mu]$. The charged pion, being minimum-ionising, is assigned the muon-hypothesis range momentum $p_\pi = \texttt{track\_range\_mom\_mu}[\pi]$. The proton momentum follows from its calorimetric kinetic energy under the proton hypothesis, $p_p = \sqrt{T_p^2 + 2m_pT_p}$ with $T_p = \texttt{track\_kinetic\_energy\_p}[p]$. Each three-momentum is $\vec{p}_i = p_i\,\hat{n}_i$, and the energies are $E_\mu = \sqrt{p_\mu^2 + m_\mu^2}$, $E_\pi = \sqrt{p_\pi^2 + m_\pi^2}$, $E_p = T_p + m_p$ (masses $m_\mu = 105.7$, $m_\pi = 139.6$, $m_p = 938.3$ MeV).

Directly measured mass $W_{\pi p}$. The invariant mass of the pion–proton four-vector sum, using no neutrino-direction assumption:

$$W_{\pi p} = \sqrt{(E_\pi + E_p)^2 - |\vec{p}_\pi + \vec{p}_p|^2}.$$

Transverse kinematic imbalance. The TKI observables treat the muon against the visible hadronic system $\vec{p}_{\rm had} = \vec{p}_\pi + \vec{p}_p$, $E_{\rm had} = E_\pi + E_p$. With the components transverse to the beam

($\hat{z}$) written $\vec{p}_T^{\mu}$ and $\vec{p}_T^{\text{had}}$,

$$\delta p_T = |\vec{p}_T^{\mu} + \vec{p}_T^{\text{had}}|, \quad \delta\alpha_T = \arccos\frac{-\vec{p}_T^{\mu}\cdot(\vec{p}_T^{\mu} + \vec{p}_T^{\text{had}})}{|\vec{p}_T^{\mu}|\,\delta p_T}, \quad \delta\phi_T = \arccos\frac{-\vec{p}_T^{\mu}\cdot\vec{p}_T^{\text{had}}}{|\vec{p}_T^{\mu}|\,|\vec{p}_T^{\text{had}}|},$$

both angles in degrees folded onto $[0, 180]$. For a stationary nucleon with no final-state interactions $\delta p_T \to 0$; Fermi motion broadens it and FSI drive $\delta\alpha_T \to 180°$.

Calorimetric mass W_{had} and struck-nucleon momentum p_n. The calorimetric available energy is $E_{\text{cal}} = E_\mu + (E_{\text{had}} - m_p) + E_b$ with the argon removal energy $E_b = 24.78$ MeV, giving the momentum transfer $Q^2 = -\left[(0, 0, E_{\text{cal}}, E_{\text{cal}}) - p^\mu\right]^2$ and

$$W_{\text{had}} = \sqrt{m_p^2 + 2m_p(E_{\text{cal}} - E_\mu) - Q^2}$$

(returned as 0 where reconstruction fluctuations make the argument negative). The inferred initial-state nucleon momentum combines the transverse imbalance with a longitudinal component p_L from energy–momentum balance, $p_n = \sqrt{\delta p_T^2 + p_L^2}$. The transverse-imbalance and calorimetric quantities are evaluated with the framework STV tool (`STVTools::CalculateSTVs`, option 1); it was formulated for a single-proton hadronic system and is applied here to the summed (πp) system with the proton rest mass, an approximation adequate for the leading-order kinematics.

The pipeline is validated end-to-end on FHC with a fake-data closure: the Poisson-thrown central-value pseudo-data are unfolded with the same Wiener-SVD procedure (§10) and compared to the A_C-smeared truth. Table 22 shows the result. The bin-by-bin χ^2/ndf is well below unity for every observable, confirming the unfolded pseudo-data are statistically consistent with truth and that the response, background subtraction and normalisation are self-consistent. The integrated-cross-section ratios (0.94–1.19) reflect the same non-norm-preserving Wiener-SVD additional smearing A_C seen in the inclusive measurement (Sec. 4.5 of the technical supplement), largest for $\delta\alpha_T$ and $\theta_{\mu\pi}$. The systematic treatment here includes the flux, cross-section and reinteraction and detector-variation terms. The detector term was added on 2026-08-31, when the detector-variation samples were referenced by this selection for the first time and all thirty-three proton-tagged universes were rebuilt; before that these results carried none.

The reverse-horn-current measurement is complete at the same finer binning (Table 23). All eleven RHC observables close positive, with σ_{int} clustered in $0.38–0.49 \times 10^{-38}$ cm^2/Ar, below the FHC band (0.43–0.57), as expected for the $\bar{\nu}_\mu$-enriched RHC flux; and $\chi^2/\text{ndf} \lesssim 1$ for almost every observable.

Reaching this point required resolving a normalisation subtlety that is worth recording, because it produced spurious *negative* cross sections for this subsample at an intermediate stage. The processed ntuples use two different POT conventions: the files behind the inclusive selection store the *total* run MC POT duplicated in every file of a multi-file run, whereas the proton-tagged files store each file's *own* POT. Normalising by the individual file POT is therefore correct for the first set but scales every split of a multi-file run to the full run exposure in the second, inflating the simulated background by the split multiplicity ($\sim 3\times$ for RHC) and driving the background-subtracted spectrum negative. Normalising by the summed per-file POT has the mirror problem. The analysis now normalises by the sum of the *distinct* per-file POT values, which reduces to the correct run total under either convention; FHC, with one MC file per run, is unaffected by the choice.

The combined-sample measurement is complete on the same binning, is included in the results tables and the data release, and covers all eleven combined proton-tagged observables.

Table 22: FHC fake-data closure for the eleven proton-tagged observables at the finer binning. $\sigma_{\rm int}$ is the integrated (unfolded) cross section over the measured phase space (10^{-38} cm^2/Ar); "unf./truth" is the ratio of integrals; χ^2 is the bin-by-bin comparison of unfolded pseudo-data to A_C-smeared truth (ndf = number of bins). The upper block is the six W/TKI observables; the two W variables are at six bins, while the four TKI variables use the coarsened two-bin scheme adopted after the finer binning was found to sit inside the resolution; the lower block is the five kinematic observables inherited from the inclusive analysis. The small χ^2/ndf validates the chain; the integral ratios are the A_C smearing effect, not a bias.

Observable	$\sigma_{\rm int}$	unf./truth	χ^2/ndf
$W_{\pi p}$	0.522	1.09	0.31/6
$W_{\rm had}$	0.483	1.21	2.52/6
δp_T	0.513	1.08	0.05/2
$\delta\alpha_T$	0.529	1.05	0.12/2
$\delta\phi_T$	0.517	1.07	0.26/2
p_n	0.555	1.10	0.21/2
p_μ	0.514	1.08	0.60/7
p_π	0.565	1.10	0.11/2
$\cos\theta_\mu$	0.441	1.09	1.13/5
$\cos\theta_\pi$	0.426	1.08	2.04/4
$\theta_{\mu\pi}$	0.501	1.11	0.56/5

Figure 24 shows the six FHC W/TKI differential cross sections at the finer six-bin binning, overlaid with all four generator predictions. The generator W/TKI predictions were regenerated at this binning (GENIE and GiBUU natively, NuWro and NEUT from their event vectors inside the SL7 container); the rebinning conserves each generator's integral to the previous three-bin value (GENIE 0.501, NuWro 0.646, GiBUU 0.735, NEUT 0.783 $\times\, 10^{-38}$ cm^2/Ar). All curves, data, tune and generators, are A_C-smeared, so the comparison is made consistently in the measurement space (Sec. 3.1 of the technical supplement). The predictions are built at truth level from each generator's event-level final state, GENIE gst events, GiBUU `FinalEvents`, and the NuWro and NEUT event vectors reprocessed in their respective SL7-container environments, by applying the inclusive signal plus a leading true proton above 0.3 GeV/c and evaluating the six observables with the same formulas as the selection (§12.1), then combining the ν_μ and $\bar{\nu}_\mu$ samples over the FHC flux to a per-argon cross section. The four generators span a factor of ~ 1.6 in the integrated proton-tagged cross section, GENIE lowest (0.50), then NuWro (0.65), GiBUU (0.74) and NEUT highest (0.78×10^{-38} cm^2/Ar), bracketing the data (~ 0.58), and they differ most in the low-δp_T / high-p_n region sensitive to the proton kinematics and final-state interactions. As throughout, the unfolded fake data sit above the A_C-smeared truth by the Wiener-SVD normalisation factor.

Table 23: RHC fake-data closure for the eleven proton-tagged observables, binning and columns as in Table 22. All observables close positive with the convention-agnostic MC POT normalisation described above; σ_{int} sits just below the FHC band, consistent with the $\bar{\nu}_\mu$-enriched RHC flux.

Observable	σ_{int}	unf./truth	χ^2/ndf
$W_{\pi p}$	0.430	1.17	0.95/6
W_{had}	0.398	1.15	1.21/6
δp_T	0.425	1.07	0.08/2
$\delta\alpha_T$	0.487	1.17	0.24/2
$\delta\phi_T$	0.442	1.10	0.06/2
p_n	0.456	1.11	0.06/2
p_μ	0.428	1.15	0.24/7
p_π	0.433	1.05	0.66/2
$\cos\theta_\mu$	0.466	1.24	1.68/5
$\cos\theta_\pi$	0.379	1.11	0.81/4
$\theta_{\mu\pi}$	0.460	1.19	0.73/5

Table 24: Combined FHC+RHC fake-data closure for the eleven proton-tagged observables at binning and columns as in Table 22. All eleven close positive and every χ^2 is comfortably below its number of bins. This completes the proton-tagged measurement in all three beam configurations. Note that the combined result is an independent extraction, not a fluence-weighted average of the FHC and RHC ones, so it is not required to lie between them; see the discussion below.

Observable	σ_{int}	unf./truth	χ^2/ndf
$W_{\pi p}$	0.466	1.08	0.35/6
W_{had}	0.428	1.19	3.24/6
δp_T	0.513	1.03	0.06/2
$\delta\alpha_T$	0.507	1.09	0.06/2
$\delta\phi_T$	0.474	1.00	0.15/2
p_n	0.494	1.09	0.11/2
p_μ	0.487	1.12	0.43/7
p_π	0.497	1.08	0.12/2
$\cos\theta_\mu$	0.457	1.13	0.56/5
$\cos\theta_\pi$	0.413	1.09	3.08/4
$\theta_{\mu\pi}$	0.481	1.11	0.10/5

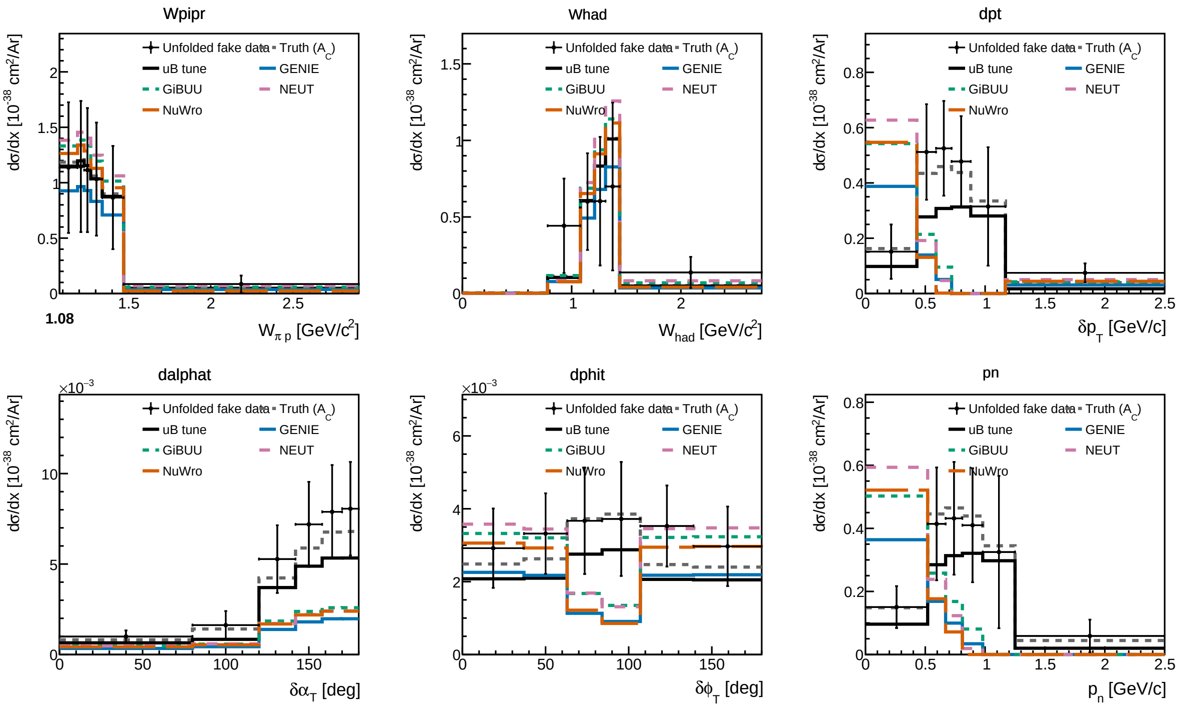


Figure 24: Proton-tagged (`CC1mu1pi1p`) differential cross sections for FHC: the πp invariant mass $W_{\pi p}$, the calorimetric hadronic mass W_{had}, and the four transverse-kinematic-imbalance observables δp_T, $\delta\alpha_T$, $\delta\phi_T$ and p_n. Unfolded fake data (black points) are compared to the A_C-smeared fake-data truth (grey dashed), the MicroBooNE tune (black), and the GENIE (blue), GiBUU (green), NEUT (purple) and NuWro (orange) predictions built for the same proton-tagged phase space. $W_{\pi p}$ falls from the $\Delta(1232)$ threshold region; $\delta\alpha_T$ rises toward 180°, the signature of final-state interactions.

With all three configurations in hand, the combined σ_{int} lies between the FHC and RHC values for all eleven observables. Earlier drafts reported four exceptions, at the few-per-cent level; they were artefacts of the pre-rebuild proton-tagged extractions and do not survive the detector-variation rebuild and the correction of the duplicated A_C.

That the combined result now brackets cleanly is reassuring but should not be over-interpreted: the combined measurement is *not* required to lie between the two single-mode ones. It is an independent extraction with its own response matrix, its own regularisation and its own A_C, not a fluence-weighted average, so excursions at the few-per-cent level would have been unremarkable against a $\sim$30% systematic uncertainty.

The spread across observables within a configuration, 0.41 to 0.51 for the combined set, is likewise not a physics spread but the residual of the additional smearing discussed in the next section: each observable's A_C rescales the integral by a different factor, so the eleven numbers are not eleven independent measurements of the same quantity.

13 Systematic Uncertainties

Full systematic covariance propagation is complete for all fifty-one extractions. Every result carries flux, cross-section-model, hadron-reinteraction, detector-variation, POT, target-count and statistical terms, propagated through the unfolding by the error propagation matrix and released per component (Sec. 11.2). The proton-tagged detector term was the last to be added, on 2026-08-31; before that those thirty-three results carried none. Table 25 lists the sources.

Table 25: Systematic sources and their bin-averaged fractional contribution, computed from the released per-component covariance rather than estimated. Ranges span observables and configurations; the two families are given separately because the proton-tagged sample is both statistically sparser and more detector-sensitive.

Source	Branch / method	Inclusive (%)	Proton-tagged (%)
Flux (PPFX multisims)	`weightsFlux`	24.0–33.4	19.5–35.5
Detector response	Dedicated samples	16.9–42.7	10.5–64.4
Cross-section model	`weightsGenie`	9.1–17.8	7.2–22.8
Hadron re-interaction	`weightsReint`	3.4–12.1	3.0–15.6
POT counting	Beam toroids	2.9–4.3	2.4–4.3
Target count	FV geometry	1.5–2.1	1.2–2.1
MC statistics	universe spread	2.0–5.9	2.1–10.4
EXT statistics	beam-off sample	1.9–6.4	2.2–9.5
Data statistics	thrown fake data	6.6–20.1	7.2–38.2
Prediction total	quadrature sum	**33.8–59.3**	**24.3–77.2**
Total	incl. data stats	**35.2–61.7**	**26.1–86.2**

Flux is the largest single term in the inclusive measurement and the detector response the largest in the proton-tagged one. Every entry is propagated into the released covariance; no source declared in the systematics configuration is left unpropagated.

13.1 Systematic breakdown by source

The framework builds a covariance matrix for every systematic; the source-group decomposition method is illustrated below on a single configuration (the definitive per-configuration breakdowns for the full FHC, RHC, and combined exposures are Table 17 of the technical supplement–Table 19 of the technical supplement). The bin-averaged fractional uncertainty from each source group (on the optimised binning) is:

Table 26: Bin-averaged fractional systematic uncertainty (%) by source group on the optimised binning.

Source	p_μ	p_π	$\cos\theta_\mu$	$\cos\theta_\pi$	$\theta_{\mu\pi}$
Total	27.2	28.3	28.8	33.8	25.1
Cross section (GENIE)	8.6	8.5	9.1	9.5	8.1
Flux, PPFX	21.8	23.6	20.7	25.4	18.7
Flux, beamline geom.	2.3	2.2	2.3	3.3	1.6
Detector	9.6	10.2	14.0	14.6	10.5
Reinteraction	3.2	2.6	3.4	3.4	2.5
MC + data stats	8.6	6.5	9.4	11.9	9.3
POT + targets	2.7	2.9	2.7	3.0	2.4

The NuMI flux uncertainty has two parts, listed separately above: the PPFX hadron-production term (the dominant systematic, 19–26%) and the beamline-geometry term. The latter is built from the 12 geometry-variation weights (`Horn_2kA`, `Horn1/2_x/y_3mm`, `Beam_spot_*`, `Beam_shift_*`, `Horns_*_water`, `Target_z_7mm`) injected into the numuMC overlay via `AddBeamlineGeometryWeights`, summed in quadrature as two-sided unisim variations. It is sub-dominant, 1.6–3.3% across the observables. (Building it required a framework fix: the covariance code dereferenced a null histogram when a systematic, here the beamline weights, present in the numuMC but not the dirt overlay, was absent from some MC sample; it now falls back to that sample's central value, i.e. unit weight.) The per-bin breakdowns are shown in Fig. 25.

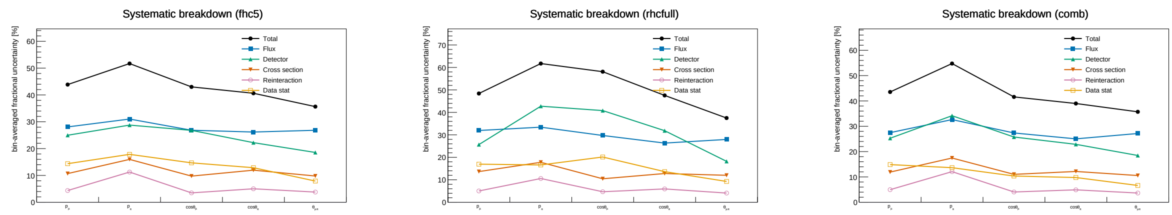


Figure 25: Bin-averaged fractional systematic uncertainty by source vs. observable, for FHC (left), RHC (centre) and combined (right); the figure form of Table 17 of the technical supplement–Table 19 of the technical supplement. Colours use the Okabe-Ito colourblind-safe palette with distinct markers per source: total (black), flux (blue), detector (green), cross section (vermillion), reinteraction (purple), data stat (orange).

13.2 Anatomy of the flux term: genuine flux vs. unfolding amplification

The flux (PPFX) term dominates the budget, and at 17–26% it is larger than the $\sim 17\%$ normalisation uncertainty usually quoted for the NuMI flux. It is therefore worth asking how much of it is the flux model itself and how much is statistical amplification introduced by the unfolding. To separate the two we evaluate the flux covariance at two stages, on the reconstructed spectrum (before unfolding) and on the unfolded result, and, at each stage, both integrated (the fully correlated, normalisation-like number, from the sum of all covariance elements) and bin-averaged (the mean of the diagonal fractional errors, i.e. the number quoted in Table 26 and Fig. 25).

Table 27: Flux (PPFX) fractional uncertainty before and after unfolding. The reconstructed-level value is $\sim 17\%$ and flat across observables; the genuine flux-normalisation uncertainty. The unfolded value is that same $\sim$17% amplified by the Wiener–SVD propagation through the finite-statistics response; the amplification tracks the data+MC statistical quality of each observable.

	p_μ	p_π	$\cos\theta_\mu$	$\cos\theta_\pi$	$\theta_{\mu\pi}$
Flux, reco (integrated / norm)	16.9	17.0	16.7	16.7	16.7
Flux, reco (bin-averaged)	17.7	16.4	16.6	17.0	17.0
Flux, unfolded (bin-averaged)	19.8	22.4	19.8	25.5	17.5
Amplification factor	1.12	1.37	1.20	1.49	1.03
Excess from unfolding (abs.)	+2.1	+6.0	+3.2	+8.5	+0.5
Data+MC stats (unfolded)	7.9	6.3	8.4	12.2	8.8

Three points follow. First, the *genuine* flux uncertainty is $\sim 17\%$ and identical for all five observables, as expected for a normalisation term. Second, it is not itself contaminated by Monte Carlo statistics: at reco level the integrated (correlated) and bin-averaged (diagonal) numbers agree to better than a percent, whereas statistical noise leaking into the covariance would inflate the diagonal well above the correlated norm. This is expected; the $\sim$600 PPFX universes reweight the *same* events, so there is no independent per-universe statistical scatter. Third, the elevated unfolded values are the $\sim$17% flux *amplified* by the Wiener–SVD propagation $A_C M\, V_{\rm flux}\, M^\top A_C^\top$ through the finite-statistics response. It is at this step, not in the flux multisim, that the response statistics transfer into the flux term: the amplification is largest exactly where the statistics are thinnest ($\cos\theta_\pi$: stat. 12.2%, amplification 1.49×, +8.5%) and negligible where the response is well conditioned ($\theta_{\mu\pi}$: 1.03×, +0.5%). The lever for reducing the flux term is therefore the response conditioning (more overlay statistics, or the observable's binning and regularisation strength), not the flux model; $\cos\theta_\pi$; the noisiest observable across the entire budget, is where it would matter most.

14 Analysis Pipeline

The primary measurement is produced by a single driver, `run_analysis.sh`, which chains the selection, the per-observable cross-section extraction, and the comparison/diagnostic stages:

```
selection/ccpi_selection.C   ->  histograms.root + unfolding_inputs.root
                                  | (5 observables x 8 methods)
xsec/ccpi_xsec.C(obs,method) ->  xsec_{obs}_{method}.root  (40 files)
xsec/compare_unfolding.C     ->  compare_plots/
xsec/compare_generators.C    ->  gen_compare/
selection/event_display.C    ->  event_displays/
```

Each output file is validated for freshness and non-zero content after the step that produces it. The official-framework cross-check of Sec. C of the technical supplement is driven separately (`xsec_analyzer/run_nuwro_observables.sh`).

15 Limitations

This section states what limits the measurement as it now stands. It is deliberately about the present, not the past: the corrections that got the analysis here, with their before-and-after values, are in the change log, and the studies behind the scope decisions are in the technical supplement. Earlier drafts carried two retrospective sections in this position, and a reader looking for the limitations found a catalogue of fixed bugs instead.

The analysis is blind, and that is the dominant limitation. No beam-on signal region has been opened. Every "data" point is Poisson-thrown from the central-value Monte Carlo, so everything demonstrated here concerns the extraction machinery rather than nature. In particular, no statement in this note tests the background model against data. At ≈55% purity, with proton-to-pion misidentification the largest neutrino background, the measurement depends on that model substantially, and fake-data closure cannot probe it: the same simulation supplies both the pseudo-data and the background estimate. This is the single largest untested assumption in the analysis.

Pion momentum is not recoverable, and that sets the scope. Charged pions interact hadronically before ranging out, so the reconstructed momentum saturates near 0.34 GeV/c and carries little information above roughly 0.4 GeV/c. Every alternative estimator was tried — golden-pion tagging, multiple Coulomb scattering, calorimetry, multivariate regression, daughter-energy recovery, the proton-tagged selection — and a truth-level golden selection performs no better than the trained classifier, which places the limit in the physics rather than the estimator. This is why p_π is reported in two coarse bins and why the differential $W_{\pi p}$ shape is withdrawn: at this exposure no binning satisfies both the migration and the statistics requirement.

The extraction carries a small negative bias. The ensemble of thirty-two throws puts the mean 2.5% below the reference, 2.1σ on the mean, with per-bin pull means consistently negative in both observables tested. It is about a fifth of the statistical uncertainty and far inside the systematic band, but it is a bias rather than a fluctuation and is not currently corrected for or assigned an uncertainty.

Coverage is demonstrated for two observables, in one configuration. The pull and coverage study covers FHC p_μ and $\cos\theta_\mu$. Both give widths consistent with unity, and the second was chosen because its A_C suppression is the most severe in the analysis, but RHC and the proton-tagged family — where the statistics are sparser — have not been tested.

The proton-tagged results are secondary, and their model comparisons are the weakest part. Their systematics are complete and their central values current, but the exposure does not support a differential $W_{\pi p}$ shape, and the sample has no dedicated control region for the $\approx 13\%$ proton mis-tag: every existing sideband applies the proton requirement rather than inverting it, so nothing directly constrains a wrong-track proton assignment, which is exactly what distorts the hadronic observables.

A_C is not norm preserving, so two different totals exist. The Wiener-SVD differential result estimates $A_C x_{\text{true}}$, and the row sums fall as low as 0.046. Summing it over bins therefore does not give a physical total cross section; the physical total is the cut-and-count value, quoted separately. Two matrices have negative row sums in their least-determined bin.

Actions required before unblinding.

- Open and validate the background control regions in beam-on data, including an inverted proton-PID region for the proton-tagged sample, before the signal region.
- Re-enable the EXT subtraction (`fake_data_ext_cancel` is a fake-data convenience and must not survive into the real-data measurement).
- Fix the permitted response to any control-region discrepancy in advance, and record it, so that no post-unblinding retuning is possible.

The automated guards that protect against regression — the normalisation manifest, the systematics/file-list coupling assertion and the staleness check — are described in Sec. 14.

16 Summary

We have developed a measurement of $\nu_\mu + \bar{\nu}_\mu$ charged-current single charged-pion production on argon using the full MicroBooNE NuMI exposure: forward horn current (8.857×10^{20} POT), reverse horn current (1.1082×10^{21} POT), and their combination (1.9939×10^{21} POT).

The analysis is blind. No beam-on signal region has been opened. Every "data" point in this note is Poisson-thrown from the central-value Monte Carlo, so the results demonstrate that the extraction works; they are not measurements, and no statement here about a generator is a statement about nature.

Measured phase space and selection. $|\mathrm{PDG}_\nu| = 14$, charged current, true vertex in the fiducial volume, exactly one muon and one charged pion, no neutral pions or kaons, any number of protons, with $p_\mu > 0.15\,\mathrm{GeV}/c$, $p_\pi > 0.175\,\mathrm{GeV}/c$ and $\theta_{\mu\pi} < 2.6$ rad. The selection reaches $\approx 15.6\%$ efficiency at $\approx 55\%$ purity. Angles and transverse observables are defined about the per-event neutrino direction, not the detector z axis.

What is measured. Fifty-one extractions: eighteen inclusive and thirty-three proton-tagged, across the three configurations. The primary scope is the total flux-averaged cross section, $d\sigma/dp_\mu$, the muon and pion angular cross sections, $d\sigma/d\theta_{\mu\pi}$, and a deliberately coarse two-bin $d\sigma/dp_\pi$. Inclusive total uncertainties run 33.8–59.3%, with the detector term 16.9–42.8% and flux the largest single contribution.

The proton-tagged subsample is secondary scope, with total uncertainties 24.3–77.2%. Its systematics are complete, but the exposure does not support a differential $W_{\pi p}$ shape at any binning, so that result is withdrawn and only the integrated proton-tagged cross section is offered.

What limits the measurement. Charged pions interact hadronically before ranging out, so the reconstructed momentum saturates near $0.34\,\mathrm{GeV}/c$. A survey of alternatives — golden-pion tagging, multiple Coulomb scattering, calorimetry, multivariate regression, daughter-energy recovery, and the proton-tagged selection — recovers none of it, and a truth-level golden selection performs no better than the classifier, which places the limit in the physics rather than the estimator. This is what reduces p_π to two bins and $W_{\pi p}$ to an integrated quantity, and it is the single most consequential limitation of the analysis.

Validation. The extraction reproduces its own input: the mean closure ratio over the eighteen inclusive extractions is 1.001 with second-derivative regularisation and 1.007 with the identity. The algebraic identity $UR = A_C$ holds to 4.4×10^{-6}, which tests the unfolding construction rather than only its behaviour on one sample. D'Agostini iterative unfolding, whose integral is flat across observables where the Wiener-SVD spread is large, establishes that the observable-to-observable spread of the Wiener-SVD integrals is the additional smearing rather than physics or acceptance.

The closure χ^2 values should not be read as agreement: their p-values cluster near unity because the measurement and the truth it is compared against share their systematics, which cancel in the residual while still inflating the covariance used to judge it.

Corrections that changed results. Nine, each recorded in the change log with before-and-after values. The largest were an obsolete flux (a factor $\sim$3.48), a double application of the central-value weight to the fake data (which moved every cross section by 3.8–12.9%), a regularisation operator that assumed a uniform grid and drove A_C towards rank one, an angular convention defined about detector z rather than the beam, and — most recently — a double application of A_C to every model prediction, which inflated the generator χ^2 and produced an apparent anomaly in the muon-angle comparisons that does not exist.

That last one was found by asking whether an external user could reproduce a published generator curve from the released matrix. Internal closure could not have found it: the data and the truth were both correct, and only the comparison was wrong.

Reproducibility from raw inputs. All fifty-one extractions were rebuilt from the processed ntuples with every cached universe deleted — new universe files, new unfolding, new covariance — into a separate tree so that a failure could not damage the release it was verifying. The rebuilt spectra agree with the released ones *bin by bin*, with no bin differing by more than 10^{-9} in any of the fifty-one. The release is therefore regenerable from the raw inputs and the recorded configuration, not merely from a preserved intermediate.

This is worth separating from the closure and ensemble tests. Those ask whether the extraction is right; this asks whether what is published is what the code currently produces. Given that this analysis has twice shipped a downstream product that lagged a fix, that is a distinct question and it now has an answer.

Before unblinding. Control regions must be opened in data first, with the permitted response to any discrepancy fixed beforehand; a clean rebuild from empty caches must reproduce the release; and the configuration-to-systematics coupling needs a check that every declared source has its expected universes, since it was exactly that interface which allowed the proton-tagged detector uncertainty to vanish silently. An ensemble of throws, rather than the single one used here, would turn the closure caveat above into a quantitative coverage statement.

A Observable Binning

Generated directly from the released `curves_*.tsv` files, so these are by construction the bin edges of the published results rather than a transcription. The binnings are identical across the three beam configurations. Earlier drafts of this appendix listed a finer exploratory binning that predates the resolution-driven optimisation; that binning is not used for any released result.

Table 28: Bin edges of every released observable, read from the data release.

Observable	Bins	Edges
Inclusive		
p_μ (GeV/c)	7	0.15, 0.35, 0.55, 0.75, 0.95, 1.25, 1.75, 3
p_π (GeV/c)	2	0.175, 0.205, 1
$\cos\theta_\mu$	5	-1, 0.45, 0.65, 0.8, 0.9, 1
$\cos\theta_\pi$	4	-1, -0.1, 0.35, 0.75, 1
$\theta_{\mu\pi}$ (rad)	5	0, 0.6, 0.85, 1.3, 1.85, 2.6
θ_μ (rad)	5	0, 0.43, 0.62, 0.84, 1.17, 3.15
Proton-tagged		
p_μ (GeV/c)	7	0.15, 0.35, 0.55, 0.75, 0.95, 1.25, 1.75, 3
p_π (GeV/c)	2	0.175, 0.205, 1
$\cos\theta_\mu$	5	-1, 0.45, 0.65, 0.8, 0.9, 1
$\cos\theta_\pi$	4	-1, -0.1, 0.35, 0.75, 1
$\theta_{\mu\pi}$ (rad)	5	0, 0.6, 0.85, 1.3, 1.85, 2.6
$W_{\pi p}$ (GeV/c^2)	6	1.08, 1.19, 1.23, 1.27, 1.34, 1.47, 2.9
W_{had} (GeV/c^2)	6	0, 0.78, 1.08, 1.21, 1.31, 1.44, 2.74
δp_T (GeV/c)	2	0, 0.3, 2.5
$\delta\phi_T$ (rad)	2	0, 50, 180
$\delta\alpha_T$ (rad)	2	0, 120, 180
p_n (GeV/c)	2	0, 0.375, 2

The upper edge of the top p_π bin is the kinematic limit of the sample rather than a chosen boundary: that bin is an inclusive high-momentum region, not a differential point, for the reasons set out in the technical supplement.

B Selection Cut Parameter Summary

Table 29: All selection threshold constants from `selection/ccpi_selection.C`.

Parameter	Value	Ntuple variable
Topological score	> 0.67	`topological_score`
Muon track score	> 0.80	`trk_score_v`
Muon vertex distance	≤ 4.0 cm	
Muon track length	> 10.0 cm	`trk_len_v`
Muon LLR PID	> 0.20	`trk_llr_pid_score_v`
Muon MIP BDT	≥ -0.10	TMVA BDT
Pion track length	> 20.0 cm	`trk_len_v`
Pion vertex distance	< 4.0 cm	
Pion LLR PID	> 0.10	`trk_llr_pid_score_v`
Pion MIP BDT	> -0.10	TMVA BDT
Pion pion-BDT	> -0.10	TMVA BDT
Pion Bragg score	≥ 0.08	`trk_bragg_pion_v`
Max. opening angle	< 2.6 rad	
Shower Y-plane hits	< 50 (if 1 shower)	
Shower vertex distance	> 10 cm (if 1 shower)	
Non-proton multiplicity	< 4	LLR PID > 0.1
Primary track multiplicity	< 5	
Cosmic BDT	> 0.0 (disabled)	`bdt_cosmic`
Software trigger	$= 1$	`swtrig`

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